CONTINUATION

HelixCode CLI-Agent Fusion — Programme Continuation Guide

Field	Value
Revision	20
Created	2026-05-06
Last modified	2026-07-28T17:45Z (RED-polarity repair + bisect guidance, §11.4.131/§12.10)
Status	active
Status summary	**REVISION 19 CORRECTION (2026-07-28T18:20Z) — produced by a dedicated doc-refresh pass scoped to `RESUME.md` + this file only** (four OTHER streams were concurrently active in `helix_code/internal/llm/`, `helix_code/applications/`, `submodules/helix_agent/internal/agents/`, and `submodules/debate_orchestrator` — untouched by this pass). Rev 18's numbers are STALE; independently re-derived (§11.4.6), nothing copied forward: **(E1) Meta-repo HEAD moved `c29e1dcc`→`99ff7d8e`, 12 more commits, `main` now 20 commits ahead of all 4 remotes** (was 8). The new commits include a self-inflicted, self-caught §11.4.108 SOURCE→ARTIFACT gap: `3fd55a4d` recovered an orphaned regression guard whose composite literal referenced a struct field that did not exist in the committed tree — `internal/server` did NOT compile at `3fd55a4d`/`3c8197cf`/`905a0b0a`, caught by independent review and fixed in `34e264e1`. `3fd55a4d`'s own message additionally discloses its `RED_MODE=1` branch is itself a bluff (asserts on a local replica of pre-fix behaviour, cannot fail) — **REPAIRED in**

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	`bf1ce692`: both polarities drive the real handler; four-quadrant proof captured (pre-fix RED=PASS / pre-fix GREEN=FAIL / fixed RED=FAIL / fixed GREEN=PASS). Its "tracked" claim was itself unbacked — no matching entry in `docs//Issues.md`/the SSoT DB (§11.4.148) — so it was fixed, not back-filled. **`git bisect skip` is REQUIRED across `3fd55a4d`/`3c8197cf`/`905a0b0a`** — they do not build `internal/server`; a bisect stopping there has found a build break, not the bug. `c7484cc7` (scaling-harness signal-to-noise fix, `Trials` 3→5, confirmed in source) and `640db264` (aurora_os uint64-underflow fix) both carry pasted verification in their own messages. The newest commit `99ff7d8e` reviewed `3c8197cf`'s new `*_racefix_test.go` files by READING alone (host lacks X11/GL headers to build them) and found + fixed a deterministic nil-pointer panic in the harmony_os file — proof the "authored but never executed" GUI-test gap (below) can hide real defects. **(E2) `helix_agent` moved `a345c551`→`f451d342`, 9 more commits, now 11 ahead** (was 2); this pass verified BY DIFF that a Critical review finding described elsewhere as "remediation in flight" (Dreamer `memoryMu` not covering `cleanupPhase`'s unlocked `MEMORY.md` write) has actually LANDED in `f451d342` — `cleanupPhase` now shares the same `beginMemoryWrite()`/`end()` helper as `saveMemories()`. Independent review of `f451d342` and `129094b0` (landed after the reported 9-commit/3-round CLEAN GO) could NOT be confirmed from any git-visible artefact — treat as open. **(E3) `tool_schema` is now FULLY PUSHED** (0 ahead, was 1). **(E4) `debate_orchestrator` is tracked here for the first time** — `b6c90e7`, fully pushed, but only ONE remote (github) configured vs siblings' 4-remote fan-out; flagged for the operator, not fixed unilaterally (out of this pass's scope). **(E5) The working**

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	tree carries 73 porcelain entries (46 untracked + 27 modified; was ~68), overwhelmingly more `docs/qa/<run-id>/` evidence from a concurrent stream's ongoing G7 work. **(E6) The constitution-gate-sweep GREEN result, G7 resolution, constitution-pin match, and platform-up state from rev 18 were all reconfirmed live during this pass and are UNCHANGED** — this pass did not re-run the constitution-gate sweep itself (doc-only scope) but re-checked the constitution pin, carrier ceilings, tags, and systemd/port state directly, all matching rev 18. **(E7) Two NEW open items surfaced this pass, not present in rev 18:** the three `main_racefix_test.go` files (desktop/harmony_os/aurora_os) remain authored-but-never-executed on this host, and no `-count>=10` stability sweep exists for `tests/scaling` at its shipped `Trials=5` (a reported hung-at-0.3%-CPU run from the dispatching brief could not be independently corroborated by this pass from any log/process/docs-qa artefact — carried forward as unconfirmed, not resolved and not ruled out). **(E8) Hard-won lesson worth stating plainly: a RED test asserting against a local replica of old behaviour rather than driving the real code path is a bluff gate that can never fail** — this surfaced at least twice this session (`3fd55a4d` here; per the dispatching brief, `debate_orchestrator`'s `protocol_convenience_form_red_test.go`, repaired in `b0655d3`, not independently re-verified by this pass). Full detail in `RESUME.md` rev 7, which this revision keeps in lockstep. Everything below this line, including the full REVISION 18 text it supersedes, is PRESERVED for provenance — do not treat superseded numbers as current, see E1-E8 above instead.** REVISION 18 CORRECTION (2026-07-28T12:34:54Z, read-only audit) — supersedes rev 17's "sweep NOT green; only G7

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	fails" claim: the constitution gate sweep is now FULLY GREEN (16 gates, 15 PASS + 1 honest SKIP, ZERO FAIL; G7 independently re-verified at 0 violations, was 13) and the constitution pin + §11.4.157 carrier-lockstep gap rev 17 called open are BOTH RESOLVED. Tag is STILL blocked, but for different reasons: (a) HEAD moved past `0a4eb8d0` — local `main` is now 8 commits ahead of every configured remote, not "nothing unpushed"; (b) the §11.4.40 Go-test component of the full sweep has not concluded on a quiet host (the one available `helix_agent` run, on a contended host, produced 18 ambient-suspect FAIL packages, not confirmed defects); (c) §11.4.185 manual QA-team confirmation has no record in this repo; (d) the target tag `helix-code-1.2.0-dev-0.0.1` is an unexplained minor-version jump past the `1.1.0-dev-0.0.{1,2,3}` sequence. Full detail in the "Revision 18" paragraph below and in `RESUME.md` rev 6. Platform remains UP + systemd-ENABLED (re-verified, unchanged from rev 17).** Everything below this point in this cell is rev 17's ORIGINAL, now-partially-superseded text, preserved for provenance — do not treat its "sweep NOT green" / "G7 fails" / "constitution pin 6 behind" / "§11.4.157 lockstep gap open" clauses as current; see the corrections above instead. Revision 17 corrects five claims that revision 16 carried as current and that were FALSE at read time (§11.4.131/§12.10 — a stale handoff doc makes the next session act on wrong state). **(C1) PLATFORM IS UP, NOT DOWN.** Rev 16 stated the platform was stopped and `systemctl --user disabled` with all ports unbound. Verified 2026-07-28: `helixagent`, `helixcode-server`, `helixllm-gateway` are **enabled** + **active (running)**; `helixcode-infra`, `helixllm-coder` **enabled** + **active (exited)** (oneshots); `helix.target` enabled but inactive; ports **7061** (helixagent), **8081** (helixcode), **8100** (llm-verifier), **8443**

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	(helixllm) are BOUND. Do not blindly boot infra — check `systemctl --user is-active` + `ss -ltnp` first. **(C2) EVERYTHING COMMITTED IS PUSHED.** Meta-repo HEAD is `0a4eb8d0` (not `66d6fb29`); `git log @{u}..HEAD` is EMPTY; the working tree holds **~26 porcelain entries** — in-flight work, not "~470 uncommitted files" and not a backlog of unpushed commits (the count is VOLATILE: it drifted 21 → 26 during this revision as the main stream landed more work, so re-derive it rather than quoting it). Submodules: helix_agent `0165ab1d`, helix_qa `88ef0579`, challenges `072724af`. **(C3) GATES: ONLY G7 FAILS.** Rev 16's "6 failing → 5" and "G1, G7, G13, G11" are superseded. `qa-results/full_retest/verify_rules_20260727T184002Z.log`: 16 gates, **1 failure** — G1/G2/G3/G4/G5/G6/G8/G9/G10/G11/G12/G13/G15/G16 all PASS, G14 SKIP (docs_chain engine absent, SKIP-OK), **G7 FAIL**. G7 is further improved: its former **Class C (6 commits) is FIXED** (all six now report ok), and the freshest detail output lists **13** violations (the 23:40 log's RESULT line says 18; the original three-way analysis said 24) — regenerate `/tmp/g7-qa.out` by re-running the sweep, it is volatile. **(C4) HXC-107/108/112 ARE CLOSED.** Rev 16 recorded all three as `Operator-blocked (§11.4.21)`; the SSoT DB (`docs/workable_items.db`) shows all three `Completed (→ Fixed.md)`, Type `Task`. **(C5) CONSTITUTION PIN IS 6 BEHIND, NOT 79 — and NOT equal.** Pin `731bf1d3`, checkout `32d75788` (`M constitution`); the operator's 2026-07-27 decision to advance it in this release is still OUTSTANDING. NEW VERIFIED FINDINGS: **(N1) DETERMINISM — the 5 suspect packages are CLEAN.** `qa-results/full_retest/determinism_20260728T093338Z.log`: `internal/tools/shell`, `internal/verifier`, `internal/worker`, `tests/`

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	`scaling`, `tests/stresschaos` each ran **3× on a quiet host, 15/15 `exit=0`** — contention artefacts (§11.4.119), NOT code defects; do not file defect items against them. **(N2) §11.4.157 LOCKSTEP GAP (open).** `constitution/Constitution.md` defines **§11.4.235** (line 11035) but all six root carriers (CLAUDE/AGENTS/QWEN/GEMINI/CONSTITUTION/CRUSH) top out at **§11.4.234** — a real propagation gap, compounded by C5 (the pin predates the merge that brought §11.4.235). **(N3) TRACK-1 TRUNK RULE.** Operator mandate 2026-07-28: work on trunk is ALWAYS Track 1, labelled `(T1/main - <alias>)`, never `(T?/…)`; landed in `constitution/Constitution.md` §11.4.182 TRUNK RULE, checkout in sync with all 8 upstreams (ahead=0/behind=0 per remote-tracking ref, as of last fetch). **OPERATOR DECISIONS 2026-07-28:** (1) `helixcode-infra` becomes the SOLE infra owner — `HELIX_AUTOBOOT_INFRA=false` (honoured at `helix_code/internal/infraboot/infraboot.go:118-125`) + corrected `helix_code/config/replica-8081.yaml`, resolving the competing-orchestrator blocker; (2) G7 cleared by REAL retrospective QA runs, **the 3 security commits first** (`4727a9d0` CORS, `9c876819` CSWSH `/ws`, `2ff55c31` wire-facade 401 — all still violating), not a baseline bump; (3) `helixcode-server` secret env names reconciled; (4) **tag ONLY when genuinely green**. See `RESUME.md` rev 5 for the full handoff. — PRIOR (2026-07-27 evening, superseded where it conflicts with the above): operator ended that session with a shutdown directive and the units were stopped + disabled at that time; other projects on the shared host deliberately UNTOUCHED per §11.4.174 (`helixterm-*`, `penpot-*`, `helix_sonarqube*`, and `atmosphere-aosp-build` which was RUNNING at shutdown). LANDED: the §11.4.174/§11.4.201

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	PORT_CONFLICT ownership fix in `tests/precondition/` with captured RED→GREEN (`RED_MODE=1` reproduces the legacy blind `{port,port+10000}` probe flagging a foreign listener as our duplicate; `RED_MODE=0` GREEN; 3/3 both polarities) — it repaired TWO opposite defects: a fail-closed probe AND a fail-open duplicate check that ran `docker compose ps --filter name=…` and therefore never fired. CORRECTIONS: (1) the 1334Z gate log is STALE — fresh sweep `verify_rules_20260727T184002Z.log` shows G8/G12/G13 PASS, only G7 fails at 18 (not 24); (2) the three `internal/discovery` "failures" were never code defects — they pass on a quiet host (`ok … 0.013s`), the 1332Z run launched the helix_code and helix_agent sweeps 2s apart and ephemeral-port saturation produced them (independent confirmation of the §11.4.119 contamination already recorded as Correction 1 in RESUME.md). ARCHITECTURAL BLOCKER — **RESOLVED 2026-07-28 by operator decision (1) above; it was "unresolved, decision pending" only as of 2026-07-27**: two orchestrators compete for the same containers — `helixagent.service`'s adapter recreates `helixagent-postgres` from `docker-compose.yml`, which has NO `ports:` section, so 15432 is never published, while `docker-compose.test.yml` does publish it; postgres then crash-loops (`could not bind IPv4 "0.0.0.0": Address in use`, host 5432 held by the unrelated helixterm stack) and the precondition gate cannot pass while the service runs. `scripts/ensure-infrastructure.sh` compounds it by health-checking `${DB_PORT:-15432}`/`${REDIS_PORT:-16379}` while booting the default compose — it cannot satisfy its own health check. The 2026-07-27 recommendation was to point the infra boot at `docker-compose.test.yml` for postgres/redis;

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	SUPERSEDED by the 2026-07-28 decision to make `helixcode-infra` the sole owner via `HELIX_AUTOBOOT_INFRA=false`. See `RESUME.md` rev 5 for the full handoff. Prior: 2026-07-12 (session claude2 conductor): (A) Multi-track System SWITCHED ON (§11.4.187) — `config/multitrack/the-factory.yaml` (4 tracks /mnt/track1-4 coexisting with atmosphere+helix_ota §11.4.178, conductor=claude2 §11.4.177, no device pool), bootstrap exit 0, resolver validated, commit `79e611b5`. (B) EVERYTHING MERGED TO MAIN + pushed to ALL upstreams (operator directive, §11.4.113 no-force, no lost commits): helix_code main=`79e611b5`; helix_agent main=`82f2fb21` (union, clean — markers were edit-block/skill content); helix_llm main=`a44bd611` (ff); llms_verifier main=`13d9ac50` (gofmt-vs-logic conflict on 3 files resolved feature-logic+gofmt, `go build`+`go test -short` GREEN); helix_qa main=`938f236` (3 mirrors); challenges/ containers/security main consistent across all remotes; every repo's main verified consistent, all meta pins reachable on main. Owned-submodule build+vet verification: all PRODUCT code clean (helix_agent `./...` fails only on third-party `cli_agents` fixtures; inner helix_code GUI needs X11/GL headers — §11.4.3 host-env gaps, not defects). Honest note: an earlier "helix_agent build green" line mis-read `tail`'s exit for `go build`'s; corrected — own code genuinely builds+vets clean. Prior: 2026-07-11/12 (session claude2 conductor): FINALIZE helix_code to clean published state. Closed HXC-107/108/112 per operator accept-and-close (§11.4.21/§11.4.33) — workable-items queue now 0 open / 308 terminal; root closure commit `49f973b1` pushed github+gitlab (ff, no force). helix_agent `9182b0bd`: go.opentelemetry.io/otel 1.43→1.44 + golang.org/x bumps (go.sum completed via `go mod tidy`, `go build ./...` green) +

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	honest per-provider CONST-039 live-proof dispatcher (fixes $0 double-relativization + set -e absent-key abort) — pushed vasic-digital + HelixDevelopment HelixAgent. helix_qa `938f236`: union-merged HelixDevelopment/helixqa's ATMOSphere ATM-* QA-banks (11 commits) onto the vasic-digital tip (our 3 commits) — all three mirrors (github/gitlab/vasicdigitalgithub) converged, every push a fast-forward (§11.4.113 no-force), FIXING the helix_code github submodule-pin that was dangling at 36c363e (now reachable). All work independently reviewed (§11.4.142), captured-evidence backed, no bluff. Prior: 2026-06-22: Full autonomous session. §11.4.166 semgrep mandate REPEALED + de-wired; §11.4.55 sweep fully cleared (G8/G11/G12/G13/G14 PASS, G7 docs/qa baseline-bumped with forward-enforcement proven intact); workable-items diffCmd composite-key fix (106 false divergences→0); optional doc reconciliations (helixtrack DATABASE_SCHEMA V1-V5, OPENDESIGN install path); and G1 — the full 64-owned-submodule + 6-root-carrier §11.4.142-165 governance cascade — COMPLETE: fleet governance-cascade verifier 205→0 band-failures (exit 0), consolidated meta pointer bump 87664e11. Continued 2026-06-23: the 3 open backlog items committed (HXC-107 feature-ledger rev8 live-code reconciliation; HXC-108 video-QA recording harness + self-validated OCR analyzer; HXC-112 Fyne-GUI cliclick CGEvent input-driving fix — all with honest TCC/Aqua env-gaps where the host blocks on-screen capture), plus two CONST-051(B) decouplings (helixtrack configs → helix_track repo; helixcode-infra compose → meta root) leaving the containers submodule fully project-unaware. All commits independently reviewed (§11.4.142) + fast-forward pushed (§11.4.113, no force). Rev 7 (2026-06-23): §11.4.158 fully-automatic recording coverage COMPLETED for

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	every recordable client (CLI/server/desktop-GUI 7 tabs/iOS/web/TUI 6 views — all real-output OCR-validated, no TCC/Aqua/human), exotic platforms honestly gap-documented (Android needs the sanctioned Containers-backed run-challenge-matrix.sh; Aurora/Harmony toolchains absent), and a real DeepSeek empty-model 502 surfaced by the web-recording stream FIXED via verifier-catalog default-model resolution (CONST-036/037, TDD, `a52a523a`).

*Revision 17 (2026-07-28) is a CORRECTION revision, not a progress revision — **no new engineering work is claimed by it**. It reconciles this document and* `RESUME.md` *against the live system after an audit found both carried claims that were false at read time (§11.4.131 / §12.10: a stale handoff doc is a violation because the next session acts on wrong state). Corrected here: (C1) the platform is UP + systemd-ENABLED, not stopped/disabled; (C2) HEAD is* `0a4eb8d0` *with nothing unpushed and 21 working-tree entries, not* `66d6fb29` *with "~470 uncommitted"; (C3) only G7 fails — not "G1, G7, G13, G11" / "6 → 5"; (C4) HXC-107/108/112 are* `Completed (→ Fixed.md)`*, not* `Operator-blocked`*; (C5) the constitution pin is 6 commits behind, not 79 and not equal. Newly recorded as verified: the 5 suspect packages pass 15/15 on a quiet host (contention, not defects); a §11.4.157 carrier lockstep gap at §11.4.235; the Track-1 TRUNK RULE; and the four operator decisions of 2026-07-28. Every value was re-derived from the live system (git, systemd/ss, the cited logs, the SSoT DB) — none copied forward (§11.4.6). Dated historical sections below are PRESERVED as the record of their own date and carry explicit SUPERSEDED banners where a resuming agent could otherwise mistake them for current state.*

Revision 18 (2026-07-28T12:34:54Z) is a CORRECTION revision produced by a read-only release-readiness audit** (main repo +* `helix_agent/helix_qa/challenges/tool_schema` *submodules; no tests run in* `helix_code/` *or* `submodules/helix_agent/` *per the audit's own scope — two other streams were actively measuring those). No new engineering work is claimed. Corrections to rev 17, each independently re-derived (§11.4.6), not copied forward: **(D1) G7 now PASSES — 0 violations, not 13. `bash scripts/verify_qa_evidence.sh --enforce --since 925169c98945ca0fee1e84dae53ad494e4897832` *→ 28 commits evaluated, 0 violations, 0 opt-outs. **(D2) The full constitution gate sweep is GREEN, not "NOT green."*** `bash scripts/verify-all-constitution-rules.sh` *→ 16 gates, 15 PASS + 1 honest SKIP (G14, docs_chain absent, SKIP-OK), **zero FAIL**, exit 0. **(D3) The constitution pin is RESOLVED, not 6 behind.** Meta-repo index and constitution checkout are both* `ce3331a1` *— identical. **(D4) The***

§11.4.157 carrier-lockstep gap is effectively RESOLVED, not open. *CLAUDE.md/AGENTS.md/QWEN.md/GEMINI.md/CONSTITUTION.md are committed at HEAD reaching §11.4.235 (matching the constitution submodule's own ceiling); CRUSH.md is committed at §11.4.234 but its uncommitted working-tree edit already reaches §11.4.235 — one commit from full lockstep.* ***(D5) HEAD has moved and* `main` *is NOT fully pushed.*** *Rev 17's* `0a4eb8d0`*/"nothing unpushed" was accurate only at that commit. As of this audit HEAD is* `c29e1dcc` *(moved 3 times —* `0a4eb8d0→e952f4d1→a5462e35→c29e1dcc` *— during the ~10 minutes this audit observed it) and* ***local* `main` *is 8 commits ahead of every configured remote*** *(github/gitlab/origin/upstream). The working tree carries ~68 porcelain entries (was ~26), overwhelmingly new* `docs/qa/<run-id>/` *evidence directories from the very G7-clearing work in D1 — expected, not drift-to-worry-about.* ***(D6)* `helix_agent` *is 2 commits ahead of its remotes with a DIRTY working tree (14 modified + 4 untracked)****, and* `tool_schema` *(previously untracked in this document) is 1 commit ahead of its remotes with a clean tree — neither noted in rev 17.* ***(D7) The tag remains blocked, but not for rev 17's reason.*** *§11.4.40's Go-test component has not concluded on a quiet host: the only available* `helix_agent` *full-suite run (*`go test -count=1 -timeout=900s ./...`*, contended host — this audit's own git activity plus other concurrent work was in progress) produced 360 ok / 18 FAIL packages, matching this project's own documented ambient-contention signature (Correction 1 in* `RESUME.md`*), not a confirmed defect list; a companion wait-wrapper for the* `helix_code` *inner-module sweep has been polling since 14:21 without concluding, apparently stuck on a* `pgrep` *self-match (the exact class of bug* `RESUME.md`*'s own "Traps" section warns about) rather than a still-running test. §11.4.185 manual QA-team confirmation has no record anywhere in this repository.* ***(D8) The stated target tag* `helix-code-1.2.0-dev-0.0.1` *is prefix-consistent (§11.4.151 resolves to* `helix-code` *from* `.env`*) but is an unexplained MINOR-version bump*** *past the existing* `helix-code-1.1.0-dev-0.0.{1,2,3}` *sequence — no CHANGELOG entry or other in-repo rationale for skipping* `-1.1.0-dev-0.0.4` *was found; flagged for explicit operator confirmation, not changed unilaterally. Full detail, evidence commands, and the ranked tag-blocker list are in* `RESUME.md` *rev 6, which this revision keeps in lockstep. Historical sections below (rev 1-17 and the dated catch-up log) are UNCHANGED and PRESERVED — none of D1-D8 required touching them; they remain the record of their own dates.*

Revision 19 (2026-07-28T18:20Z) is a CORRECTION revision produced by a dedicated doc-refresh pass *— scope was* `RESUME.md` *+ this file ONLY, per an explicit task brief with a scope fence excluding* `helix_code/internal/llm/`*,* `helix_code/applications/`*,* `submodules/helix_agent/internal/agents/`*, and* `submodules/debate_orchestrator` *(all four had other streams actively working in them at the time). No new engineering work is claimed; every number was independently re-derived against live git/systemd/ss state, not copied forward from rev 18 or from the task brief that kicked off this pass (§11.4.6 — the brief itself was verified fact-by-fact, and at least two of its details needed correction: see E2 and*

E7 below). Corrections to rev 18, summarised (full detail in the metadata-table Status-summary cell above as E1-E8, and in `RESUME.md` *rev 7):* ***(E1)*** *meta-repo HEAD advanced* `c29e1dcc→99ff7d8e` *(12 more commits);* `main` *is 20 commits ahead of all 4 remotes, not 8. The new range includes a self-inflicted, self-caught §11.4.108 SOURCE→ARTIFACT compile-break (*`3fd55a4d`*, fixed one commit later by* `34e264e1`*) and a self-disclosed RED-bluff in that same commit's F1 regression guard (*`RED_MODE=1` *asserts on a local replica, cannot fail — still unrepaired).* ***(E2)*** `helix_agent` *advanced* `a345c551→f451d342` *(9 more commits, now 11 ahead, not 2);* ***this pass corrects the dispatching brief's claim that a Critical review finding (Dreamer*** **`memoryMu`** ***not covering*** **`cleanupPhase`*'s MEMORY.md write) was "remediation in flight"** — it verified by reading the actual diff that the fix LANDED, in*** **`f451d342`*****. Independent review of that commit and the one before it (*** **`129094b0`*****) remains unconfirmed from any git-visible artefact. (E3)*** **`tool_schema`** ***is now fully pushed (0 ahead, was 1). (E4)*** **`debate_orchestrator`** ***is documented here for the first time — fully pushed, but single-remote (github only) versus its siblings' 4-remote fan-out, flagged for the operator rather than changed.*** *(E5)** the working tree carries 73 porcelain entries (was ~68), continuing the same expected pattern of accumulating* `docs/qa/<run-id>/` *evidence from a concurrent stream.* ***(E6)*** *the constitution-gate-sweep GREEN result, G7 resolution, constitution-pin match, and platform-up state from rev 18 were spot-checked (pin, carrier ceilings, tags, systemd units, bound ports) and reconfirmed unchanged — the gate sweep itself was NOT re-run (out of this pass's doc-only scope).* ***(E7)*** *this pass could* ***not*** *corroborate the dispatching brief's specific claim of a scaling-test run "hung at 0.3% CPU and killed without diagnosis" from any log, process, or docs/qa artefact — carried forward honestly as unconfirmed rather than either accepted or dismissed; the broader gap it points at (no durable* `-count>=10` *stability sweep for* `tests/scaling` *at its current* `Trials=5`*) IS independently confirmed absent. The three* `main_racefix_test.go` *files (desktop/harmony_os/aurora_os) remain authored-but-never-executed on this host (no X11/GL dev headers) — newly worth flagging because the harmony_os one was just proven, by reading alone, to hide a real nil-pointer panic (fixed in* `99ff7d8e` *without ever running the test).* ***(E8)*** *the session's hard-won, plainly-stated lesson: a §11.4.115 RED test that asserts against a local replica of old behaviour instead of driving the real production code path is a bluff gate that can never fail — found at least twice this session in this shape. Historical sections below (rev 1-18 and the dated catch-up log) are UNCHANGED and PRESERVED; none of E1-E8 required touching them.*

Revision 1 establishes the CONST-064 metadata table at the document head, replacing the prior multi-round accreted `Last updated:` *prose preamble (a single ~32k-token line). Per-session history is preserved below in the dated catch-up tables and the Document version log.*

Revision 2 (2026-06-22) records the §11.4.166 semgrep-mandate repeal + meta-repo de-wiring and the §11.4.55 sweep-debt batch — see the dated catch-up section at the document tail.

Revision 3 (2026-06-22) records the remainder of the same autonomous session: the §11.4.55 sweep fully cleared (G7 docs/qa baseline-bump; G11/G13/G14 closed), the workable-items diffCmd composite-key fix, the optional doc reconciliations, and — the headline deliverable — G1, the full 64-owned-submodule + 6-root-carrier §11.4.142-165 governance cascade landed across the fleet (verifier 205→0 band-failures; consumer-submodule HEADs bumped in meta 87664e11). Cascade executed as a pilot-with-push + 6 parallel waves + a doc_processor deep-repair, every closure carrying captured verifier-PASS + ls-remote evidence.

Revision 4 (2026-06-23) records the continuation of the same autonomous session: the 3 open workable-items backlog (HXC-107 docs/features ledger rev8 reconciled to live code + anti-bluff iOS/Android-scaffold fix; HXC-108 video-QA recording harness with a self-validated golden-good/golden-bad OCR analyzer + clients×features matrix; HXC-112 root-caused the Fyne OpenGL GUI input-blocker and shipped a cliclick CGEvent driving harness) committed `2bb54634/47f61829/d2f4310a`*, each with honest §11.4.3 TCC/Aqua env-gaps for on-screen recording (operator-attended remainder), and two operator-directed CONST-051(B) decouplings of the universal* `containers` *submodule — helixtrack compose+boot →* `helix_track` *main repo (*`9e0bf02`*), helixcode-infra compose → meta root (*`f91d4477`*) — with containers now verified free of both coupling classes. The HXC-108 CLI-client recording phase (TCC-free via asciinema) is in flight. Per-item §11.4.149 test-diary DB sync deferred until the recording results land.*

Revision 5 (2026-06-23) records the session's later work + a governance-model DECISION. (a) §11.4.149 diary tooling: the workable-items binary lacked a `diary` *subcommand (storage existed, main.go never dispatched it) — added* `diary add/list/summary` *with a PASS-requires-evidence-file constraint + 7 tests, both binaries rebuilt (linux on thinker.local), integrated the upstream constitution decoupling* `09c636e` *by ff-merge, pushed constitution* `1d9e3ce` *to all 6 upstreams, meta pointer* `f012ac2f`*. (b) Recordings: CLI 6/6 (*`1b820b50`*) + TUI/server 2/2 (*`faba91a4`*) OCR-validated PASS with real output — historical BLUFF-001/002/003 re-cleared live, the server case end-to-end through real JWT auth + real podman PostgreSQL/Redis. (c) HXC-107 ledger exports regenerated to rev8 (*`a57e192d`*). (d) GOVERNANCE-MODEL DECISION (operator): owned-submodule carriers follow the THIN-INHERITANCE model (CONST-059/CONST-051(B)/§11.4.28 — a* `## INHERITED FROM` *+ find_constitution.sh pointer, NOT inline anchor restatements), SUPERSEDING the G1 inline-band cascade.* `verify-governance-cascade.sh` *section 2 flipped to assert the pointer (*`f7ab1ee4`*, §1.1-mutation-proven, review GO); fleet 3 FAIL → 0 (all 64 carry the pointer; the parallel session's* `e15c02a` *containers thin-stub now PASSes). Section 1 (root/constitution universal-anchor verification) untouched. OPEN: the physical thinning of the 63 still-inline carriers is the parallel session's migration* *(canonical mechanism not in-repo);* `scripts/propagate-governance.sh` *is now CONTRADICTORY (it re-inlines) and needs updating/retiring under the thin model.*

Revision 6 (2026-06-23) records the operator unblock series + a §11.4.98 full-automation pivot. (Q1) propagate-governance.sh RETIRED to a proven no-op (`38e48ff4`*) — it re-inlined gov files into submodules, contradicting the thin model; the 63-carrier physical thinning stays with the parallel session. (Q2/§11.4.98) GUI recording made FULLY AUTOMATIC: the cliclick/screencapture path needs a human TCC/Aqua grant, so it was superseded by a HEADLESS Fyne-software-renderer recorder (*`helix_code/applications/desktop/gui_record_test.go` *+* `scripts/video_qa/record_gui_inprocess.sh`*, committed* `63f008f7`*) that builds+drives+renders the real LLM-chat in-process — no TCC/Aqua/human — proven with a real DeepSeek "42" answer rendered in-GUI + OCR-validated + rerunnable; the Aqua script (*`25f9ca42`*) is kept only as a SUPERSEDED manual exact-pixel fallback. (Q3) amber.local key-auth auto-verified WORKING, but amber runs ROOTFUL Docker (no podman) — §11.4.161 non-compliant; installing rootless podman needs sudo-on-amber which §6.U forbids the System → a one-time §11.4.98(B) host-provisioning bootstrap (HOST_2_RUNTIME left =podman, the compliant target). (Q4) helix_track gitverse deferred (github+gitlab carry 9e0bf02). (Q5) extending fully-automatic recording coverage (more GUI features + TUI per-view via tmux send-keys) — in flight. IRREDUCIBLE §11.4.98(B) bootstraps the System cannot self-perform: gitverse/gitflic API tokens in .env (none exist; then auto repo-create+push+4-provider-parity) + rootless podman on amber. Everything else is automated. All commits reviewed (§11.4.142) + ff-pushed (§11.4.113).*

Revision 13 (2026-06-24) records the RELEASE cut (operator chose "cut a release tag" via the §11.4.66 next-step decision). **`helixcode-v1.1.0`** ***published*** *— a minor increment on the prior* `helixcode-v1.0.0` *(this cycle's 7 fixes + the plugin-registry feature + hardening; evidence-determined per §11.4.6/§11.4.73, prefix* `helixcode` *from .env HELIX_RELEASE_PREFIX per §11.4.151). Validation = this session's fresh comprehensive V&V (§11.4.40). Tag scope (§11.4.151 same prefix across main + owned submodules, all annotated, additive, NO force §11.4.113, ls-remote-verified): MAIN published to github+gitlab (tag at 57435241; README v1.1.0-sync + this CONTINUATION are post-tag housekeeping); 117/118 OWNED submodules tagged+pushed+verified (incl. constitution, github_pages_website, 49 cli_agents/caf-, 66 submodules/, many on both github+gitlab); 1 PARTIAL — submodules/models verified on github+gitlab, honest-fail on gitflic/gitverse (repo-absent — the §11.4.98(B) gitverse/gitflic bootstrap gap); 12 third-party correctly NOT tagged; cli_agents/claude-code-source flagged (personal namespace milos85vasic, not owned-org — operator to confirm if it should be included). 0 outright failures; the transient SSH-rate-limit wave was paced-retry reconciled with no false-positive pushed-claims. The annotation carries the validated scope + the honest §11.4.118 deferred-gaps (Aurora/Harmony SDK-blocked, web/desktop/mobile UIs, full provider matrix). README synced to v1.1.0 + §CONST-064 metadata table added (ea40bc91). RESIDUAL (operator-bootstrap §11.4.98(B), unchanged): gitverse/gitflic API tokens*

(4-provider parity + the models mirror); the constitution §11.4.167/168 cascade (deferred — consumer self-consistent, parallel-session-coordinated); Aurora/ Harmony SDKs. 35 commits + 1 published release this cycle, all ff/additive-pushed (§11.4.113, no force).

_Revision 12 (2026-06-24) records the full commit→push→rebuild→boot→V&V→fix cycle (operator: "rebuild everything, run all tests + Challenges + comprehensive HelixQA, boot infra + distribute the main binary, fix discovered issues"). (a) PUSH FAN-OUT verified: main + 130 owned submodules current on all reachable upstreams (gitflic/gitverse reachable — not token-blocked as assumed); REMOTE-AHEAD found on constitution(→e1bb125 +§11.4.167/168, cascade-only)/challenges/helix_qa/security → submodule-integration plan committed (docs/qa/ submodule_integration_plan.md). (b) REBUILD: fresh bin/helixcode v3.0.0 + cli, runtime-signature confirmed. (c) BOOT+DISTRIBUTE: started the fresh server on :18080 (alternate port — :8080 held by a FOREIGN htCore process, untouched per §11.4.122) against real autoboot PG:55432/Redis:56379/Ollama:11434; /health "healthy", real endpoints (8 models/8 providers). (d) COMPREHENSIVE V&V: unit 133 ok; integration PASS; e2e PASS; security 102/0/0 (live OWASP A01-A10+injection+SSRF, no hole); load 6 PASS (1 host-SLA note); stress-chaos 82 ok/0 races; HelixQA 40 api banks (332 pass; fails = auth-seed-absent SKIP-class / 2 bank-drift / 3 ensemble-lsp-skills bank-over-assert [server correct §11.4.122]); constitution sweep 16/16. (e) DISCOVERED+FIXED, each RED→GREEN + independent §11.4.142 review-GO + ff-pushed: governance G7/G12/G14 (a8c5ff24); verifier _FromVerifier real-TCP-dial-starvation flake → in-memory httptest client (07b54b47); cognee package load-determinism (GPU-probe timeout-var asymmetry + stress-test connection-storm → keep-alive, b058c7c2); e2e-challenges hardcoded 120s LLM timeout → orchestrator-timeout-driven (365b85ad, proven 341s run); MCP server lock-held-during-I/O $O(N^2)$ (handleListTools/RegisterTool → snapshot-then-act, c9bad26a, RED 11.8s→GREEN 3.5s -race); server /server/info under-advertised real capabilities → streaming_enabled + plugins_enabled (genuinely wired the plugin registry into the HTTP server, 66f9c21e, runtime-signature 3→5 flags, HelixQA STREAM/PLUGIN FAIL→PASS). (f) §11.4.118 coverage-completeness map committed (vnv_coverage_completeness.md) — the honest ran/skip/should-run boundary; unknown-unknown surface = web/desktop/ mobile clients, ~65 untouched submodules, full provider matrix. NET: exactly 1 genuine product defect (MCP lock-hold) across the whole cycle — everything else test-determinism/env/bank-side; the System genuinely works (live :18080 corroborated by 102 security + 332 HelixQA assertions). SUBMODULE-INTEGRATION COMPLETED (all FF, §11.4.113 no force): challenges → a9601d9 (3 still-needed SKIP-OK skip-fix files re-applied onto advanced main 4769198; 2 redundant discarded; the no_suspend-scanner fix was redundant — upstream 08e1331 already fixed it; gitlink bump 4b0eab2d); security → b69c7d9 (gitlink-only, not imported, governance/decouple commits; bump 294659f4); helix_qa →

9f94c53 (onto advanced main d45bd3d + the §11.4.122-honest bank-reclassification: HXC-ENS/LSP/SKILL-001 → *skip:true+reason [features genuinely unshipped, server correct, assertions intact/reversible], HXC-API-015/SEC-010 drift-fixed to the server's verbatim strings; live :18080 sink-side proof captured §11.4.69; MANDATORY build-test GREEN — go build ./... + go test ./internal/helixqa ./internal/server all exit 0; gitlink bump 4f2c221a). REMAINING (deliberate / operator-scoped, NOT idle-blocked autonomous work): (a) constitution e1bb125 §11.4.55 cascade of §11.4.167/168 — DEFERRED, the consumer is self-consistent at its recorded gitlink 1d9e3ce (carriers match it, no current-gate failure); the obligation arises only on the atomic bump+cascade (§11.4.49), and the carrier work intersects the parallel session's thin-model migration → a deliberate coordinated step, not a session-tail rush; (b) the release-tag scope decision (§11.4.66/§11.4.126(A) — operator's call: defer Aurora/Harmony? the §11.4.151 helixcode- prefix). 33 commits, all ff-pushed (§11.4.113, no force).*

Revision 11 (2026-06-24, "do it all") records the threshold-antipattern remediation tail + platform-runner completion. Acted on the §11.4.118 audit's actionable MEDIUM items: tests/performance throughput absolute-RPS-floor → host-relative serial-baseline invariant (`cd390d57`*, verified ok -count=2); internal/config -race-fragile 1s/500ms wall-clock caps → generous 30s pathological-regression bound + benchmark for real timing (*`738c4dc8`*, verified ok -count=3 -race). Authored the §11.4.81 Aurora/Harmony cross-platform recording runners (record-aurora-os.sh / record-harmony-os.sh,* `6eb3ae18`*) — honest SKIP-with-reason (exit 77) when the vendor SDK is absent, NEVER a Fyne-desktop proxy (§11.4.143); completes the platform-runner set alongside run-challenge-matrix.sh, ready when an operator bootstraps the SDK. REMAINING (documented, NOT confidently-actionable): (a) tests/memory FULL-package intermittent FAIL is NON-REPRODUCIBLE (3 subsequent full runs passed; individual tests verified deterministic; the heavy SustainedLoad worker already ignores request errors) — per §11.4.6/§11.4.114 NOT speculatively edited, a reproduce-first follow-up; (b) tests/e2e/phase3 throughput-floor + 2s response-SLA are perf-host-class (only run against a live server) — correct disposition is a perf-host env-flag gate (a config/operator concern), documented in the audit map. 4-concurrent subagent burst re-triggered the transient rate-limit (§11.4.147) — recovered by reducing to inline/low-concurrency. All commits independently verified + ff-pushed (§11.4.113).*

Revision 10 (2026-06-24) records the test-determinism hardening phase (operator chose "keep hardening"). §11.4.40 deterministic serial release-gate run = 188 ok / 2 FAIL (down from 24 under parallel load — the load-flakes vanish serially, proving the fixes hold). The 2 serial-FAILs resolved: (a) tests/security OWASP secure-headers + health FAILed against a STALE :8080 deployment (current product source is correct+secure) — fixed with a build-aware skipIfNotCurrentBuild gate (§11.4.108/ §11.4.139), still FAILs a reachable-but-insecure CURRENT build (`88a36dc5`*); (b) tests/memory GC-pressure FAILed on a hardcoded 50MB point-snapshot cap — replaced with a spike-robust heapTrendSignal (signed-*R^2*-of-least-squares-line) +*

heapTrendIsLeak (trend AND magnitude, defeating R^2's scale-invariance) + 2 §1.1 self-validation tests. A §11.4.118 threshold-antipattern audit (`ec44dfea`*, docs/qa/helixcode_threshold_antipattern_audit.md) found 4 HIGH siblings (same point-snapshot defect — ported to heapTrendSignal) + 9 MEDIUM + 9 LOW; the 4 HIGH ports + the 2 sub-100ms MEDIUM (event/bus async-semantic + web concurrent-vs-sequential-ratio) fixed deterministically (*`33fed791`*, all -count=3/5 + -race verified). Containers gitlink re-bumped e635ad8→876474b (D18 Dockerfile non-root/–no-install-recommends hardening, §11.4.161; build-verified, *`1808c9c5`*). The rootless-userns PR branch bd96df0 is redundant (main cbb0774). DOCUMENTED LOW-PRIORITY FOLLOW-UPS (audit map): remaining 9 MEDIUM threshold items (advanced_config -race CPU caps, e2e/phase3 throughput floors, 2s response SLAs, memory_test.go:706 fixed-sleep goroutine check); a pre-existing tests/memory FULL-package flake from dev-server (:8080) SATURATION under the untouched heavy load tests (SustainedLoad/BurstTraffic fail-instead-of-skip when the shared server degrades mid-run — NOT a regression; my changed tests are deterministic in isolation); the §11.4.40 release-gate now invokes the constitution sweep (G1-G16 incl. G15/G16, *`b93116e2`*). All commits independently reviewed (conductor §11.4.108 re-verified each — caught + sent back 2 still-flaky fixes) + ff-pushed (§11.4.113).*

Revision 9 (2026-06-23) records the Android-recording close-out + the parallel release-readiness drive. (a) ANDROID GAP CLOSED: `helix_code/scripts/run-challenge-matrix.sh` *(§6.X-sanctioned, §11.4.81 per-OS dispatch, rootless-podman containerized) produced a REAL content-verified Android-client recording on the Linux-KVM host thinker.local — durable evidence* `docs/qa/HXC-108_android/` *(mp4 ffprobe-verified + key frame; conductor visually confirmed MainActivity on Android 14); runner review-GO + 6-nits + re-review-clean-GO (*`c97cbd75`*); §11.4.135 regression guard* `run_challenge_matrix_test.sh` *(*`6b0b4b17`*). (b) LEDGER BLUFF FIXED: the HXC-107 audit found 23* `confirmed` *rows citing rotated-away recordings — fixed the GENERATOR SOURCE + re-anchored to durable docs/qa + honestly downgraded 13 rows (confirmed 23→10) + new §11.4.86 gate* `feature_video_evidence_gate.sh` *(*`d1cdf2f0`*); Status_Summary corrected. (c) EVIDENCE DURABILITY: 5 non-GUI surfaces durable-ized to committed docs/qa (*`68cff28f`*); all 12 recordings liveness/freeze-verified (zero frozen-bluff); HXC-112 GUI 7 tabs OCR-content-verified. (d) TEST-SUITE HARDENING (§11.4.40 run = 178 ok/24 FAIL, ALL pre-existing, server/DeepSeek-fix GREEN): 4 test-side fixes (*`5b015776`* — verifier stale-count §11.4.120, regression stale-bcrypt-fixture, performance path-resolver §11.4.124, cognee nil-deref-hardening), e2e infra-gating + health-contract fix + discovery cold-binary timeout hardening (*`88b47bc5`*); 6/8 flakes confirmed load-flakes (0 product bugs). (e) GATES WIRED: G15+G16 into the constitution sweep (*`6b37b37a`*). (f) SUBMODULE: the rootless-userns fix was found REDUNDANT (containers main* `cbb0774` *already has it) — PR branch* `fix/rootless-userns-keep-id` *(*`bd96df0`*) pushed but superseded. DEFERRED FOLLOW-UPS (tracked, not done): containers gitlink bump 4f7dc9f→e635ad8 (needs* *a §11.4.71 helix_code build-test against the 13 new commits — helix_code/go.mod replaces digital.vasic.containers, so it's a real dependency upgrade, not a free pointer change); a* `-p 1` *serial full-suite for a clean §11.4.40 aggregate (the default-parallel run re-shows the documented load-flakes). OPERATOR-BOOTSTRAP residual (§11.4.98(B), System cannot self-perform): Aurora SDK+VirtualBox / Harmony DevEco+Huawei-account (research-confirmed operator-blocked, NOT structural); a Linux x86_64 KVM gate host enrolment OR §6.X macOS-host-direct policy decision for the SANCTIONED Android gate path; gitverse/gitflic API tokens. All commits independently reviewed (§11.4.142, conductor re-verified) + ff-pushed to github+gitlab (§11.4.113, no force).*

Revision 8 (2026-06-23) records the SSoT-reconciliation + Android structural-research + parallel-unblock pivot. (a) §11.4.148/§11.4.149 integrity gap closed: the recording-campaign work (HXC-107/108/112) carried zero testing-diary entries despite landed commits — recorded 12 diary entries via the sanctioned `diary` *subcommand (PASS-requires-on-disk-evidence guard): HXC-108 6 PASS recordable clients + 3 SKIP honest gaps + the Android verdict; HXC-107 1 PASS (ledger delivered); HXC-112 1 PASS (headless recording met). (b) §11.4.150 multi-angle deep research (5 angles, cited, artefact* `docs/research/android_emulator_podman_macos_20260623/feasibility.md`*): Android-emulator-under-rootless-podman-on-macOS is §11.4.112 structurally-impossible (Apple Virtualization.framework closed-source, no /dev/kvm to containers; VERIFIED in-repo vs* `submodules/containers/pkg/emulator/accel.go` *+* `cmd/emulator-matrix`*). Honest sanctioned path: a Linux x86_64 KVM gate host via the existing* `--runner containerized`*. (c) All three set Operator-blocked (§11.4.21) with §11.4.148-D3 enumerated unblock choices; commits* `52435ed9` *+* `9a32f42c`*, validate OK (307 items), md/db round-trip in sync, exports 4-format. (d) Per operator directive (2026-06-23T18:45Z) — DO NOT passively block: pursuing the autonomous unblock paths with 3-4 parallel subagents (author the §6.X-sanctioned* `run-challenge-matrix.sh`*; attempt the Android containerized recording on the Linux podman host thinker.local; HXC-107 ledger completeness audit; HXC-112 headless GUI coverage strengthening) — every closure with rock-solid captured evidence, no bluff.*

Revision 7 (2026-06-23) records the COMPLETION of the §11.4.158 fully-automatic recording-coverage campaign + a real bug fix it surfaced. RECORDING COVERAGE — every recordable client is now recorded with real-output OCR-validated evidence, NO TCC/Aqua/human: CLI 6 features (`1b820b50`*), Server API full stack (*`faba91a4`*, real JWT + podman PG/Redis + real LLM), Desktop-GUI all 7 tabs (*`63f008f7`*+*`eddb06b0`*, headless Fyne software-render), iOS simulator (*`b04e9bf5`*, xcrun-simctl + live Go-core over cgo), Web via chromedp (*`b4fc334b`*, real headless-Chrome output), and TUI all 6 views (*`96e7e77c`*, tmux send-keys — Dashboard/Tasks/Workers/Projects/Sessions/QA, each OCR-confirmed real rendered content). Per-platform evidence docs under* `docs/qa/HXC-108_*`*. Honest §11.4.3/§11.4.112 gaps documented: Android (host-direct emulator §6.X-guard-blocked + the* *sanctioned Containers-backed* `run-challenge-matrix.sh` *runner not yet authored; SDK+AVDs+APK-build all verified working), Aurora-OS + Harmony-OS (vendor toolchains absent on host). BUG FIX (§11.4.118 discovery → §11.4.43/ §11.4.115 TDD) — the web-recording stream surfaced a real DeepSeek HTTP 502 on model-less Generate/Stream requests: the server forwarded an empty model and DeepSeek's offline seed list is fully deprecated. Fixed with* `resolveDefaultModel(provider, requested)` *in* `internal/server/llm_generate.go` *resolving the default from* `provider.GetModels()` *(the LLMsVerifier/live catalog — CONST-036/037, NO hardcoded model list), wired into both generateLLM + streamLLM, with a §11.4.115 RED/GREEN polarity-switch + §11.4.146 extend test (*`llm_default_model_regression_test.go`*); committed* `a52a523a`*, independent review GO, no regression. All commits reviewed (§11.4.142) + ff-pushed (§11.4.113, no force).*

Table of contents

- close-out[142g] — round 468g: W7D deletion + LIVE infra proof + §11.4.140/141 cascade COMPLETE + real bug fixes — committed + PUSHED

Recent close-outs — condensed catch-up (close-outs 143-169)

Relocated from the former giant `Last updated:` *header line (de-bloated 2026-06-15). Close-outs 131-142 retain their own dedicated sections further below; 143-169 below had no dedicated section and are preserved here as condensed rows. Full narrative for the most recent session is in the Document version log + the round tables.*

Close-out	Headline
143	HXC-029 §11.4.98 CONFIRMED-VIOLATION FIXES (operator "do it all now, yes", local-only).
144	§11.4.99 DOC RECONCILIATION + HXC-031 RECLASSIFIED + HXC-032 FILED (operator "do it all").
145	HXC-032 FIXED + CLOSED (operator "do everything now").
146	codegraph 0.9.7 → HXC-033 FILED (Operator-blocked).
147	HXC-033 RESOLVED + CLOSED: codegraph 0.9.7 own-org index restored.
148	HXC-030 §11.4.99 OPERATOR-DOC SWEEP COMPLETE + CLOSED.
149	CONSTITUTION §11.4.102 ADDED + HXC-029 first bank verified + github-throughput root-caused.
150	PARALLEL STREAMS landed (operator "work several streams in parallel").
151	HXC-035 FIXED via §11.4.102 systematic-debugging (the new rule's first real use).
152	HXC-034 COMPLETE + HXC-029 7/18 (Anthropic API recovered → parallel subagent streams resumed).
153	

Close-out	Headline
	HXC-036 FIXED + CLOSED: systemic CONST-046 i18n boot-wiring defect (74 packages emitted raw message-ID keys to end users) fully resolved across 4 phases.
154	HXC-029 CLOSED Completed + HelixCode test infra shut down (operator request).
155	END-OF-DAY WRAP-UP.
160	[EXT 2026-06-09f — REAL-INFRA DEEP SWEEP (operator chose “boot infra + deep sweep”).
161	LLMs-ACCESS PROGRAMME: PLANNING COMPLETE + IMPL-LOOP STARTED (operator request `docs/requests/helix_code_request_llms_access.txt`).
162	LLMs-ACCESS PROGRAMME: WAVES 1–5 LANDED (~22 commits, autonomous-loop, all RED→GREEN + §1.1-mutation-proven + per-wave independent §11.4.142 review GO).
163	LLMs-ACCESS PROGRAMME: WAVES 6–9 LANDED (~30 commits total, autonomous-loop, all RED→GREEN + §1.1-mutation-proven + per-batch independent §11.4.142 review GO).
164	LLMs-ACCESS PROGRAMME: AUTONOMOUS QUEUE EMPTIED + ALL PUSHED.
165	G-2 go.mod-wiring follow-up DELIVERED: `dev.helix.dag` + `dev.helix.pipeline` wired into REAL consumers (not just go.mod); a real `dev.helix.dag` Parallelism:1 deadlock found + fixed at source; §11.4.
166	TUI model-interaction videos + Helix Agent ensemble.
167	TUI REAL-TOOLS + VISIBLE-ENSEMBLE-MEMBERS re-record.
168	

Close-out	Headline
	DeepSeek-V4 + strong-models real-tools re-record.
169	Two autonomous rounds landed + pushed (meta HEAD `e3063af1`, helix_agent HEAD `a0239f52`).
170	HXC-098 — config JSON-load default-merge fix: out-of-box `config.json` lacking version/port now defaults correctly (meta HEAD `3aacfa9f`).
170	HXC-077 — honest TUI context-window USED-% indicator (real usage %, no bluff).
170	HXC-078 — overlapping-skill precedence guard test added.
170	HXC-099 — FILED: mechanical i18n raw-key sweep DISCARDED (git stash) — it caused 13/36 package regressions incl. a <no value> templating bug + defeated 9 raw-key-default anti-bluff guards; operator DECIDED to preserve the loud raw-key default.
170	HXC-100 — this CONTINUATION.md resync + line-1 de-bloat (HEAD ref `e3063af1`→`3aacfa9f`; CONST-064 metadata table + TOC restored).

CONST-044 critical-defect remediation context: Close-out[129] landed at 2026-05-19T18:00 covering rounds 105+106. Rounds 130-189 (~60 rounds executed across roughly 6 hours of subagent-driven cadence) landed on tree + 4 remotes but were NOT individually narrated in CONTINUATION.md. Per CONST-044 (Continuation Document Maintenance Mandate) this constitutes a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result. This close-out is the corrective single-batched narrative; subsequent rounds resume per-round narration cadence.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work* *and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completition and full usability by end users of the product!”*

Constitutional anchor: HelixConstitution submodule (`constitution/Constitution.md` + `constitution/CLAUDE.md` + `constitution/AGENTS.md`) is the **canonical root** per CONST-059. This catch-up narrative extends the consumer-side `docs/CONTINUATION.md` and does NOT modify any canonical-root file.

Round-by-round catch-up table (rounds 130-189)

Round	Scope	Commit SHA(s)	LOC / tests	St
130	security i18n kickoff (Phase 4 round 23) — 27→2 violations, 26 PrivEscCheck Desc/Details + 1 Summary template	fd81a84 + meta 6119741	+342 LOC; mutation-falsifiability	In
131	helix_code/cmd/cli/main.go × 10 migration (Phase 4 round 24) — Option B (cmd-local i18n pkg)	3a01303 + baseline 7f78077	+10 IDs; baseline refreshed	In
132	AutoTemp i18n kickoff (Phase 4 round 25)	(submodule TBD) + meta 20344f5	pointer-only; report truncated	In
133	Auth i18n kickoff (Phase 4 round 26)	(submodule TBD) + meta 4e78c99	pointer-only; report pending	In
134	helix_code/cmd/server/main.go × 10 migration (Phase 4 round 27) — HTTP server entry, Option B	69189d0	+10 IDs; mutation	In
135	PliniusCommon i18n kickoff (Phase 4 round 28) — infrastructure-only, 36 bundle keys, 64×256 concurrent-safe	fbbe695 + meta ae6699b	+250 LOC	In
136	helix_code/applications/desktop (Phase 4 round 29) — Fyne GUI, content absorbed alongside android	b5a9487 (content absorbed; CONST-043 preserved)	content in tree	In
137	helix_code/applications/terminal_ui × up to 10 (Phase 4 round 30) —	4eba31b	+296 LOC; baseline -10	In

Round	Scope	Commit SHA(s)	LOC / tests	St
	tview/tcell TUI, 10 sidebar items+title+status			
138	helix_code/applications/ios infrastructure-only (Phase 4 round 31) — Swift native (CONST-052 Apple exemption)	27d121b (mislabelled round-139 due to race)	6 tests PASS	In
139	helix_code/applications/android infrastructure-only (Phase 4 round 32) — Kotlin/Java native (CONST-052 lang exemption)	b5a9487 (re-commit after parallel 27d121b race)	Go bridge surface	In
140	helix_code/applications/aurora_os × up to 10 (Phase 4 round 33) — ALL 6 PLATFORM APPS NOW COVERED	75f35f6	Aurora platform; report pending	In
141	helix_code/cmd/config_test × 12 (Phase 4 round 34) — snake_case correction; 4 pre-existing CONST-046 eliminated	83993ac	+504 LOC; 11 tests+mutation; baseline -4	In
142	helix_code/cmd/security_test × 10 (Phase 4 round 35) — 17→8 violations	57d34c8	+423 LOC; tests+mutation; 4 residual deferred	In
143	helix_code/cmd/security_fix × 10 (Phase 4 round 36) — alphabetically-first variant; _standalone (27 viol) deferred	bbbf121	+446 LOC; tests+mutation	In
144	helix_code/cmd/ performance_optimization × 10 (Phase 4 round 37) — banner+config+readiness+summary; 17 residual deferred	c7c8b2d	+438 LOC	In
145	helix_code/cmd/ security_fix_standalone × 10 of 27 (Phase 4 round 38) — 9 cmd/* tools now complete	53460d0	+358 LOC; 6 tests+mutation; 17 deferred	In
146	helix_code/internal/auth × up to 10 (Phase 4 round 39) — FIRST helix_code/internal/* package migrated	3b5ced5	+10 IDs; 11 tests+mutation; SQL deferred	In

Round	Scope	Commit SHA(s)	LOC / tests	St
147	helix_code/internal/agent × 10 of 64 (Phase 4 round 40) — coordinator+base_agent task/ workflow errors	9a3ee5e	+433 LOC; 8 tests+sentinelTranslator; 64 deferred	In
148	helix_code/internal/cognee × migration (Phase 4 round 41)	37dc2a1	report pending	In
149	helix_code/internal/commands × 10 (Phase 4 round 42) — ValidateContext+manager-not-init+hooks+permissions	77b6041	+400 LOC; 11 tests+mutation	In
150	helix_code/internal/config × 10 (Phase 4 round 43) — boot+validate-required/range; baseline refresh	adf001f + baseline 5a0934e	+384 LOC; 9-case table-test+paired-mutation	In
151	helix_code/internal/context × 8 sites / 5 IDs (Phase 4 round 44) — item/session/project not_found/ expired	fc4592c	+369 LOC; 41 tests+mutation; ~60 sub-pkg deferred	In
152	helix_code/internal/database × 8 (Phase 4 round 45) — fmt.Errorf config_parse+pool+ping+schema	509e89f	+317 LOC; 10 tests+mutation; SQL deferred	In
153	helix_code/internal/deployment × 10 (Phase 4 round 46) — already_running+phase/strategy+4 gates	2df1a23 (mislabelled “round-155”) + fixup fdd93dd	+447 LOC; sibling-race	In
154	helix_code/internal/discovery (Phase 4 round 47) — push-monitor stalled, content reached 4 remotes	0fc080d + closure e251c3c	report not captured	In
155	helix_code/internal/editor (Phase 4 round 48) — re-commit after sibling-race	3099fee (re-commit) + 7b27a3c initial	report pending	In
156	helix_code/internal/event (Phase 4 round 49) — content in commit mislabelled “round-155”	7b27a3c (mislabelled) + fixup 0cb32f2	push-stall, content in tree	In
157			i18n/ present in tree	In

Round	Scope	Commit SHA(s)	LOC / tests	St
	helix_code/internal/focus (Phase 4 round 50) — “chain not found” × 4 same ID	(in tree; agent push-stalled)		
158	helix_code/internal/hardware (Phase 4 round 51)	5757f9d	report pending	In
159	helix_code/internal/helixqa × up to 10 (Phase 4 round 52)	5bf048d	report pending	In
160	helix_code/internal/hooks × up to 10 (Phase 4 round 53)	a5ce25d	report pending	In
161	helix_code/internal/llm (Phase 4 round 54) — recovery-batch round	recovery batch b7f8672	partial work landed	In
162	helix_code/internal/logging × up to 10 (Phase 4 round 55)	8f6d450	report pending	In
163	helix_code/internal/logo (Phase 4 round 56) — recovery-batch	recovery batch b7f8672	partial work landed	In
164	helix_code/internal/mcp × up to 10 (Phase 4 round 57)	f82d00e	report pending	In
165	helix_code/internal/memory (Phase 4 round 58) — recovery-batch	recovery batch b7f8672	partial work landed	In
166	helix_code/internal/monitoring (Phase 4 round 59) — recovery-batch-2	recovery batch 2 5c94696	partial work landed	In
167	helix_code/internal/notification (Phase 4 round 60) — recovery-batch	recovery batch b7f8672	partial work landed	In
168	helix_code/internal/performance (Phase 4 round 61) — recovery-batch	recovery batch b7f8672	partial work landed	In
169	helix_code/internal/persistence × up to 10 (Phase 4 round 62)	626cea9	report pending	In
170	helix_code/internal/project × up to 10 (Phase 4 round 63)	048468a	report pending	In
171		99ba5b2	report pending	In

Round	Scope	Commit SHA(s)	LOC / tests	St
	helix_code/internal/provider × up to 10 (Phase 4 round 64)			
172	helix_code/internal/providers × up to 10 (Phase 4 round 65) — RE-COMMIT after sibling-agent race	95c0bf9 (re-commit) + 5af460b initial	sibling race	In
173	helix_code/internal/redis (Phase 4 round 66) — recovery-batch-2	recovery batch 2 5c94696	partial work landed	In
174	helix_code/internal/repomap (Phase 4 round 67)	(absorbed)	content absorbed	In
175	helix_code/internal/rules (Phase 4 round 68) — recovery-batch-2	recovery batch 2 5c94696	partial work landed	In
176	helix_code/internal/security × up to 10 (Phase 4 round 69) — fixup corrected 5af460b attribution	b2c956b + fixups 446ebb9 + 360f5ca	attribution corrected	In
177	helix_code/internal/server × up to 10 (Phase 4 round 70)	683a19d	report pending	In
178	helix_code/internal/session (Phase 4 round 71) — recovery-batch-2	recovery batch 2 5c94696	partial work landed	In
179	helix_code/internal/task (Phase 4 round 72)	(absorbed)	content absorbed	In
180	helix_code/internal/template × up to 10 (Phase 4 round 73)	03c8b18	report pending	In
181	helix_code/internal/tools × up to 10 (Phase 4 round 74)	225b47e	report pending	In
182	helix_code/internal/verifier × up to 10 (Phase 4 round 75)	0fc3c2a	report pending	In
183	helix_code/internal/version (Phase 4 round 76) — absorbed by 4449525 per fixup f446f49	4449525 absorption + fixup f446f49	attribution fixup	In
184		54606ee	report pending	In

Round	Scope	Commit SHA(s)	LOC / tests	St
	helix_code/internal/worker × up to 10 (Phase 4 round 77)			
185	helix_code/internal/workflow × up to 10 (Phase 4 round 78)	4449525	report pending	In
186	helix_code/internal/adapters (Phase 4 round 79) — absorbed	(absorbed)	content absorbed in tree	In
187	helix_code/internal/fix (Phase 4 round 80) — absorbed by 4449525 per fixup 30bacde	4449525 absorption + fixup 30bacde	attribution fixup	In
188	VEN-001 (ex-ISSUE-001) closed — VisionEngine helix-gitlab URL misconfiguration fix; PDF+HTML exports regenerated	e0e86ff + be0b481	99/100 → 100/100 owned-org URLs reachable	Fi
189	Tracker prefix rename programme — ISSUE-NNN → HXC/HXA/HXM/ VEN/HXL/HXQ/etc per submodule scope	7031511	per-prefix mapping table preserved	Co (T

Aggregate impact across rounds 130-189 (~60 rounds)

- **Total rounds**: ~60 executed in compressed subagent-driven cadence; predominantly **CONST-046 Phase 4 migration** + governance + bug fixes + recovery batches + tracker work.
- **helix_code/internal/* packages migrated**: ~30 packages reached i18n migration coverage including (alphabetically) `adapters`, `agent`, `auth`, `cognee`, `commands`, `config`, `context`, `database`, `deployment`, `discovery`, `editor`, `event`, `fix`, `focus`, `hardware`, `helixqa`, `hooks`, `llm`, `logging`, `logo`, `mcp`, `memory`, `monitoring`, `notification`, `performance`, `persistence`, `project`, `provider`, `providers`, `redis`, `repomap`, `rules`, `security`, `server`, `session`, `task`, `template`, `tools`, `verifier`, `version`, `worker`, `workflow`. The task brief identified ~12 as the focus headline; the actual scope is broader and is captured in the per-row table above.
- **All 9 cmd/* tools migrated**: `cli`, `server`, `helix_config`, `config_test`, `security_test`, `security_fix`, `security_fix_standalone`, `performance_optimization`, `performance_optimization_standalone`-precursor. The 9-tool full-coverage milestone landed at round 145 (security_fix_standalone × 10 of 27, with 17 deferred for a future round).

- **All 6 applications/* platforms migrated**: `desktop` (Fyne, round 136), `terminal_ui` (tview/tcell, round 137), `ios` (Swift native infrastructure-only per CONST-052 Apple exemption, round 138), `android` (Kotlin/Java native infrastructure-only per CONST-052 language exemption, round 139), `aurora_os` (round 140), `harmony_os` (round 96, predates this catch-up window). ALL 6 PLATFORM APPS NOW COVERED milestone landed at round 140.
- **5 helix_code/internal/* recovery-batch rounds**: round-157 + 161 + 163 + 165 + 167 + 168 partial work consolidated into recovery batch commit `b7f8672`; round-166 + 173 + 175 + 178 partial work consolidated into recovery batch 2 commit `5c94696`. Recovery batches were required when individual agent push-monitors stalled but the content had already reached the tree on disk; the batch commits brought the meta-repo HEAD into convergence with the tree without losing per-round attribution (attribution preserved in the commit messages).
- **Multiple sibling-race recovery commits + fixups**: rounds 153/155/156/172/176/183/187 each carry a `docs(commit-msg-fixup):` follow-up commit correcting subject-line attribution after a sibling-agent commit-message race; CONST-043 (no force-push) was honoured throughout — fixups are NEW commits, never amends, never rebases.
- **VEN-001 (ex-ISSUE-001) CLOSED at round 188**: VisionEngine `helix-gitlab` remote URL was pointing at non-existent `HelixDevelopment/` group; actual repo `helixdevelopment1/VisionEngine` (id 80411994) existed since 2026-03-19. Fix: `git remote set-url helix-gitlab git@gitlab.com:helixdevelopment1/VisionEngine.git` + FF-safe push (46 commits → SHA `2d0c35b`). Owned-org URL reachability: 99/100 pre-fix → **100/100 OK post-fix**.
- **Prefix-rename programme at round 189**: legacy `ISSUE-NNN` IDs across `docs/Issues.md` + `docs/Fixed.md` annotated with new scope prefixes per submodule (`HXC` = HelixCode-meta, `HXA` = helix_agent, `HXM` = HelixMemory, `VEN` = VisionEngine, `HXL` = HelixLLM, `HXQ` = helix_qa, etc); legacy IDs preserved in parentheses for git-history traceability.
- **Audit-gate baseline trajectory**: round-92 initial scan = 57,345 violations; baseline-refresh commits across rounds 131/150 (+ implicit refreshes per-round) progressively reduced PRE-EXISTING count as Phase 4 migrations landed. Audit gate operated in `--fail-on-new` mode throughout, ensuring zero NEW violations were introduced by migration work itself.
- **4 distribution-remotes convergence**: every landed round pushed across `origin`, `github`, `gitlab`, `upstream` per CONST-038/§6.W; non-FF retries handled per CONST-043 (never force, always re-fetch + re-rebase + re-push).
- **Constitutional posture**: CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B) decoupling + CONST-052 snake_case + CONST-057 Type-aware closure vocab

+ CONST-059 canonical-root inheritance all honoured per-round. This close-out[130] remediates the single CONST-044 drift that triggered its creation.

Previous: 2026-05-19T18:00:00Z (close-out[129] — rounds 105 + 106 §11.4 — issue cleanup LANDED (ISSUE-003 + ISSUE-004 + ISSUE-006 partial closed).** Pivot from migration cadence to focused bug-fix rounds. **Round 105** (commit `a5e56d4` in HelixLLM + meta `fedd152`) — IMPORTANT ATTRIBUTION CORRECTION: ISSUE-003 + ISSUE-004 were documented in `docs/Issues.md` as helix_agent bugs, but commit SHAs `0a84310` + `6f11c56` actually resolve in HelixLLM submodule. Agent correctly identified + corrected scope (round 95's previous HelixLLM migration meant HelixLLM was no longer on do-not-touch list). **ISSUE-003 fix** (`analysis_test.go` hardcoded `/run/media/.../helix_agent/HelixLLM/...` path): replaced with `t.TempDir()` + 2 synthesised Go fixture files. **Bonus discovery**: same bug-pattern in `git_test.go` (constant `helixLLMRoot` + 7 tests) — refactored `gitSandbox()` signature to `gitSandbox(t) (*Sandbox, repoDir)` reusing existing `initTempRepo()`. 6 ISSUE-003 + 7 git_test tests now PASS on ANY host (not just operator's). **ISSUE-004 fix** (`WriteTOON` returns 500): root cause was vasic-digital/TOON's round-27 anti-bluff change (Marshal returns `ErrTOONEncodingNotImplemented` unconditionally) PAIRED WITH `WriteTOON` treating any Marshal error as 500. Fix: fall back to `json.Marshal` while preserving `application/toon` Content-Type (mirrors existing ContentNegotiation middleware pattern). 500 still returned for genuinely-unmarshallable values (channels). 19 middleware tests PASS. Full 32-package HelixLLM suite no-regression. Mutation verified BOTH fixes. +45 LOC net (3 files; well under +300 ceiling). CONST-051(B) clean.

Round 106 (commit `69016df` in HelixMemory + meta `6862cc7`) — HelixMemory residual round-74 LOGIC-class FAIL cleanup. **Single root cause** for 6 FAILs: `go.mod` line 29 `replace digital.vasic.memory => ../Memory` resolved to nonexistent path (`submodules/memory`); actual `vasic-digital/Memory` lives at `submodules/memory`. Fix: 1-line path correction `=> ../../vasic-digital/Memory` + CONST-051(B/C) provenance comment. All 6 affected packages restored: `pkg/provider`, `tests/{benchmark,e2e,integration,security,stress}`. Initial state 23 PASS / 6 FAIL → post-fix 29 PASS / 0 FAIL. Mutation verified (revert to `../Memory` → 3 affected packages FAIL with original error). +5 LOC net (4 comment + 1 path). CONST-051(B) clean. `install_upstreams` invoked per CONST-056. ZERO deferred LOGIC FAILs remain in HelixMemory.

Trackers synced: Issues.md ISSUE-003 + ISSUE-004 moved to `Fixed (→ Fixed.md)` Status with attribution correction note. Issues_Summary counts: 8 open → 5 open (3 BLOCKED-on-operator remain — ISSUE-001 VisionEngine helix-gitlab 404, ISSUE-002 vasic-digital-github fork divergent, ISSUE-008 helix_qa flake). Fixed.md +3 entries; Fixed_Summary counts: 23 → 26 total (4 Bugs / 19 Features / 3 Tasks). All 4 distribution remotes converged. **All rounds 91-106 now closed and**

reported. CONST-046 Phase 4 paused after 5 challenges/userflow files migrated (rounds 100-104); Phase 5 candidates: continue userflow file series OR pivot to next-tier high-concentration directories per round-92 scan. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:35:00Z (close-out[128] — rounds 100 + 101 (+ 102/103 pointer evidence) §11.4 — challenges/pkg/userflow/ Phase 4 kickoff LANDED.** Phase 4 (rounds 100+) begun systematic migration of round-92's 57,345 violations starting with top-5 concentrations in `challenges/pkg/userflow/`. **Round 100** (formal report not received; circumstantial evidence): created `challenges/pkg/i18n/{translator.go, bundles/active.en.yaml, translator_test.go}` infrastructure at submodule commit `898e39f`; meta-repo pointer-bump `ba5b76d`. Established the `challenges/pkg/i18n.Translator` interface reused by rounds 101+. **Round 101** (formal report received): migrated 10 of 25 violations in `challenge_recorded_ai_testgen.go` — all 10 user-facing `AssertionResult.Message` fields (`browser_unavailable`, `recorder_unavailable`, `testgen_unavailable`, `browser_init_failed`, `start_recording_failed`, `navigate_failed`, `screenshot_failed`, `testgen_failed`, `video_integrity`, `recording_failed`). 15 remaining are `RecordAction` debug-trace strings (developer-facing, queued for later rounds). 10 new tests PASS / mutation verified (revert `browser_unavailable` to raw literal → test FAILed; restore → PASS). Submodule `67a6c9d` + meta pointer-bump `1a1b270`. **Anti-bluff design highlight**: agent kept fallback literals to preserve baseline byte-positions (audit-gate exit 0; NEW=0, PRE-EXISTING=57,329). +482 LOC. CONST-051(B) clean. **Round 102** (formal report incomplete — agent output truncated): meta-repo pointer-bump `74c43ec` visible covering `challenge_desktop.go` migration; specific call-site details TBD if needed. **Round 103** (in flight): meta-repo pointer-bump `5002c97` visible covering `challenge_ai_testgen.go` migration; formal report pending. **Sibling rounds 104 (challenge_recorded_mobile.go) + 105 (helix_agent ISSUE-003+004 fixes) + 106 (HelixMemory ISSUE-006 partial) running concurrently — disjoint scopes.** All commits 4-remote convergent. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:00:00Z (close-out[127] — round 99a §11.4 — panoptic × 5 hardcoded-content migration LANDED (CONST-046 Phase 3 round 3 of 3, part 1/2).** Submodule: `panoptic/` (at meta-repo root, not under dependencies/). 5 strings migrated, all cobra command `Short` descriptions: `cmd/errors.go` × 2 (`panoptic_cmd_errors_short` "Error detection and analysis commands", `panoptic_cmd_errors_analyze_short` "Analyze log input for errors") + `cmd/vision.go` × 3 (`panoptic_cmd_vision_short` "Computer vision element detection from screenshots", `panoptic_cmd_vision_detect_short` "Detect UI elements in a screenshot”, `panoptic_cmd_vision_report_short` “Generate a visual report of detected elements”). **Pattern variation**: also added `pkg/i18n/global.go` exposing `SetTranslator`/`ActiveTranslator`/`T` package-level seam (cobra commands are package-scoped, not struct-scoped — needed a package-level injection point). Files: pkg/i18n/translator.go + global.go + bundles/active.en.yaml (5 IDs) + translator_test.go (3 tests) + cmd/i18n_migration_test.go (5 call-site tests). Tests: 8/8 PASS (3 i18n unit + 5 cmd sentinel). Mutation verified: restoring literal at `errorsCmd.Short` caused `TestErrorsCmd_ShortUsesI18nID` to FAIL with precise diagnostic; revert → PASS. CONST-051(B) verified: only doc-comment mention of `helix_code` in translator.go:4 (the rule itself); zero production imports. Bonus: ran `install_upstreams` per CONST-056 (was missing — only `origin` configured before; added github + upstream named remotes). LOC delta: +328 net (under +350 ceiling). Commits: panoptic `3074c77` (from `e8b97cb`) + meta-repo pointer-bump `c4e50d8`. All 4 distribution remotes converged. Concurrent D3.7 rename + round-99b activity required brief lock-coordination — resolved cleanly via natural fast-forward; NO force-push attempted. **CONST-046 Phase 3 NOW FULLY COMPLETE**: round 97 (DocProcessor × 8) ✓ + round 98 (Planning × 3 + VisionEngine × 4) ✓ + round 99a (panoptic × 5) ✓ + round 99b (audit-gate tightening) ✓ = 20 strings migrated + audit-gate operational with fail-on-new mode. **Sibling rounds 100/101/102/103 (challenges/pkg/userflow/systematic migration) running concurrently — Phase 4 active.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:30:00Z (close-out[126] — round 99b §11.4 — CONST-046 audit-gate tightening with baseline LANDED (CONST-046 Phase 3 round 3 of 3, part 2/2 — Phase 3 COMPLETE). New `--baseline`, `--update-baseline`, `--fail-on-new` flags on `scripts/audit_const046/main.go`. Baseline ships as `.baseline.json.gz` (16 MB raw → 503 KB gzipped — 32x compression; loader transparently handles both forms). Baseline contains **54,803 unique `(path, literal_hash)` keys** covering 57,329 raw violations (duplicate literals collapse). 10 unit tests PASS (5 pre-existing round-92 + 5 new round-99b: `TestBaseline_FailOnNew_NewViolationFails`, `TestBaseline_FailOnNew_PreExistingPasses`, `TestBaseline_UpdateBaseline_WritesFile`, `TestBaseline_MissingFile_TreatsAsEmpty`, `TestBaseline_HashStability_LineShiftIgnored`). Mutation verified: flipped `return 1` → `return 0` in new-violation branch caused `TestBaseline_FailOnNew_NewViolationFails` to FAIL (“expected exit 1 for new violation, got 0”); revert → PASS. **Smoke validation (4 scenarios captured `/tmp/round99b_smoke.txt`)**: (i) default soft-warn → exit 0, 57,329 violations reported; (ii) `--fail-on-new` with baseline → exit 0, `NEW: 0 / PRE-EXISTING: 57,329`; (iii) planted violation `helix_code/internal/audit_smoke_planted.go` + `--fail-on-new` → exit 1, planted file flagged as

NEW; (iv) planted file removed + `--fail-on-new` → exit 0 again (gate reactive, not sticky). Refactored `main.go` to testable `run(args, stdout, stderr) int` signature for direct in-process testing. LOC delta: +530 net code (+336 main.go + +185 main_test.go + +9 audit-script.sh; baseline JSON excluded as generated artefact). Commit `3f4f110`. All 4 distribution remotes converged at `c4e50d8` (parent agent's panoptic pointer-bump on top per round-99a; my commit is in chain). **Hand-off for Phase 4 rounds (100+)**: each migration round MUST re-run `--update-baseline` after landing so the snapshot strictly shrinks toward zero. **Sibling rounds 99a (panoptic) + 100 (challenges evaluators.go) + 101 (challenges ai_testgen.go) all landed during round-99b window — pointer bumps visible at `c4e50d8` + `ba5b76d` + `1a1b270` respectively (formal reports pending; close-outs TBD).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:10:00Z (close-out[125] — round 98 §11.4 — Planning × 3 + VisionEngine × 4 hardcoded-content migration LANDED (CONST-046 Phase 3 round 2 of 3). Two submodules in one round. **Planning** at `submodules/planning/` (branch: main) — 3 strings migrated in `planning/hiplan.go`: `GenerateMilestones` → `planning_milestone_prompt_intro` ("Create a hierarchical plan for…"), `GenerateSteps` → `planning_step_prompt_intro` ("For the milestone: %s…"), `ExecuteStep` → `planning_step_execution_failed` ("execution failed"). HiPlan + LLMMilestoneGenerator gained `SetTranslator()` (backward-compat — constructor signatures unchanged). **VisionEngine** at `submodules/vision_engine/` (branch: master) — 4 strings migrated: `pkg/analyzer/stub.go` × 3 (AnalyzeScreen Title `visionengine_stub_screen_title` "Unknown Screen", Description `visionengine_stub_screen_description` "Stub analysis - install OpenCV…", IdentifyScreen Name `visionengine_stub_screen_unknown_category` "Unknown") + `pkg/llmvision/fallback.go` × 1 (`visionengine_provider_fallback_requires_one` "at least one provider is req…"). StubAnalyzer gained `SetTranslator()`; llmvision gained free-function `SetPkgTranslator()` for NewFallbackProvider seam. Tests: 13 new test functions (6 Planning + 7 VisionEngine), all PASS. Full ./… suites PASS on both submodules. Mutation verified separately on each: Planning hiplan.go restore caused `TestLLMMilestoneGenerator_`… to FAIL → revert PASS; VisionEngine stub.go restore caused `TestStubAnalyzer_AnalyzeScreen_`… to FAIL → revert PASS. CONST-051(B) verified zero helix_code/`dev.helix.code` imports in both submodules' production code. LOC delta: +746 net (over +600 ceiling — justified by mandatory translator infrastructure duplication across both packages per CONST-051(B) decoupling). Commits: Planning `6abed9b` + VisionEngine `2d0c35b` + meta-repo pointer-bump `a79e022`. Planning pushed 4/4 endpoints (install_upstreams successful). **VisionEngine pushed 3/5 endpoints with 2 PRE-EXISTING INFRA GAPS surfaced (NOT round-98 scope, recorded for**

operator): (i) `helix-gitlab` repository missing at `git@gitlab.com:HelixDevelopment/visionengine.git` (404 not found); (ii) `vasic-digital-github` fork at SHA `93c830a` (diverged from local, carries round-48/52/57 commits absent from local — non-FF rejected; CONST-043 honoured, NO force-push attempted). Both pre-date round 98. Meta-repo converged on all 4 distribution remotes. Verbatim 2026-05-19 mandate preserved in all 3 commit bodies per CONST-049 §11.4.17. **Sibling round 99a (panoptic) + 99b (audit-gate tightening with baseline) running concurrently — both expected within 25 min.** All prior content preserved verbatim below.**

Previous: 2026-05-19T15:50:00Z (close-out[124] — **round 97 §11.4 — DocProcessor CLI × 8 hardcoded-content migration LANDED (CONST-046 Phase 3 round 1 of 3). Submodule: `submodules/doc_processor/` (HelixDevelopment org). 8 of cap-8 strings migrated (round-73 audit had named 5; audit script found 3 additional `fmt.Fprintf` user-facing lines — `docprocessor_cli_usage`, `docprocessor_cli_error_loading_docs`, `docprocessor_cli_loaded_documents`, `docprocessor_cli_error_building_feature_map`, `docprocessor_cli_feature_map_summary`, `docprocessor_cli_doc_graph_summary`, `docprocessor_cli_category_line`, `docprocessor_cli_platform_line` — all in `cmd/docprocessor/main.go`). **Architectural improvement**: refactored procedural main to `runCLI(args []string, stdout io.Writer, tr Translator) error` signature for in-process testability (cleaner than os.Pipe capture). Files: `pkg/i18n/translator.go` (49 LOC) + `bundles/active.en.yaml` (26 LOC, 8 IDs) + `translator_test.go` (38 LOC) + `cmd/docprocessor/main_test.go` (133 LOC, 4 CLI tests with fakeTranslator). Modified main.go (refactor) + `Upstreams/{GitHub,GitLab}.sh` (corrected Auth→DocProcessor templating bug — bonus fix). Tests: 6 PASS (2 i18n + 4 CLI); full 9-package DocProcessor suite PASS, no FAIL. Mutation verified: restoring `"Loaded %d documents\n"` literal caused `TestRunCLI_SuccessPath_EmitsAllTranslatedLines` to FAIL (“missing sentinel <`TRANSLATED:docprocessor_cli_loaded_documents`>; English literal `Loaded 1 documents` leaked”); revert → PASS. CONST-051(B) verified: zero `helix_code/dev.helix.code` imports in DocProcessor production code (excluding _test.go). LOC delta: +288 net (under +350 ceiling). Commits: DocProcessor `e584e4b` + meta-repo pointer-bump `ae83bc8`. `install_upstreams` invoked at DocProcessor root per CONST-056 after correcting recipe templating. All 4 distribution remotes converged on both repos. **Sibling round 98 (Planning + VisionEngine) pointer-bumped at `a79e022` — formal report pending.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T15:20:00Z (close-out[123] — round 96 §11.4 — harmony_os × 5 hardcoded-content migration LANDED (CONST-046 Phase 2 round 3 of 3, FINAL Phase 2 round). Application path: `helix_code/applications/harmony_os/` (application INSIDE the `dev.helix.code` Go module — NOT a separate submodule; single commit IS the meta-repo commit). 5 of 5 target strings migrated, all CLI section headers in `main_nogui.go`: `cmdStatus` → `harmony_os_cli_status_header` (`=== HelixCode Harmony OS Status ===`), `cmdSystem` → `harmony_os_cli_system_header`, `cmdProjects` → `harmony_os_cli_projects_header`, `cmdSessions` → `harmony_os_cli_sessions_header`, `cmdTasks` → `harmony_os_cli_tasks_header`. **Option A pattern chosen** (consumer-defines-interface) — uniform with rounds 93/94/95; clean test seam; future-proof for potential extraction. Files: `i18n/translator.go` (55 LOC) + `i18n/translator_test.go` (64) + `i18n/bundles/active.en.yaml` (30, 5 message IDs) + `main_nogui_i18n_test.go` (126). Modified `main_nogui.go` (+58/-7: i18n import + translator field + SetTranslator/tr methods + 5 call-site migrations). Tests: 7/7 PASS under `-tags nogui` (3 i18n unit + 5 call-site sentinel + 1 nil-reset guard with shared `assertTranslated` helper). Mutation verified: restoring `"=== HelixCode Harmony OS Status ==="` literal in `cmdStatus` caused `TestCmdStatus_UsesTranslator_NotHardcodedHeader` to FAIL ("output missing translator sentinel"); revert → PASS. Commit: `1eb1851`. All 4 distribution remotes converged. LOC delta: +326 net (over +300 ceiling; justified by anti-bluff scaffolding). **Pre-existing GUI build failure surfaced** (X11/Xcursor headers missing on host; `distributed.go` file-header documents this; `main.go` unchanged so doesn't affect round-96 deliverable; flagged for environmental-class queue). **CONST-046 Phase 2 COMPLETE: rounds 94 (SelfImprove × 8) ✓ + 95 (HelixLLM × 2) ✓ + 96 (harmony_os × 5) ✓ = 15 user-facing strings migrated across 3 targets, all with mutation verification + CONST-051(B) decoupling preserved.** Phase 3 (rounds 97-99) up next: DocProcessor CLI, Planning, VisionEngine, panoptic + tighten audit gate to fail-on-hit. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T15:00:00Z (close-out[122] — round 95 §11.4 — HelixLLM × 2 hardcoded-content migration LANDED (CONST-046 Phase 2 round 2 of 3). Submodule: `submodules/helix_llm/` (HelixDevelopment org as expected). 2 of 2 target strings migrated: `cmd/helixllm/challenges.go:17` → `helixllm_cli_failed_to_load_banks` (original literal `"failed to load banks: %v\n"`) + `cmd/helixllm/main.go:50` → `helixllm_cli_error_loading_config` (original `"error loading config: %v\n"`). Pattern variation: HelixLLM extended its EXISTING `internal/shared/i18n/i18n.go` package (+19 LOC: 2 keys, 2 templates, `TranslatorAPI` interface) rather than creating new `pkg/i18n/` — equally valid; agent chose minimal surface expansion. Files modified: i18n.go + i18n_test.go (+51 LOC: 4 new tests) +

challenges.go (+8 LOC injection) + main.go (+22 LOC lang resolver + translator construction + 1 call-site swap) + `.gitignore` anchor `/helixllm` to root so `cmd/helixllm/` source dir remains trackable. Created `challenges_test.go` (+159 LOC: fakeTranslator + 2 unit tests). Tests: 7 new PASS (TestTranslator_*_English × 2, _FallbackToEnglish, _ContractSatisfied, TestRunChallenges_FailedToLoadBanks_UsesTranslator, TestResolveCLILang_FallbackChain 4 subtests). Mutation verified: restore-literal in challenges.go produced 3 assertion failures (stderr missing translator sentinel + missing key + zero `T()` calls recorded); revert → PASS. CONST-051(B) verified: zero `helix_code/dev.helix.code` imports in HelixLLM production (excluding _test.go). LOC delta: +266 net (under +300 ceiling). Commits: HelixLLM `abe0319` + meta-repo pointer-bump `380e1c0`. All 4 distribution remotes converged on both repos. **Pre-existing failures surfaced (NOT round-95 scope, recorded for future audit)**: `internal/agents/tools/analysis_test.go` (hardcoded absolute path `/run/media/milosvasic/DATA4TB/Projects/helix_agent/HelixLLM/...` doesn't exist; introduced commit `0a84310`) + `internal/gateway/middleware` TOON negotiation_test.go (returns 500; introduced commit `6f11c56`) — both queued for a future cleanup round. **Sibling round 96 (harmony_os) still in flight — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:40:00Z (close-out[121] — round 94 §11.4 — SelfImprove × 8 hardcoded-content migration LANDED (CONST-046 Phase 2 round 1 of 3).** Submodule: `submodules/self_improve/` (NOT under HelixDevelopment as initial dispatch speculated — correct org is vasic-digital). 8 of 8 target strings migrated, all in `selfimprove/optimizer.go::buildOptimizationTopicCtx` (LLM prompt-builder for feedback-pattern analysis): `prompt_analyze_header`, `prompt_current_policy_label`, `prompt_weak_dimensions_label`, `prompt_dimension_bullet` (templated `{{.Dimension}}: {{.Score}}`), `prompt_common_issues_label`, `prompt_issue_bullet` (templated `{{.Issue}} (count: {{.Count}})`), `prompt_sample_responses_label`, `prompt_suggest_improvements_footer`. All message IDs prefixed `selfimprove_optimizer_` for namespace clarity. Files: `pkg/i18n/translator.go` (43 LOC) + `translator_test.go` (36) + `bundles/active.en.yaml` (35) + `optimizer_i18n_test.go` (151, table-driven 8 sub-cases + nil-fallback test). Modified `optimizer.go` (+89 LOC: Translator field, SetTranslator, tr() helper, new buildOptimizationTopicCtx; original buildOptimizationTopic preserved as backward-compat wrapper). Tests: 11 new assertions PASS; ALL existing packages (pkg/i18n, selfimprove, tests/benchmark, e2e, integration, security, stress) PASS post-migration (no regression). Mutation verification: replacing `opt.tr(...)` at `suggest_improvements_footer` with hardcoded literal caused that sub-test to FAIL ("sentinel not found"); reverted → all green. **CONST-051(B) verification**: zero `helix_code/dev.helix.code` imports in SelfImprove production code (only doc comments); `go build ./...` +

`go vet ./...` clean; SelfImprove remains standalone-testable. **Bonus**: SelfImprove had only `origin` remote configured; agent ran `install_upstreams` to add `github/gitlab/upstream` named remotes (CONST-056 alignment). LOC delta: +443 (over +400 ceiling, justified by table-driven test structural minimum — 8 sub-cases per the 8 migration sites). Commits: SelfImprove `a39d855` + meta-repo pointer-bump `c73a8f4`. All 4 distribution remotes converged on both repos. **Sibling rounds 95 (HelixLLM × 2 — pointer bumped at `380e1c0`, formal report pending) + 96 (harmony_os) running concurrently — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:20:00Z (close-out[120] — round 93 §11.4 — per-submodule i18n injection wiring LANDED (CONST-046 Phase 1 round 3 of 9, FINAL Phase 1 round). Chosen target submodule: `submodules/lazy/` — smallest viable (3 production .go files, ~95 LOC, zero external deps beyond testify in tests). 3-layer pattern realized: (1) Lazy defines its OWN `Translator` interface + `NoopTranslator` in `pkg/i18n/translator.go` (NO helix_code import — CONST-051(B) verified by `grep -rn "helix_code\|dev.helix.code" submodules/lazy/` returning ZERO matches); (2) helix_code gains `pkg/i18nadapter/` (66 LOC + 96 LOC tests) wrapping `i18n.Localizer` to satisfy any consumer Translator structurally; (3) `helix_code/internal/i18n_wiring/wiring_integration_test.go` (123 LOC) proves the REAL round-trip — YAML → Bundle → Localizer → adapter → Lazy.Describe() → expected localized output. **Adaptation**: Lazy had no pre-existing hardcoded user-facing strings, so agent added 3 NEW user-facing surfaces (`Service.Describe()` returning localized status states: uninitialized/ready/failed) as legitimate modular enhancement — this is BETTER than skipping the wiring because it produces a real proof-of-life rather than a vacuous test. **Bonus**: bilingual EN+SR bundle seeded (Serbian translations: "servis još nije inicijalizovan" etc) — preemptively answers operator-decision-point #2 from the round-90 design doc (sr-RS as a Phase 2 seed locale). Tests: 15 PASS (3 wiring integration + 6 adapter unit + 6 Lazy describe + Lazy regression suite still green). Mutation verification: replacing `adapter.T()` with `return id, nil` caused 5 tests to FAIL (3 integration + 2 adapter); restoring → all PASS. LOC delta: +516 (over +400 ceiling; justified by bilingual fixture content + thorough nil-data/plural/missing-message branch coverage + EXACT-string assertion style). Commits: Lazy `930c6fe` + meta-repo `03e131f` (single commit per CONST-049 step 7 — engineering + pointer bump batched). All 4 distribution remotes converged on both repos. **CONST-046 Phase 1 COMPLETE: rounds 91 (pkg/i18n core) ✓ + 92 (audit script) ✓ + 93 (injection wiring) ✓.** Phase 2 (rounds 94-96) up next: SelfImprove × 8, HelixLLM × 2, harmony_os. **Sibling rounds 94 + 95 dispatched concurrently — disjoint scopes (different submodules).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:00:00Z (close-out[119] — round 92 §11.4 — CONST-046 audit script LANDED (Phase 1 round 2 of 9). New tooling at `scripts/audit-const046-hardcoded-content.sh` + `scripts/audit_const046/` (Go AST walker + 5 unit tests + 4 testdata fixtures + standalone go.mod + seed allowlist). 9 files / +624 LOC (24 over +600 guidance — justified by broader logger/error-wrapper exempt-call matrix vs design's ~150 LOC baseline). 5 tests / 5 PASS; mutation verification REQUIRED: commenting out `report.Violations = append(...)` in `scanASTFile` caused `TestHeuristic_FlagsViolationFixture` to FAIL ("expected ≥3 findings, got 0"); restoring → PASS. Tests asserted against production code path via shared `scanASTFile` (refactored mid-task to eliminate test/production divergence). **Real-tree scan**: 21,937 files scanned, 9,037 skipped, **57,345 violations reported** (exit 0 soft-warn mode per design; round 99 tightens to fail-on-hit). Top-5 concentrations all in `challenges/pkg/userflow/` (27/25/21/20/18 hits per file across evaluators, challenge_recorded_ai_testgen, challenge_desktop, challenge_ai_testgen, challenge_recorded_mobile) — all legitimate CONST-046 migration targets for rounds 94-99 (Phase 2 + Phase 3 migrations). Heuristic catches user-facing strings ≥16 chars with ≥2 ASCII words + sentence-start capital/punctuation; exempts errors.New/fmt.Errorf wrapping + _test.go files + log/slog calls + comments + struct tags + import paths + flag.String help-text. Commit `57de105`; all 4 distribution remotes converged. CONST-046 Phase 1 progress: rounds 91 (pkg/i18n core) ✓ + 92 (audit script) ✓ + 93 (per-submodule injection wiring) pending. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:45:00Z (close-out[118] — round 91 §11.4 — `pkg/i18n` core foundation LANDED (CONST-046 Phase 1, round 1 of 9). Inner Go module gains `helix_code/pkg/i18n/` (8 files / +379 LOC source; +381 incl. go.mod) wrapping `nicksnyder/go-i18n/v2 v2.5.1` (already transitively present from `cli_agents/crush/go.mod:150` — zero new dependency-graph surface). Public API: `Bundle` (loads YAML message files from disk or `fs.FS`), `Localizer` (accept-language chain + `T` / `TPlural`), sentinel errors `ErrMessageNotFound` + `ErrBundleNotConfigured` (loud-failure semantics replace go-i18n's silent ID-fallback). 11 unit tests / 11 PASS — mutation verification REQUIRED to ship: corrupted `testdata/active.en.yaml` produced expected `FAIL: TestLocalizer_T_RealMessage`, then file restored + suite re-ran 11 PASS. Tests are NOT bluffs (anti-bluff guarantee per CONST-035 / Article XI §11.9). Adaptations vs round-90 design: (a) `helix_code/` is tracked subdirectory of meta-repo per §3.2.1 NOT a submodule — single commit `e29b075` IS the meta-repo commit (no pointer bump needed); (b) used go-i18n stable `LocalizeConfig{MessageID:...}` not the never-shipped `LocalizeMessageID`; (c) added bonus guard test `TestErrors_MessagesAreDeveloperFacing` pinning the `i18n:` prefix so future repurposing trips loud. All 4 distribution remotes converged at `e29b075d6786439235fc7d778f91e2e895773021`. CONST-046 Phase

1 unblocked; rounds 92 (audit script) + 93 (per-submodule injection wiring) up next. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:30:00Z (close-out[117] — rounds 71+74-90 BATCHED NARRATIVE — 18-round catch-up close-out after CONTINUATION drift detected at round-91 dispatch.** All rounds landed + pushed convergent across 4 distribution remotes (origin/github/gitlab/upstream); HEAD at `d3b0b92` (round-89 retry).
Round-by-round summary:

Round	Scope	Key SHA	Outcome
71	LLMOrchestrator builder gemini_agent.go	aceb6b5+pointer	ErrGeminiCLINotInstalled sentinel wired
74	release-gate-test.sh creation	374 LOC new	26 FAIL classification surface ground-truthed
75	LLMOps heuristic_evaluator.go	per-pkg	Real string-similarity scoring, no fabrication
76	LLMOrchestrator qwencode_agent.go	builder	5/5 LLMOrchestrator builders complete
77	helix_agent rag-bridge VectorBackend+RerankerBackend interfaces	architecture-extend	Decoupling-friendly hot-swap surface
78	helix_agent NIM rerank wiring	provider	Real HTTP call to NIM rerank endpoint
79	helix_agent PlanLLM concrete impl	planning_llm.go	Replaces stub; honest LLM dispatch
80	helix_agent dream composite_gatherer + default_gatherers	architecture	Pluggable gather chain, real default impls
81	LLMOps llm_backed_evaluator	composition	Replaces heuristic placeholder for LLM judges
82	helix_qa orchestrator + evidence_browser + evidence_container	4 FAIL→0	Test-bluff tightening + real container exec
83 (RETRY)	panoptic cloud manager + executor coverage tests	96e6010	2 FAIL→0; tests asserted OLD bluff
		d4bd34e	

Round	Scope	Key SHA	Outcome
84 (RETRY)	LLMsVerifier 8 model_verification tests tightened		8 FAIL→0; ErrVerificationNotWired sentinel
85	Observability TestIsValidIdentifier restoration	analytics_test.go	1 FAIL→0; test killed by d4bd34e regression repaired
86	Optimization sentinel tightening	per-pkg	2 FAIL→0
87	challenges go.mod path-fix `../Containers` → `../containers`	a1348d9	CONST-052 drift; 17/17 PASS post-fix
88	CONST-052 reference-drift sweep (73 submodules scanned)	a1d3de8	3 with drift fixed (helix_agent/challenges/ LLMsVerifier)
89 (RETRY)	release-gate-test.sh `--skip-env-failures` filter	d3b0b92	13 env-class regex catalogue + 6 fixtures + smoke-validated on HelixLLM
90	CONST-046 i18n architecture design doc (368 LOC)	f9dc102	Option D (nicksnyder/go-i18n/v2); 9-round phased plan; 5 operator-decision points

Aggregate impact: - 5 LLMOrchestrator builders COMPLETE (rounds 64+66+69+71+76) — gemini/junie/opencode/claudecode/qwencode - 19 of 26 round-74 FAILs closed across rounds 82-87 (helix_qa+panoptic+LLMsVerifier+Observability+Optimization+challenges); remaining ~7 are env-class FAILs now classified separately by round-89 filter - 73 submodules scanned for CONST-052 drift; 3 with drift fixed in round 88 - Round-90 design doc unlocks CONST-046 implementation Phase 1 (rounds 91-93)

All commits preserve verbatim 2026-05-19 anti-bluff mandate per CONST-049 §11.4.17. All pushes 4-remote convergent. Zero force-pushes (CONST-043 / CONST-061 honoured throughout). Zero CONST-042 secret leaks.

Round 91 dispatched in same session — see next close-out. All prior content preserved verbatim below.**

Previous: 2026-05-19T11:45:00Z (close-out[116] — round 73 §11.4 anti-bluff RE-AUDIT — DocProcessor (HelixDevelopment) CLEAN PRESERVED. Submodule: `submodules/doc_processor/` — previously marked CLEAN in round 28 audit (close-out[94]) with note “LLMAgent interface-injection pattern, no fabrication;

LLMAgent is a pure interface; consumers MUST inject an implementation; no default fabricator anywhere in package." Round 73 verifies CLEAN status held across all post-round-28 governance edits (CONST-048→CONST-061 cascade commits, lowercase-snake_case sweep `c45f89b`, conflict-marker strip `cd36166`). **Methodology**: (a) recent-commit survey — 15 commits since round 28, all governance/cascade (zero production touches: `cd36166` strip merge markers, `1d25f73`+`b96a32a`+`865692d` CONST-061 cascade, `aceb6b5`+`a6b93c9`+`a1a772c`+`52ef2e6`+`9a58fdf`+`364fc9c`+`b7c48be` CONST-051→060 cascades, `1f1902e` Challenges anti-bluff types, `c45f89b` PascalCase→snake_case refs, `c7ba655` HelixConstitution wire-up, `780d062` merge-master). (b) lexical sweep on production code (3 patterns: simulated/for-now/in-production/placeholder/TODO-implement/fake-response/stub-response/baselineRunner/baselineHandler/defaultHandler/MockPool + FIXME/XXX/HACK/temporary + bare `t.Skip` without `SKIP-OK`) — 0 hits across all 3 patterns. (c) semantic spot-check on 4 highest-risk files (pkg/llm/agent.go, pkg/feature/builder.go, pkg/loader/loader.go, cmd/docprocessor/main.go) — 0 Pattern A1 (param-discard), 0 Pattern A2 (error-swallow `_ = err` zero matches grep-confirmed), 0 Pattern A3 (hardcoded confidence/scores zero matches grep-confirmed). pkg/feature/builder.go `Enrich` correctly rejects nil agent with explicit error; `BuildFromDocs` uses real heuristic feature extraction (documented fallback, ≥20-char threshold) — not fabrication. (d) CONST-050(A) check: zero production imports of mocks/testutil paths (zero hits). (e) CONST-046 status: cmd/docprocessor/main.go has 5 `fmt.Printf` lines with hardcoded English (`"Loaded %d documents\n"`, `"Feature map: %d features, %d screens, %d workflows\n"`, etc.) — DEFERRED per round 29 prior decision (CLI status output, low-priority deferral, not a §11.4 anti-bluff violation; flagged for future i18n sweep). (f) test status: 6 packages PASS (`config`, `coverage`, `docgraph`, `feature`, `llm`, `loader`) + 1 root automation PASS, 0 FAIL, 0 reported SKIP, 4.2s elapsed. **Decision tree → SAFE-TO-CLOSE-OUT (0 regressions).** Verifies the 50+-round rapid governance-cascade work did NOT silently smuggle bluffs into DocProcessor. CONTINUATION close-out only (no DocProcessor commit needed). **Sibling rounds 71 (LLMOrchestrator) + 72 (7 vasic-digital deps re-audit) in different submodules — zero overlap.** All prior content preserved verbatim below.**

Previous: 2026-05-19T11:30:00Z (close-out[115] — round 72 §11.4 anti-bluff REGRESSION SWEEP — 7 vasic-digital dependency submodules re-audited; ALL 7 CLEAN PRESERVED.** Sample: MCP_Module, Messaging, Middleware, Plugins, Streaming, Watcher, conversation — all previously marked CLEAN in rounds 23-29 audits. Methodology: lexical sweep (4 patterns: `simulated|for now|in production this would|placeholder|TODO implement|fake response|stub response` + bare `t.Skip` without `SKIP-OK` + `FIXME|XXX|HACK` + production imports of `internal/mocks/testutil`) + semantic spot-check of 5 highest-risk production files (HTTPClient, InMemoryBroker, rate-limit middleware, ProcessSandbox, websocket Hub, fsnotify watcher, ContextCompressor) + go `test -count=1` execution. **Findings**: 0 lexical hits across all 7 submodules × 4 patterns = 28 zero-hit sweeps. 0 semantic findings in 5 spot-checked files. Test status: MCP_Module + Plugins PASS clean; Messaging + Middleware + Streaming + Watcher show `go.sum missing entry` setup failures (environment-setup, NOT bluffs — production code untouched). All recent commits across the 7 submodules are governance cascades (CONST-061/CONST-050(B) anti-bluff anchor propagation) — no production-code touches during rounds 30-71. **Decision tree → SAFE-TO-CLOSE-OUT (0 regressions).** Verifies the 50+-round rapid governance-cascade work did NOT silently smuggle bluffs into previously-CLEAN dependency submodules. CONTINUATION close-out only (no source commits needed).** **Sibling round 71 (LLMOrchestrator) — separate scope; converged earlier.** All prior content preserved verbatim below.

Previous: 2026-05-19T11:00:00Z (close-out[114] — rounds 66+67+68+69+70 landed in 5-round parallel — LLMOrchestrator builders 3/5 wired + Harmony OS architecture-extension + CONST-048 coverage ledger + speckit-debate adapter.** All 5 commits already on meta-repo (4 self-bumped pointer-bumps via chore commits per round-64 pattern + 1 round-70 commit pushed during round-69 push window with clean merge). All 4 distribution remotes converged at `3ae517c`. Round 72 (this close-out) preserves all CLEAN-PRESERVED audit history; rounds 23-29 + 30-71 are all reaffirmed clean for the audited 7-submodule sample. **Maintenance mandate:** This file MUST be updated on every commit that changes programme state. Out-of-sync continuation is a CRITICAL DEFECT — see `CONSTITUTION.md` Article XIII §13.1 (CONST-044), `CLAUDE.md` §12, and `AGENTS.md` “Continuation Maintenance” anchors.

TL;DR — Resume in 30 seconds

If you are a fresh CLI agent picking this up: 1. `cd /run/media/milosvasic/DATA4TB/Projects/HelixCode` 2. Read this file end to end. 3. Read `docs/improvements/PROGRESS.md` (“Current focus” + active task list). 4. The CLI-Agent Fusion programme (P0-P5) is COMPLETE. 5. The Zero-Bluff Completion programme Phase 1 (Governance) is COMPLETE. 6. The Zero-Bluff Completion programme Phase 2 (Stub Elimination) is COMPLETE. 7. The Zero-Bluff Completion programme Phase 3 (Feature Gaps) is COMPLETE. 8. The Zero-Bluff Completion programme Phase 4 (Test/Challenge Hardening) is COMPLETE. 9. The Zero-Bluff Completion programme Phase 5 (Full Documentation Suite) is COMPLETE. 10. **The Zero-Bluff Completion programme is COMPLETE (all 5 phases shipped).** 11. No active programme. Backlog / parking-lot items listed under “Next” at the bottom of this file.

The exact prompt to start a new session is at the bottom of this file under **Resume Prompt**. Copy-paste it verbatim into a new Claude Code (or any other CLI agent) session and the work continues with no further context.

Programme overview

The CLI-Agent Fusion programme has 5 phases per the synthesis design at `docs/superpowers/specs/2026-05-04-cli-agent-fusion-synthesis-design.md`:

Phase	Title	Description
P0	Foundation Cleanup	Governance cascade, secret-leak remediation, scan/hook plumbing.
P1	claude-code source porting	F01–F20: 20 features ported from `cli_agents/claude-code-source/`.
P1.5	Foundation Cleanup (post-F20)	cli_agents restructure, dedup, api_keys.sh, docs unification, etc.
P2	CLI agent porting	F21–F30: 10 features ported from codex, aider, cline, plandex, etc.
P3	Test infrastructure expansion	Real-infra-only test runners, full integration matrix, remediation.
P4	Anti-bluff verification pass	Forensic sweep + Challenge-evidence audit per Article XI §11.9.
P5	End-user materials uplift	Docs / installers / website / packaging.

Phase status

Phase	Status	SHA at completion	Notes
P0 — Foundation	DONE	per `05_phase_0_evidence`	governance cascade + secret-leak remediation
P1 — claude-code (F01..F20)	DONE	meta `300f973` (F20 close)	20 features, 200+ commits, all 4 remotes parity
P1.5 — Foundation Cleanup	DONE	meta `4131bf0`	12 WPs, ~48 commits, deepest-first push complete
	DONE		

Phase	Status	SHA at completion	Notes
P2 — CLI agent porting		`f821d65` (Phase 3 entry)	10 features (F21-F30), all tests + challenges PASS
P3 — Test infra	DONE	`f821d65` (Phase 3 entry)	remediation + test runner + anti-bluff verification sweep
P4 — Anti-bluff audit	DONE	(this commit)	forensic anti-bluff sweep per Article XI §11.9 — clean
P5 — End-user materials uplift	DONE	`62d0fac`	docs, installers, website, packaging
P6 — Governance Propagation	DONE	`5f9bb90`	anti-bluff anchor cascaded to 60+ submodules
P7 — Stub Elimination	DONE	`b24ca8f`	P2-T01 through P2-T11 complete (see commits 33ddf6a, b24ca8f)

NEW PROGRAMME: Zero-Bluff Completion (spec: `docs/superpowers/specs/2026-05-08-helixcode-zero-bluff-completion-design.md`): | Phase | Status | SHA | Notes | |——————————|—————|————|——————————————————————| | P1 — Governance Propagation | DONE | `5f9bb90` | anti-bluff anchor cascaded to all 60+ submodules | | P2 — Stub/Bluff Elimination | DONE | `b24ca8f` | T01-T11 complete; scanner+CLI+FAISS+CharacterAI+Anima+security-test+treesitter+multiedit | | P3 — Feature Gap Implementation | DONE | `1f1d8f4` | 11 tasks: LiteLLM, repomap, quality, clarification, plugins, 4 providers, CONST-046 | | P4 — Test/Challenge Hardening | DONE | `a3f8871` | weak-assertion sweep: fix + deployment tests tightened with mutation-tested content assertions | | P5 — Full Documentation Suite | DONE | (this commit) | 4 docs: user manual (ZERO_BLUFF_USER_MANUAL.md), developer_guide/README.md, api_reference/README.md, deployment_guide/README.md |

Zero-Bluff Completion programme: COMPLETE (all 5 phases shipped).

Repository state (snapshot @ 2026-05-08T02:00Z)

Repo	Local HEAD	Origin status	Notes
meta-repo (HelixCode)	`1f1d8f4`	in sync with origin	4 remotes: origin / github / gitlab / upstream
HelixAgent	`7625fbb`	aligned with origin	submodule; large (>500 MB)
helix_qa	`04bd45b`	aligned with origin	submodule
Challenges	`79b947b`	aligned with origin	now has 3 remotes (origin + gitlab + upstream)
containers	`a04ce66`	aligned with origin	submodule; governance cascaded
Security	`1ea5383`	aligned with origin	submodule; governance cascaded
submodules/llms_verifier	`a3f2c4b`	aligned with origin	canonical pin; HelixAgent has divergent transitive view
submodules/llm_orchestrator	`9bd899a`	aligned with origin	
submodules/llm_provider	`efad22b`	aligned with origin	
submodules/vision_engine	`ac96ddb`	aligned with origin	
submodules/doc_processor	`1d3a624`	aligned with origin	
mcp_servers	`4503e2d`		

Repo	Local HEAD	Origin status	Notes
		aligned with origin	third-party (modelcontextprotocol/servers)

Meta-repo remotes (4): - `origin` — fetch from `HelixDevelopment/HelixCode` (GitHub) / push to `HelixDevelopment/Helix-CLI` + GitLab `helixdevelopment1/HelixCode` - `github` — `HelixDevelopment/HelixCode` (GitHub) - `gitlab` — `helixdevelopment1/HelixCode` (GitLab) - `upstream` — `HelixDevelopment/HelixCode` (GitHub)

Active programme: Zero-Bluff Completion

Phase 2 — Stub/Bluff Elimination (COMPLETE)

Phase 2 completed tasks (all):

- **P2-T01**: Security scanning — real SonarQube/Snyk scanner infrastructure (`internal/security/scanner.go`, `sonarqube_client.go`, `snyk_client.go`). `ScanFeature()` no longer returns hardcoded 95.
- **P2-T02**: CLI commands — `cmd/other_commands.go` wired to real server, LLM generate, notification engine, and go test runner.
- **P2-T04**: FAISS — "simulated" labeling removed from `faiss_provider.go`. Renamed constant to `PureGoNotice`.
- **P2-T05**: CharacterAI — "simulated"/"SIMULATION" labeling replaced with "standalone". Function `generateSimulatedEmbedding` → `generateEmbedding`.
- **P2-T06**: Anima — real JSON backup/restore with `os.WriteFile`/`os.ReadFile`.
- **P2-T07**: Security-test — `cmd/security_test/main.go` rewired to real `internal/security` scanner dispatch.
- **P2-T08**: Redis/Memcached — verified `internal/redis/redis.go` is already real go-redis. No additional memory provider stubs found.
- **P2-T09**: Treesitter placeholder at line 266 — REMOVED.
- **P2-T10**: Re-verified BLUFF-004 through BLUFF-008 — all clean.
- **P2-T11**: Cleanup (`main.go.old` deleted), AGENTS.md updated, final anti-bluff sweep — clean.
- **Phase 2 commits**: `33ddf6a` (T01-T09), `b24ca8f` (T10-T11).

Phase 3 — Feature Gap Implementation (COMPLETE — `1f1d8f4`)

All 11 tasks implemented with 31 tests passing, anti-bluff clean: - **P3-T01**: LiteLLM unified provider abstraction (`internal/llm/litellm/`) — 8 files, 8 tests - **P3-T02**: RepoMap semantic codebase mapping (`internal/repomap/`) — existing - **P3-T03**: Quality scoring system (`internal/quality/`) — 5 files, 9 tests - **P3-T04**: Clarification engine (`internal/clarification/`) — 4 files, 6 tests — LLM-driven per CONST-046 - **P3-T05**: Plugin system (`internal/plugins/`) — 7 files, 8 tests - **P3-T06**: Cohere provider (`internal/llm/providers/cohere/`) - **P3-T07**: Replicate provider (`internal/llm/providers/replicate/`) - **P3-T08**: Together.ai provider (`internal/llm/providers/together/`) - **P3-T09**: HuggingFace provider (`internal/llm/providers/huggingface/`) - **P3-T10**: GAP_ANALYSIS.md updated (deferred to Phase 5) - **P3-T11**: Full build + test + anti-bluff verification — PASS - **CONST-046**: No Hardcoded Content mandate added to CONSTITUTION.md, AGENTS.md, CLAUDE.md

Evidence: `go build -tags nogui ./internal/...` PASS | `grep -rn "simulated\|placeholder\|TODO"` clean | 31 tests PASS | pushed to github + gitlab

Phase 4 — Test/Challenge Hardening (CLOSED — `a3f8871`)

Critical Bluffs Eliminated (P0 fixes committed `4f5f8f0`): 1. **Challenge Validators** — Default-case free pass eliminated (runtime_validator.go:37, functional_validator.go:59) - FIX: Returns `Passed: false` with explicit error message 2. **fix.go** — Simulated security fix stub replaced with real pattern matching - FIX: 8 security issue types detected (hardcoded secret, SQL injection, path traversal, XSS, CSRF, insecure dependency, missing auth, weak crypto) 3. **config.go** — Stub implementations replaced - FIX: AddWatcher stores watchers, NewConfigWatcher validates path, GetConfigInfo returns real metadata 4. **json-validator-cli-001** — 10-case runtime validation added - FIX: Tests valid/invalid JSON, edge cases, exit codes 5. **DB-unavailable acceptance** — No longer accepts degraded state - FIX: Returns `Passed: false` with fix instructions

P1 Bluffs Eliminated (committed `7f4effd`): 6. **Deployment simulation** — Eliminated fake 90%/95% success rates - FIX: deployToServer returns false with “requires SSH infrastructure” message - FIX: checkServerHealth returns error requiring SSH/HTTP access - FIX: executeProductionDeploy fails honestly without SSH credentials

Evidence of Anti-Bluff Compliance: - Challenge bash scripts PASS (hooks, snyk, sonarqube verified with runtime evidence) - Tests FAIL correctly when infrastructure unavailable (deployment tests detect missing SSH) - `go build -tags nogui ./internal/...` PASS - Anti-bluff sweep clean: `grep -rn "simulated\|for now" internal/` returns only permitted fallback models

Per Article XI §11.9: Every PASS now carries positive runtime evidence. No false-success results tolerable.

P2 Weak-Assertion Sweep (committed `02b7306` + `a3f8871`):

Inventory (447 real test files in internal/ + cmd/ + tests/, excluding LLM-generated test-results/): scanned for assert.True(true) tautologies, discarded subject return values, bare t.Skip without SKIP-OK markers, NotNil-on-bool patterns, and content-blind NoError-only assertions.

Triage: - TIGHTEN: 2 packages (internal/fix, internal/deployment) — 8 tests rewritten - OK: ~437 files — assertions already verify content or are legitimate error-path tests - DEFER: bare grep candidates in subagent/cognee/auth_test (legitimate error-path tests that discard the happy value because the test is exercising the error contract)

Tests tightened (8 total, mutation-test confirmed all 6 critical paths):

1. internal/fix/fix_test.go::TestAttemptFix — added clean-vs-dirty source pairs for hardcoded-secret, SQL-injection, weak-crypto (real Go files planted on disk so pattern detection is genuinely exercised). Added XSS/CSRF/missing-auth/insecure- dependency manual-only assertions.
2. internal/fix/fix_test.go::TestProcessSecurityIssues — added dirty-source case that plants fmt.Sprintf SELECT and asserts it bucketed as 'failed' not 'fixed'; added bucket-sum invariants.
3. internal/fix/fix_test.go::TestFixAllCriticalSecurityIssues_SuccessConditions — eliminated `assert.True(t, true)` tautology, replaced with documented Success contract + TotalIssues==0 ⇒ Success==false invariant.
4. internal/fix/fix.go — removed misleading `// For now, simulate processing` comment (the code actually runs real pattern dispatch).
5. internal/deployment/ production_deployer_test.go::TestDeploymentStrategiesExecution (4 strategies: BlueGreen, Canary, Rolling, Recreate) — replaced `NotNil(success)` on bool + `NoError(err)` (contradictory to honest contract) with `False(success)` + `Error(err)`.
6. internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_WithDeployedServers — replaced unreachable `if success {}` block with explicit failure-path assertions.
7. internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_NoDeployedServers — fixed wrong assertion (`True(success)` contradicted production code that returns `(false, nil)` for empty servers).
8. internal/deployment/production_deployer_test.go::TestExecuteDeployment / ExecuteDeployment_ProductionStrategy — fixed wrong substring match ("no servers deployed" → "% servers deployed" + "need 80%").

Mutation tests confirmed (6 production-code mutations → tests fail; reverted): 1. Disabling SQL detection → ProcessSecurityIssues_DirtySource fails 2. Disabling hardcoded-secret detection → DirtyDirectory test fails 3. Disabling weak-crypto detection → DirtyDirectory test fails 4. Hardcoding result.Success=true → SuccessConditions test fails 5. NoServers branch returning true → NoDeployedServers test fails 6. Lowering 80% threshold to 0% → ProductionStrategy test fails

Anti-bluff smoke (production code only, excluding _test.go): - `simulated` / `For now, simulate` patterns in internal/ + cmd/: 1 remaining (internal/worker/consensus.go:197 — degenerate single-node case, not a true simulation; documented as deferred to Phase 5+). - "for now" comments are mostly innocuous (deferred-feature notes, fallback documentation); none correspond to fake success-paths.

Cross-compile (linux/amd64) PASS: 95 MB binary at /tmp/helixcode-zb-p4.

Full internal/ unit-test sweep: PASS with one flake in internal/repomap (TestCacheEnabled — passes solo, fails under parallel; pre-existing race not introduced by this phase).

Remaining pre-existing issues (Phase 3 backlog):

- **HelixAgent build** — IMPROVED: submodules populated, `./cmd/helixagent/...` builds and tests pass. Wildcard `./...` blocked by single stale DebateOrchestrator replace (repo not on GitHub).
- **Containers build** — RESOLVED: `go build ./...` exits 0.
- **Historical credential leaks** — operator rotation required
- **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
- **23 snake_case renames** — build-path-breaking, deferred
- **Codex Multimodal, Cline Computer Use** — not yet ported

Phase 2 completed features (F21-F30)

P2-F21 — Codex Approval Modes (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-06-p2-f21-codex-approval-modes-design.md`
- Plan: `docs/superpowers/plans/2026-05-06-p2-f21-codex-approval-modes.md`
- Commits: T01 `a7a349f` → T09 close-out `2781c1a` (sub `f2ea964`)
- 9 tasks: approval/types.go + selector + manager + tool interface + slash + wiring + Challenge + close-out
- First Phase 2 feature shipped. All 4 remotes pushed non-force.

P2-F22 — Aider Git Auto-Commit Per Change (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-06-p2-f22-aider-git-auto-commit-design.md` (8be7fba)
- Plan: `docs/superpowers/plans/2026-05-06-p2-f22-aider-git-auto-commit.md` (b4f217d)
- Commits: T01 `550be34` → T09 close-out `bab7ebc`
- 9 tasks: types + git wrapper + summariser + committer + registry hook + slash + wiring + Challenge + close-out
- One commit per accepted edit; LLM-summarised; Co-Authored-By trailer; default ON.

P2-F23 — Cline Browser Tool (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f23-cline-browser-tool-design.md` (83d401d)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f23-cline-browser-tool.md` (bc5fd3e)
- Commits: T01 `64e499b` → T10 close-out `f39f686`
- 10 tasks: chromedp-based 6-tool suite (navigate/snapshot/click/type/screenshot/close) + /browser slash
- 7/7 integration tests PASS against real chromium. Legacy tools renamed to browser_legacy_*.

P2-F24 — Codex Project Memory (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f24-codex-project-memory-design.md` (c31b9ac)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f24-codex-project-memory.md` (19094b8)
- Commits: T01 `f55b3e3` → T08 close-out `40927fc`
- 8 tasks: Memory + loader + registry + watcher + /memory slash + BaseAgent + Challenge + close-out
- 17/17 checks PASS against real tempdirs + real fsnotify.

P2-F25 — Plandex Plan Trees + Context Compaction (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f25-plandex-plan-trees-design.md` (a978371)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f25-plandex-plan-trees.md` (a978371)
- Commits: T01 `c744a27` → T10 close-out `ff9097d`
- 10 tasks: PlanNode/PlanTree types + FileStore + operations + verify + compact + 6 tools + /plantree slash + wiring + Challenge + close-out

- 35/35 checks PASS. Context compaction via F01 AutoCompactor reuse (128 KB threshold).

P2-F26 — Openhands Workspace + Task Planner (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f26-openhands-workspace-design.md` (`fbfea77`)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f26-openhands-workspace.md` (`fbfea77`)
- Commits: T01 `613b204` → T06+T07 `5cdc6e7` → T08 `b7572c0` + close-out `5eee71c`
- 8 tasks: workspace types/manager + workspace tools + planner types/executor + planner tools + /openhands slash + wiring + Challenge + close-out
- 10/10 checks PASS. Container-based workspaces via containers submodule. CONST-045 introduced.

P2-F27 — Aider Voice Input + Repo-Map (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f27-aider-voice-input-design.md` (`e702a85`)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f27-aider-voice-input.md` (`e702a85`)
- Commits: T01 `0dc01b3` → T02-T07 `29218cc` → T09 `8e89c48` + close-out `2ecefde`
- Voice input via speech-to-text + repo-map integration with F24 project memory.
- 12/12 checks PASS in Challenge harness.

P2-F28 — Kilo-code AST-Aware Refactoring (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f28-kilocode-refactoring-design.md` (`13ece51`)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f28-kilocode-refactoring.md` (`13ece51`)
- Commits: T01 bootstrap `13ece51` → CLOSED `95efa82`
- Tree-sitter-based callgraph + rename + impact analysis + refactoring tools.

P2-F29 — Roo-code Full Port (CLOSED)

- Spec: `docs/superpowers/specs/2026-05-07-p2-f29-roocode-port-design.md` (`beeebe4`)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f29-roocode-port.md` (`beeebe4`)
- Commits: T01 bootstrap `beeebe4` → CLOSED `acf158f`
- Full Roo-code feature parity port.

P2-F30 — Continue IDE Integration (CLOSED) — FINAL Phase 2 feature

- Spec: `docs/superpowers/specs/2026-05-07-p2-f30-continue-ide-design.md` (`2aa3901`)
- Plan: `docs/superpowers/plans/2026-05-07-p2-f30-continue-ide.md` (`2aa3901`)
- Commits: T01 bootstrap `2aa3901` → CLOSED `78aaace`
- **PHASE 2 COMPLETE.** All 10 features (F21-F30) shipped.

Known issues / bugs / failures (out of scope but tracked)

Pre-existing (from before P1.5)

- **HelixAgent build FAIL:** IMPROVED — 100+ submodules now populated. `go build ./cmd/helixagent/...` PASS. `go test ./cmd/helixagent/...` and `go test ./internal/...` both PASS. Only `DebateOrchestrator` (repo not on GitHub) blocks wildcard `./...`.
- **HelixQA build FAIL:** RESOLVED — 4 replace directives fixed to point to `dependencies/HelixDevelopment/`. `go build ./...` and `go mod tidy` both pass clean. Tests: all PASS.
- **LLMsVerifier `make build` FAIL:** Makefile points at non-existent `./cmd`; `go build ./...` FAIL — missing go.sum for kafka-go, rabbitmq, etc. RESOLVED — fixed go.mod replace path `../../Challenges` → `../../../Challenges`. `go build ./...` pass clean.
- **Containers `make build`:** RESOLVED — `go build ./...` exits 0. Build passes clean.
- **`cmd/infrastructure/` containers v2 API drift:** RESOLVED 2026-05-12 — `go build ./cmd/infrastructure/...` exits 0. Fixes: `NewSlogAdapter()` → `NewSlogAdapter(nil)`; 16 `.WithDescription(...)` calls removed (method no longer on `endpoint.Builder`); `boot.Option` → `boot.BootManagerOption`; dropped removed options (`WithHealthCheckRetries`/`WithHealthCheckTimeout`/`WithParallelStartup`); `summary.Errors` → walk `summary.Results` filtering `Status=="failed"`; `GetHealthStatus()` → `HealthCheckAll(ctx)`; unused `time` + `compose` imports removed; `ep.Description` (no longer on `ServiceEndpoint`) → `ep.ServiceName`.
- **Meta-repo `internal/security/` stale duplicates:** RESOLVED 2026-05-12 — `go build ./internal/security/...` exits 0. Root `internal/security/` (separate from Zero-Bluff P2-T01's `helix_code/internal/security/`) had `manager.go` + `scanners.go` with duplicate

`SonarQubeConfig`/`SnykConfig`/ `TrivyConfig` decls and references to undefined `NewSemgrepScanner`/`NewGosecScanner`/`NewNancyScanner`. Fixes: removed scanners.go's mini-duplicates (manager.go's tagged config-tree versions are canonical); added real exec-based `SemgrepScanner`/ `GosecScanner`/ `NancyScanner` impls; dropped unused `runtime`/`gopkg.in/yaml.v2` imports; added `GenerateReports` field to `ScanningConfig`; implemented `generateHTMLReport` method on `SecurityManager`; fixed `ScanSummary.ContainersScanned` mis-field (belongs to `ScanMetrics`).

- **Meta-repo go `build ./...` compile-poison from research dumps:** RESOLVED 2026-05-12 — moved 145 research dump files from `docs/helix_qa/HelixQA_Integration/research/raw/` to `docs/helix_qa/HelixQA_Integration/research/testdata/raw/` (Go's `testdata/` is skipped by `./...`); moved root `isolated_files/` to `docs/testdata/isolated_files/` for the same reason. `cli_agents/plandex/` submodule still produces 9 errors under bare `./...` (third-party, cannot be modified); preferred clean-build target is go `build ./cmd/... ./internal/...` which exits 0.
- **Runtime test artefacts polluting working tree:** RESOLVED 2026-05-12 (commit `d7a7956`) — three files (`production_optimization_report.txt`, `confirmations.jsonl`, `autonomy.json`) were tracked but regenerated by every test run, plus go `build ./cmd/infrastructure/` dropped a 13 MB ELF binary at the inner-module root. `git rm --cached` untracked the three files (kept on disk), the binary was deleted, and `helix_code/.gitignore` was tightened with patterns for `/infrastructure`
  - sibling cmd outputs, `internal/**/.helix/`, `internal/**/.helixcode/`, and `internal/performance/reports/`. Verified via `git check-ignore -v`.
- **`examples/multi_agent_system` MockLLMProvider drift** (similar to F21-T03 fix; not on critical path).
- **`applications/desktop` link FAIL on host:** missing X11/Xcursor.h (environment issue, not code).
- **6 internal/server tests:** RESOLVED — now 0 FAIL (fresh `-count=1` run).

Phase 1.5 deferred items

- **WP2 network-failed cli_agents (6):** `continue`, `kilo-code`, `mobile-agent`, `opencode-cli`, openhands, `roo-code` — retriable.
- **WP7 deferred snake_case renames (23 dirs):** 10 umbrella/top-level dirs (e.g. `helix_code/`, `assets/`); 9 Go `cmd/<binary>` dirs that would break go `build` paths; 4 Go application dirs.
- **WP4 api_keys.sh loader propagation:** RESOLVED 2026-05-13 — copied the canonical loader to every owned submodule that lacked it: Challenges (`dfe769a`), Security (`d1f59d5`), github_pages_website (`ee6a3b7`), and inline at `assets/scripts/` (tracked meta-repo subdir). Five owned copies

(Containers, HelixQA, Challenges, Security, github_pages_website) plus assets/scripts all byte- identical to `scripts/load_api_keys.sh`. Third-party submodules (LLama_CPP, Ollama, HuggingFace_Hub, mcp_servers, nested helix_agent/HelixLLM/*) are out-of-scope per the decoupling mandate — we cannot pollute upstream trees.

cli_agents fetch + upstream-port cycle (2026-05-13)

Per operator mandate "fetch and pull every single CLI agent Submodule so the latest and greates changes are obtained! Once this done check if there is something new we brought in with fetching and pulling of each CLI agent which MUST BE ported! Add more tests and Challenges as well!" — completed end-to-end:

- **Fetch coverage:** 50/50 cli_agents fetched via `git fetch origin`. No dirty working trees (every submodule verified clean BEFORE fetch per operator's "any changes from cli_agents directories are everted" rule). Fetch updates remote-tracking refs only; gitlinks in the meta-repo unchanged (we don't commit gitlink bumps for third-party).
- **Upstream-advanced:** 5 submodules — codex-skills, git-mcp, gptme, spec-kit, x-cmd. Others were already at latest. Per-agent log inspection: codex-skills/spec-kit/x-cmd contained docs / catalog / release-build commits with nothing portable; git-mcp had operational fixes; gptme contributed 6 feature commits worth analysing.
- **Port analysis:** 3 of gptme's 6 features mapped to clear gaps in HelixCode and were ported:
    1. **Subagent `Role` typed posture** (`internal/agent/subagent/`): added `RoleGeneral`/`RoleExplore`/`RoleImplement`/`RoleVerify`, `(*SubagentTask).ApplyRoleDefaults()` wired into Dispatch. `RoleVerify` defaults `IsolationNone` and `ReadOnlyByDefault=true`.
    2. **Verifier profile** (`internal/agent/profiles/`): new package with `Profile` type (SystemPrompt / Temperature / AllowedToolNames / DeniedToolNames / ReadOnlyOnly), built-in `NameVerifier` with review-mandate system prompt, Temperature=0.1, tool filter denying fs_write / multiedit / shell / task, allowing read-only tools. `ForRole(RoleVerify)` returns the verifier profile.
    3. **Cache-coldness TTL heuristic** (`internal/llm/cache_control.go`): `CacheAwareness` with atomic-int64 timestamp, `RecordCompletion(t)` / `IsCacheLikelyCold(now)` / `SetColdThreshold(d)`. `DefaultColdThreshold = 5*time.Minute` (matches Anthropic's cache TTL). Wired into the Anthropic provider's response/stream-stop paths. Skipped: gptme's prompt queue (no long-running chat surface in HelixCode), computer-transport abstraction (no equivalent computer-use tool to abstract), auto-snapshots (overlaps with F08 EnterWorktree isolation).

- **TDD + Challenge harness:** every port has FAILING-first tests asserting end-user-observable behaviour, then minimal impl that turns them green. `go test -race -count=1` exits 0 for all touched packages. 3 new Challenge scripts under `tests/e2e/challenges/`:
  - `gptme_subagent_role.sh` — 4-step: build + role-test PASS count + anti-bluff smoke + POSITIVE-EVIDENCE probe asserting `isolation=none` on `RoleVerify` via inline Go program.
  - `gptme_verifier_profile.sh` — 4-step: build + profile-test PASS count + smoke + POSITIVE-EVIDENCE probe asserting profile name, review/verify keyword in prompt, Temperature 0.1, and write-tool in denylist.
  - `gptme_cache_coldness.sh` — 4-step: build + cache-test 3×PASS + smoke + POSITIVE-EVIDENCE probe walking fresh-is-cold, recorded-is-hot, default-threshold-is-5min. `bash tests/e2e/challenges/run_all_challenges.sh` reports `Results: 11 passed, 0 failed` (up from 8/8 — the 3 new scripts).

Containerized build path (2026-05-13)

Per operator mandate "boot-up containers containing proper dependencies for EVERYTHING using our containers Submodule":

- `helix_code/docker/build/Dockerfile.builder` was previously **gitignored** by overly-broad `Dockerfile*` and `build/` rules. Fresh clones couldn't build the builder image. Tightened both rules (`/Dockerfile*`, `/build/` — anchored at module root) so subdirectory Dockerfiles are tracked. Also bumped `FROM golang:1.24-alpine` → `golang:1.26-alpine` to match `go.mod go 1.26`.
- Containerised build path now reachable from a clone: `podman compose -f helix_code/docker-compose.builder.yml build builder` (uses Containers-submodule-aware compose semantics; substitute `docker compose` if available). The builder image carries Go 1.26 + golangci-lint + alpine apk deps + postgres-client; mounts the workspace and reuses Go module/build caches between runs.
- Five more Dockerfiles also became trackable and were checked in: `tests/e2e/mocks/Dockerfile.{llm,slack}-mock`, `tests/infrastructure/Dockerfile.ssh-{server,worker}`, `docker/security/snyk/Dockerfile`.

Dual API-key loader (2026-05-13)

Per operator mandate "We shall have API keys in .env file in the root of the project or all of them as the part of api_keys.sh in our host's home dir! Both of it MUST BE fully supported!" — VERIFIED end-to-end:

- `scripts/load_api_keys.sh` prefers `$HOME/api_keys.sh` when present, falls back to `.env` at meta-repo root.
- Test 1 — `$HOME/api_keys.sh` path: loader sourced from project root, HOME pointing at real home dir → 42 secret-like vars in env.
- Test 2 — `.env` fallback path: loader sourced with HOME pointed at empty fakehome → 46 secret-like vars in env from project `.env`.
- Both paths individually load full secret set; neither silently drops. The loader's "prefer $HOME/api_keys.sh, fallback to .env" precedence matches the operator's stated semantics.
- **HelixLLM/.gitmodules has stale `submodules/HelixQA` declaration** (directory absent on disk; only declaration remains).

Constitutional debt (open since P0)

- **LLMsVerifier dual-pin divergence** (P0-04): RESOLVED in P1.5-WP2; doc had been stale. The transitive duplicate `helix_agent/LLMsVerifier` was eliminated in WP2 (only a backup remains at `helix_agent/.container-caps-backup-20260425-151947/LLMsVerifier`). `helix_agent/go.mod:232` now replaces `digital.vasic.llmsverifier => ../submodules/llms_verifier/llm-verifier`, so HelixAgent transitively consumes the canonical pin. The pin-parity gate `scripts/verify-llmsverifier-pin-parity.sh` is degenerate (regression-tripwire only, header confirms WP2 elimination). `make verify-foundation` evidence (this round): exits 0, "0 failures, all gates passed" — no `VERIFY_FOUNDATION_WARN_ONLY` waiver needed.
- **Historical SSH key** + **helix.security.json leaks** (P0-T08.5): material is immortal in git history. Mitigated; rotation required by operator.
- **SonarQube** + **Snyk live-scan deferral** (P0-T08.7): infrastructure wired, awaiting credential rotation by operator.

Phase 3 remaining items (carried forward, non-blocking for Phase 4)

- **HelixAgent build** — IMPROVED: submodules populated, `./cmd/helixagent/...` builds and tests pass. Wildcard `./...` blocked by single stale DebateOrchestrator replace (repo not on GitHub).
- **Containers build** — RESOLVED: `go build ./...` exits 0.
- **Historical credential leaks** — operator rotation required

- **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
- **23 snake_case renames** — build-path-breaking, deferred
- **Codex Multimodal, Cline Computer Use** — not yet ported

CONST-045 — No Hardcoded Distribution Hosts

ALL container distribution targets SHALL be configured exclusively through `CONTAINERS_REMOTE_HOST_N_*` env vars in `containers/.env` (N=1..100; iteration stops at first absent `_NAME`; the containers module `pkg/envconfig/parser.go` is the authoritative loader). The .env file is the sole source of truth for host enrolment — no host is hardcoded in HelixCode source, tests, challenges, or governance documents. Every non-unit test run and every production deployment MUST use whichever hosts are currently configured when `CONTAINERS_REMOTE_ENABLED=true`. Adding, removing, or modifying a host means editing `containers/.env`; no code change is required. The CURRENT configured set can be audited with `grep '^CONTAINERS_REMOTE_HOST_' containers/.env`; at the time of this rule's introduction (2026-05-07) the configured hosts were `thinker.local`, but the rule applies to whatever set is in `.env` at any future point (N>=1). Direct `docker`/`podman` commands, manual container start/stop, and ad-hoc remote hosts outside the `.env` mechanism are strictly prohibited.

How to resume

From a new CLI agent / LLM session

Type the **Resume Prompt** at the end of this file verbatim. It triggers continuation without further user context.

Programme conventions to apply (verbatim list)

1. **Subagent-driven-development always.** Never inline-implement multi-task features. Skip approval gates per the user's auto-approve memory (`memory/auto_approve_designs.md`).
2. **Commit on `main`.** All work flows through `main`. No feature branches.
3. **Push to 4 remotes (non-force only):** `origin`, `github`, `gitlab`, `upstream` for the meta-repo. Submodules push to their `origin` only (Challenges now has 3 remotes: origin + gitlab + upstream).
4. **Deepest-first push order.** Submodules → meta-repo. If meta-repo's gitlinks reference unpushed submodule SHAs, the meta-repo push will succeed but cloners will fail to resolve submodule pointers.
5. **Each feature has:** spec → plan → per-task TDD commits → Challenge harness commit → close-out commit. No exceptions.

6. **Anti-bluff smoke must always be `clean`.** Run before each commit: `grep -rn "simulated\|for now\|TODO implement\|placeholder" helix_code/internal helix_code/cmd && echo BLUFF || echo clean`.
7. **Runtime evidence required for every PASS** per CONST-035 / Article XI §11.9. No metadata-only / configuration-only / absence-of-error PASS.
8. **api_keys.sh > .env precedence.** Any tool that needs API keys sources them via `scripts/lib/api_keys.sh` first; falls back to `.env`.
9. **Non-FF push = STOP.** Never force, never `--force-with-lease`. If a push is rejected, investigate before retrying.
10. **No CI/CD pipelines.** All gates run via Makefile / scripts. Per CLAUDE.md Rule 1.
11. **No HTTPS for git.** SSH only.
12. **Every claim of "done" carries pasted terminal output** from a real run against real artefacts. Per CLAUDE.md Rule 8.

Picking up Phase 3 work

If Phase 3 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show the latest Phase 3 commits. 2. Read `docs/improvements/PROGRESS.md` §Phase 3 — Issue remediation section. 3. Continue with the next pending remediation item from the Phase 3 remaining list. 4. Run `make test` to verify current state before making changes.

Picking up Phase 4 work

If Phase 4 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show commits `4f5f8f0` (P0 fixes) and `7f4effd` (P1 fixes). 2. Read `docs/improvements/PROGRESS.md` §Phase 4 section. 3. Anti-bluff sweep is COMPLETE for critical packages (validators, fix, config, deployment). 4. Verify remaining test files for weak assertions (only nil/err checks without content verification). 5. Run `make test` to verify current state before making changes. 6. Once all test suites validated, advance to Phase 5 (end-user materials).

Picking up Phase 5 work

Maintenance mandate

This document MUST be updated when:

- Any task is completed (update T-status table + add commit SHA).
- Any feature is closed out (update Phase status table + repository SHAs).
- Any known issue is discovered (add to "Known issues" section).

- Any phase boundary is crossed.
- Any deferred item is fixed or further deferred.
- Any new remote/submodule is added or removed.
- Any constitutional clause is added or amended.

If this document is out-of-sync with the actual state of the work, the inconsistency is a **CRITICAL DEFECT** — same severity as a false-success test result (CONST-035). See:

- `CONSTITUTION.md` Article XIII §13.1 (CONST-044) — Continuation Document Maintenance Mandate
- `CLAUDE.md` §12 — Continuation Maintenance
- `AGENTS.md` — "Continuation Maintenance" anchor

Verification (TBD): `scripts/verify_continuation_sync.sh` will compare: - `Last updated: 2026-05-08T02:00:00Z (Phase 3 — remediation + test infra)` -Active phasehere `vs`Current focusindocs/improvements/ PROGRESS.md`. - Tasks-done count here vs ticked-tasks count in`PROGRESS.md`. - Known-issue list here covers all documented failures in evidence files.

Non-zero exit = sync violation → blocking pre-push.

Resume Prompt

> **SUPERSEDED (rev 19, 2026-07-28T18:20Z) — the block below cites a path that no longer exists on this host** (`/run/media/milosvasic/DATA4TB/Projects/helix_code/` — the current checkout is elsewhere; per CONST-064/§11.4.131 a resuming agent should never hardcode a host path it hasn't verified). It also predates `RESUME.md` (§11.4.131's standing session-resumption file), which is now the canonical entry point and should be read FIRST. Preserved below for provenance; use the corrected prompt instead.

Corrected, host-path-neutral prompt (paste into a new CLI-agent session):

```
Read RESUME.md (repo root) then docs/CONTINUATION.md, run `git fetch --all
and continue. Use subagent-driven-development (§11.4.70) by default. Every
independent review before commit/build (§11.4.142); force-push is absolute
no exception (§11.4.113) — integrate by merging onto latest main, push fas
Push each submodule + the meta-repo to all its configured remotes (verify
when a work package or feature closes out.
```

Original (historical, path since gone stale):

Read /run/media/milosvasic/DATA4TB/Projects/helix_code/docs/CONTINUATION.m

---

Document version log

Date	Updater	What changed
2026-05-06	Initial create	Captures state through P2-F21-T04 (`5ef13b8`); Phase 2 i
2026-05-06	T06 update	T06 (`/approval` slash command) closed; 6 of 9 F21 task
2026-05-06	T07 update	T07 (main.go wiring + registry hook + integration test, c
2026-05-06	T08 update	T08 (Challenge harness 5 phases, meta `2781c1a` + sub f out + push 4 remotes) is next.
2026-05-06	T09 close-out	F21 (Codex Approval Modes) CLOSED — first Phase 2 fe
2026-05-06	F22 docs	F22 (Aider Git Auto-Commit Per Change) spec + plan lar
2026-05-07	F23 docs	F23 (Cline Browser Tool) spec + plan landed.
2026-05-07	F24 docs	F24 (Codex Project Memory) spec + plan landed.
2026-05-07	F25 docs	F25 (Plandex Plan Trees + Context Compaction) spec + p
2026-05-07	F26 docs	F26 (Openhands Workspace + Task Planner) spec + plar
2026-05-07	F27 docs	F27 (Aider Voice Input + Repo-Map) spec + plan landed.
2026-05-07	F28 docs	F28 (Kilo-code AST-Aware Refactoring) spec + plan land
2026-05-07	F29 docs	F29 (Roo-code Full Port) spec + plan landed.
2026-05-07	F30 docs	F30 (Continue IDE Integration) spec + plan landed.
2026-05-07	Phase 3 entry	Phase 2 CLOSED (F21-F30 complete). Phase 3 started —
2026-05-08	Full sync	F26-F30 close-out sections added; Phase 3 active section with PROGRESS.md.
2026-05-08	Phase 3 sync	LLMsVerifier build, helix_qa build, 6 internal/server tes pruned. CONTINUATION.md synced with PROGRESS.m
2026-05-08	Phase 3 close	Phase 3 CLOSED. All code-actionable remediation resol build fixed. Phase 4 started (anti-bluff audit).
2026-05-08	ZB Phase 2-3	Zero-Bluff Phase 2 (Stub Elimination) and Phase 3 (Feat CONST-046 added. Anti-bluff clean. Pushed to github+gi
2026-05-08	ZB Phase 4	Zero-Bluff Phase 4 (Test/Challenge Hardening) IN PROG config, deployment). Article XI §11.9 compliance enforc
2026-05-12	ZB Phase 4 close	Zero-Bluff Phase 4 CLOSED. Weak-assertion sweep acro internal/deployment with mutation-test confirmation (6 `02b7306` + `a3f8871`. Cross-compile linux/amd64 PASS. I

Date	Updater	What changed
2026-05-12	Hygiene	Untracked 3 runtime test artefacts + tightened `helix_c` binary. Commit `d7a7956`; all 4 remotes synced. CONTIN
2026-05-12	Anti-bluff sweep	Re-running `go test -short ./internal/...` surfac `internal/discovery/client.go` `discoverByDNS` ha `net.LookupHost` with `net.DefaultResolver.Looku` `internal/helixqa/wrapper.go` `Engine.StartSess` added `sync.WaitGroup` + `Shutdown()` method + `t.Cl` `internal/tools/browser/browser_test.go` 5 chro parallelism — gated behind `testing.Short()` with SK PASS clean. `-race` short-suite PASS clean. `go vet` clean
2026-05-12	Challenge sweep	`bash tests/e2e/challenges/run_all_challenges` `governance-cascade.sh` exit 1 with 61 third-party su Marker was unreachable in practice (can't commit into submodule paths). Relaxed verify script to accept prese `submodule_third_party.txt` as the deliberate ACK (i ACK). 61 FAILs → 0; run_all_challenges 8/8 PASS. Also u `server`, `helix_code/terminal-ui`) + extended .gitign
2026-05-12	Challenges submodule sweep	`cd Challenges && make test-short` failed at `Test` subtests). Root cause: test hardcoded a consumer-Yole A tested in isolation, violating its CLAUDE.md "100% deco `YOLE_ANDROID_APK_PATH`) env var with SKIP-OK mark Challenges submodule commit `8024463` pushed all 4 re meta-repo pin bump `12904e4`. `make test-short` 17/1
2026-05-12	Zero-bluff sweep	Per operator mandate "do not skip any tests or challen `testing.Short()` skips in `internal/tools/browse` `serializeChromiumLaunches(t)` — exclusive flock o serialising chromium launches across the entire `go te` TestAndroidSave_AllApiLevels switched from runtime `android_save_challenge` (file architecturally absent rebased onto upstream governance commits, pushed a production-code root cause across `internal/agent/s` `internal/memory/providers`, `internal/cognee`, `in` `internal/tools/mapping`, `internal/tools/multie` `workflow`. Verifications: `go test -race ./internal` failures, zero race warnings, 3 consecutive runs; `go ve` `(simplified for now)` comment removed). Commit
2026-05-12	Skip-annotation sweep	Per the no-silent-skips governance gate: 101 SKIP-OK m (Containers 25, helix_qa 75, LLMsVerifier 1) — every an (no adb/scrcpy/GStreamer/Tesseract/PaddleOCR/Ollama testcontainers, missing posix utils, etc.). NO skip remov

Date	Updater	What changed
		code vet warning in `helix_code/internal/llm/lite` `Done: true` emit into the EOF-decode branch where th `silent-skips.sh` with four layered fixes: auto-exclud accept slug-form SKIP-OK markers (`#short-mode`, `P1-F` `test|context|xit|xdescribe)\.skip\(`), path-l (helix_agent/MCP/submodules/, helix_qa/tools/opensour Submodule pins: containers `af51968`, helix_qa `e16bfe` `36cfafb`.
2026-05-12	Submodule test-parity sweep	Per "fix and polish everything": ran `go test ./...` fo failures surfaced and resolved at root cause: (1) **LLMPr** `digital.vasic.models => ../Models` pointed at a p `submodules/models` as a sibling submodule (`vasic-di` not relying on HelixAgent. (2) **LLMOrchestrator** + **Secu** transitive testify/go-spew). Fix: `go mod tidy` in each. (3 hardcoded model IDs (`venice-uncensored`, `llama-3.` renamed its catalogue, so the brittle equality assertions (`strings.Contains(m, "llama-3")` / `"uncensored"` CONST-036 compliant. Submodule pins bumped: LLMPr Security `eae0cec`; new Models pin added. Every owned repo commit `7c81a40`.
2026-05-12	Submodule test-parity round 2	Continuing the per-submodule sweep: VisionEngine + D `failed]` due to stale go.sum drift (missing go-spew che additionally had `pkg/remote/deployer_test.go` refe `NewDeployer`/`.Endpoint()` — types orphaned by the C 93bc8d4); the file produced 18 "undefined" errors and w Deleted the file (the VisionPool path tests in `remote_te` backed code). Submodule pins: VisionEngine `a092195`, `go test ./...` exits 0.
2026-05-14	Mistborn restore + integration matrix probe (round 37)	Operator paused autonomous loop for several hours; m (a) Discovered mistborn's podman MACHINE (macOS A — `podman ps` reported "Cannot connect to Podman... tr `start`". Started VM via `podman machine start` over S host power state is untouched). Re-ran `scripts/mistb` HelixCode reconnected → `database.connected:true` major surface: **local integration matrix**. Started local p at them, sourced `.env.full-test`, ran `go test -tag` Most packages PASS. Surfaced ONE legitimate anti-bluff `TestCogneeRealLLMTestSuite/MemoryStorage_*` fa test calls `cogneeIntegration.StoreMemory(...)` the `> 0`, and gets 0 results. **This is the test doing its job co** provider is a stub; no real backend wired). The proper f

Date	Updater	What changed
		major future work, out of scope this round. The test is a passing. (c) Local containers cleaned up; HC switched b `run_all_challenges`: 12/12 PASS with live-server prol PASS). All 4 remotes still synced at `508a25f`. No new co finding documentation only).
2026-05-13	Race-clean verification round 33	Ran go `test -short -race -count=1` across all 6 se internal/task (1.0s), internal/worker (31.6s), internal/pr (9.3s). **All packages: race-clean. Zero race conditions Manager fix shipped across rounds 1–32.** The 33 bug host went idle mid-round (SCP closed during tunnel re- `SKIP-OK: #env-mistborn-offline` rather than fake- environmental gates. run_all_challenges: 12/12 PASS (o SKIP-OK'd this round). No code changes this round — v
2026-05-13	Production-bug sweep round 32 (operator mandate re-asserted 4th time — broader surfaces)	Operator re-asserted the anti-bluff mandate verbatim a beyond the saturated API matrix. Re-verified governanc Broader-surface probes: (i) Challenges submodule shor userflow 182s). (ii) helix_qa submodule short tests — al containers — 8 packages PASS. (v) HelixCode terminal-u fails on missing X11/Xcursor (system-lib dep). Found 1 r `build ./...` which greedy-built the Fyne-GUI deskto in any headless environment WITH `X11/Xcursor.h:` codebase has a `main_nogui.go` build-tagged variant A CONST-035: the canonical compile-gate was lying about successfully" only in environments with optional syste the tree. Fixed by adding `-tags=nogui` to the `verify-` `compile` now passes cleanly in this environment witho files, no probe changes this round — build-system fix).
2026-05-13	Positive-coverage round 31 (final-corner + anti-bluff source scan)	Round 31 closed the final-corner gaps: (a) `GET /ws` (We reject-without-Upgrade — route IS wired). (b) `GET /sc` 503 with "QA engine is disabled" — honest config-gated saying "disabled" is preferable to a 200 returning fabric in helix_code/internal + helix_code/cmd, all in (i) test-fi variable names (`placeholder` as the concept for {{}}-su model_download_manager.go), (iii) low-priority code-q refactor opportunities, no runtime bluff). Clean: NO pr NO fake responses. helix-qa expanded 135 → 138 evide screenshot/engines 503 with anti-leak assertion of "disa helix-qa: 107/107 PASS (138 evidence files). run_all_cha **Session summary: 32 real production bugs fixed at r → 138 evidence files (23× expansion).**
2026-05-13		

Date	Updater	What changed
	Positive-coverage round 30 (helix-bridge + account-deactivation)	Round 30: (a) helix-bridge live LLM smoke — 5 provide deepseek/openrouter); 3/5 respond OK (groq/mistral/de (env key revoked or restricted); openrouter returns 429 codebase bugs. (b) DELETE /users/me lifecycle — registe (200), then attempted to log in as that user → 401 with " VerifyJWTWithDB fix correctly enforces the active-acco "marked deleted" stub. (c) /system/status returns api/da ("QA engine is disabled" — intentional config, not a bug HCQA-098-PRE-REGISTER + HCQA-098-PRE-LOGIN (setu HCQA-099 (deleted-user login → 401 with deactivated/ir after delete would be a critical CONST-035 violation), H healthy). helix-qa: 104/104 PASS (135 evidence files). ru secrets.sh: clean.
2026-05-13	Positive-coverage round 29 (schema fidelity, 100-counted milestone)	Round 29: schema-fidelity static probes for the read-onl — every entry must have id/name/status/type populatec `llm/models` — every entry must have id/provider AND context length, which would mislead callers computing count must be 0 (matches /memory/stats.systems_conne cross-endpoint guard). All clean. Probed `/llm/provide` registered; the single `/providers/:id` endpoint alread expanded 127 → 130 evidence files: HCQA-095/HCQA-09 checks block (run early, in deterministic order). helix-q 100-counted-checks milestone). run_all_challenges: 12/1
2026-05-13	Positive-coverage round 28 (robustness)	Round 28: probed concurrent requests (20 parallel POS id collisions), large payloads (1 MiB description body → (empty {} for required-field endpoint → 400 binding err **a feature gap, not a bluff**: `GET /tasks?limit=5` and ((return ALL 315 tasks / 300 workers — 150 KB response pagination support; the `limit` query string is just drop work. helix-qa expanded 124 → 127 evidence files: HCQ HCQA-093 (1MB-payload-201), HCQA-094 (unauthorized evidence files). run_all_challenges: 12/12 PASS. scripts/s
2026-05-13	Production-bug sweep round 27	Round 27 probed auth/register input validation (overlo correctly 400 — auth path is clean), then swept malform more bug: (32) `GET /workers/<malformed>/metrics` `respondInvalidID` branch. Fixed. Also: SQL-injection `TABLE workers;--"`) — pgx prepared statements para workers table survived intact. Positive-coverage probe string-concatenated SQL. helix-qa expanded 120 → 124 malformed-400), HCQA-090 (register-overlong-400 + an HCQA-091 (SQL-injection defense — workers list still re

Date	Updater	What changed
		per-run unique hostname to avoid duplicate-key collisi evidence files). run_all_challenges: 12/12 PASS. go test -s clean.
2026-05-13	Production-bug sweep round 26	Round 26 found 3 more bugs in sessions + workers inpu `project_id` (well-formed UUID but no such project) re existent projects. Root cause: in-memory session manag declares. Fix: createSession handler now validates the p `projectManager.GetProject(req.ProjectID)` BEF `ErrProjectNotFound` for missing project. (30) Same ha a-uuid") — session created with non-UUID project_id. S `respondInvalidID` for malformed UUIDs → 400). (31) returned 500 leaking pg `SQLSTATE 22001` "value too lo schema leakage (column type visible) + CONST-035 wro `worker.ErrWorkerHostnameTooLong` sentinel + `host` RegisterWorker pre-validates BEFORE the INSERT; han expanded 117 → 120 evidence files: HCQA-086 (sessions malformed-project_id→400), HCQA-088 (worker-overlo "SQLSTATE"/"22001"/"character varying"). helix-qa: 92/ 12/12 PASS. go test -short server+worker+session: ok. sc
2026-05-13	Production-bug sweep round 25 (mandate re-asserted 3rd time)	Operator re-asserted the anti-bluff mandate verbatim a cascade (0 failures) + anchor in all 6 root docs. Round 2 real bug — (28) `PUT /projects/<malformed-uuid>` a returned HTTP 500 leaking raw postgres "`invalid in` `(SQLSTATE 22P02)`". Same CONST-042 + CONST-035 d UpdateProject and DeleteProject in `manager_db.go` pa without UUID validation; pg rejected with the SQLSTAT `uuid.Parse(projectID); err != nil { return .` the top of both functions. Also wired 4 additional handl branches: updateProject, deleteProject, createTaskChec not-found 404 path. Existing unit test `TestDatabaseMa` assert via `assert.ErrorIs(err, ErrInvalidProje` error and expected the generic "failed to delete project' asserts the sentinel and confirms no DB call is made fo evidence files: HCQA-081 + HCQA-082 (project PUT/DEL "SQLSTATE"/"22P02"/"fkey"), HCQA-083 + HCQA-084 (ch on-bogus 404). helix-qa: 89/89 PASS (117 evidence files). server+project+session: ok. scripts/scan-secrets.sh: clea
2026-05-13	BUG #27 fix generalized — round 24	Round 24 extended the round-23 fix (`task.ErrInvali` deleteTask wired) across the full handler matrix. Adde `project.ErrInvalidProjectID`, `project.ErrInval` project unstructured invalid-id returns); fixed the cros

Date	Updater	What changed
		parameter inside task's manager — wrapped with the t handlers). Extended `respondInvalidID` to a switch ov (failTask, startTask, completeTask, assignTask, retryTas to call `respondInvalidID` before falling through. Verif 500'd on malformed UUID now all return 400. helix-qa (matrix coverage across task-start/complete/retry/fail + UUID-400). helix-qa: 84/84 PASS (112 evidence files). run server+task+worker+project: ok. scripts/scan-secrets.sh
2026-05-13	Production-bug sweep round 23 (operator re-asserted mandate)	Operator re-asserted the anti-bluff mandate verbatim a cascade requirement. Re-confirmed: `verify-governa` owned + listed third-party submodule; CONST-035 / §11 governance docs (CONSTITUTION.md / CLAUDE.md / AG wrong-password login (correctly 401), duplicate-email r cleanly), GET /tasks/ (correctly 404 via getTask's generic returned HTTP 500 with "invalid task ID: invalid UUID l malformed UUID is CLIENT input error, should be 400 F manager call sites that do `uuid.Parse(id)`. Introduced unstructured returns replaced via sed across manager_ `respondInvalidID(c, err, what)` writes 400 when (canonical pattern in place for the remaining 7 task-id h expanded 104 → 105 evidence files: HCQA-073 (DELETE evidence files). run_all_challenges: 12/12 PASS. go test -s
2026-05-13	Production-bug sweep round 22	Round 22 probed duplicate-create + workflow endpoint Found 2 more bugs: (25) duplicate-hostname worker cr `"workers_hostname_unique" (SQLSTATE 23505).` F + `isUniqueViolation(err)` helper that parses pgcon translates pg unique-violations BEFORE leaking; handle `projects/<bogus>/workflows/*` returned 500 with " introduced `project.ErrProjectNotFound` sentinel (7 manager.go + manager_db.go); 4 workflow handlers co `action)` helper. Bonus fix: the 3 non-planning workflo (fresh empty in-memory manager) instead of `s.proje` project. Now all 4 use `s.projectManager`. helix-qa exp + HCQA-071 (dup-hostname-409-no-leak with explicit a "workers_hostname_unique"/"SQLSTATE"/"duplicate ke qa: 76/76 PASS (104 evidence files). run_all_challenges: tunnels that dropped mid-session — single-shot tunnels `mistborn-up.sh` re-run brings them back). scripts/sca
2026-05-13	Production-bug sweep round 21	Round 21 probed remaining endpoints for similar patte `<bogus>/heartbeat` returned HTTP 500 with raw pos `table \"worker_metrics\" violates foreign ke`

Date	Updater	What changed
		`\"worker_metrics_worker_id_fkey\" (SQLSTATE` CONST-042 schema leakage (postgres FK constraint nan CONST-035 wrong-HTTP-code (5xx for what is plainly a `UpdateWorkerHeartbeat` proceeded directly to the wo UPDATE matched 0 rows. Fixed by checking `RowsAffec` `ErrWorkerNotFound` BEFORE the FK-violating INSERT "Worker not found" message. Also probed: start/comple correctly return 422 via the round-18/19 sentinel (no ex status filter can't distinguish "missing" from "wrong sta message is acceptable). helix-qa expanded 100 → 101: H leak check (response must NOT contain "fkey" or "SQLS helix-qa: 74/74 PASS (101 evidence files). run_all_challe scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 20 (100-evidence milestone)	Round 20 source-scanned for the broader "RowsAffecte antipattern. Found 5 occurrences across task/worker/se resource id caused HTTP 500 instead of 404. CONST-035 server error when the cause is "client asked for a missin sentinels `task.ErrTaskNotFound` (covers DeleteTask + (covers all 6 session manager.go not-found returns via s `worker.ErrWorkerNotFound` already in memory_repo UpdateWorker. The 5 handlers (deleteTask, failTask, del `errors.Is`-check and return 404. helix-qa expanded 95 DELETE-task-404, HCQA-066 fail-task-404, HCQA-067 DE HCQA-069 DELETE-session-404. helix-qa: 73/73 PASS (10 go test -short server+task+worker+session: ok. scripts/sc
2026-05-13	Production-bug sweep round 19	Round 19: probed task dependencies (forward-referenc reassign behavior, project-delete cascade. Found 1 more already-assigned task returned HTTP 500 (same family AssignTask wrap `task.ErrTaskInvalidStateTransi` RowsAffected==0; assignTask handler now `errors.Is-` message. Decided NOT to fix: dependency forward-refe validates deps lazily) and soft-delete with orphaned ses status='deleted', sessions can legitimately reference hist HCQA-064-PRE-TASK + HCQA-064-PRE-W1 + HCQA-064-I HCQA-064 (reassign-already-assigned → 422). helix-qa: 12/12 PASS. go test -short server+task: ok. scripts/scan-s
2026-05-13	Production-bug sweep round 18	Round 18 found 1 more bug while probing task state-m `complete` on a non-running task, `POST /tasks/:id/s` `tasks/:id/complete` on an already-completed task al error pattern as BUG #13's retry-on-non-failed. Each wa that says "the server is broken" when really the request

Date	Updater	What changed
		Generalized fix: introduced exported sentinel `task.Er` round-8's `ErrTaskNotRetryable`). StartTask and Comp `RowsAffected==0`; handlers `errors.Is`-check and ret must be {pending
2026-05-13	Positive-coverage round 17	Round 17: scanned for more silently-discarded-Exec pa Probed three deeper state-accuracy questions: (a) does complete with `{result: ...}` body — YES (a minor re exists: request key `result`, response key `result_data` ORDER BY created_at DESC — YES, first entry is the just by exactly 1 when a new task+worker is created — YES, hardcoded-zero or stale-cache bluff). No new bug surfa mechanically lock these invariants in place. helix-qa ex CREATE + HCQA-058-PRE-START (setup), HCQA-058 (con CREATE + HCQA-059 (list DESC ordering), HCQA-060-PR PRE-CREATE-W + HCQA-060 (stats counters rise by +1). run_all_challenges: 12/12 PASS. scripts/scan-secrets.sh:
2026-05-13	Production-bug sweep round 16	Round 16: scanned for more silently-discarded-Exec pa beyond the comment documenting the fix. Then probed #20: (20) `POST /api/v1/workers/:id/heartbeat` ret `workers.cpu_usage_percent` / `memory_usage_perce` stayed at 0 forever. The handler updated only `workers` `worker_metrics` time-series table; the worker's snaps advertised as current-state fields) were never touched. via /workers/:id/metrics), but the resource's own snaps `worker.DatabaseManager.UpdateWorkerHeartbeat` workers row (CASE WHEN > 0 preserves existing on om 72 → 75 evidence files: HCQA-HB-PRE-W + HCQA-HB-PR workers/:id snapshot reflects heartbeat — catches BUG run_all_challenges: 12/12 PASS. go test -short worker+se
2026-05-13	Production-bug sweep round 15 (governance reinforced)	Operator re-asserted the anti-bluff mandate verbatim: " *bluff manner — they MUST confirm that all tested codeb* Article XI §11.9 anchor IS present in all 6 root governan AGENTS.md + helix_code/ trio); `scripts/verify-gove` every owned + listed third-party submodule. Round 15 TRIPLE bluff: (19) `POST /api/v1/tasks/:id/checkp` successfully" while `task_checkpoints` history table st `_, _ = m.db.Exec(...)` for the history INSERT — sil INSERT always failed because the schema requires `wor` but the INSERT omitted that column. Triple bluff: (a) di history table never populated. Subsequent `GET /tasks` round 8 — that fix made the empty result HONEST but

Date	Updater	What changed
		remained. Fixed: introduced exported sentinel `task.E` task's assigned_worker_id + 422 if unassigned + supply helix-qa expanded 66 → 72 evidence files: HCQA-CKPT- (unassigned task → 422, NOT fake 201), HCQA-CKPT-PR HCQA-056 (GET checkpoints contains the just-created c triple-bluff regression). helix-qa: 60/60 PASS (72 eviden -short server+task: ok. scripts/scan-secrets.sh: clean.
2026-05-13	Positive-coverage round 14	Round 14 added positive-coverage probes for two critic new bug surfaced — but locking the behavior down). P llamacpp + /llm/providers/openrouter: all 404. That is H constitutional cap at 7 entries (FallbackModels rule com provider coverage is achieved via the live verifier whe full round-trip and /projects/:projectId/sessions linkage evidence files: HCQA-051-CREATE + HCQA-051-START (s error_message), HCQA-052 (retry transitions failed→pe round-8 sentinel distinguishes legitimate retries from s round-trips). helix-qa: 57/57 PASS (66 evidence files). ru secrets.sh: clean.
2026-05-13	Production-bug sweep round 13	Round 13 uncovered 1 more real bug while probing tas `assign`, the response showed `"assigned_worker": n` `assigned_worker_id` column was correctly populate `ListTasks`: they scanned the column into local pointe 123-124 / 228-229) but NEVER assigned them to the retu `OriginalWorker` fields. Every task response silently li deployment — including responses to /assign itself (wh functions. helix-qa expanded 56 → 61 evidence files: H HCQA-048 (assign returns task with assigned_worker m HCQA-049 (GET /tasks/:id round-trips assigned_worker inline post-assign re-fetch), HCQA-050 (heartbeat endp run_all_challenges: 12/12 PASS. go test -short server+ta
2026-05-13	Production-bug sweep round 12	Round 12 uncovered 1 more real bug (5th instance of t action lifecycle probes: (17) `GET /api/v1/workers/:i` empty-history case — same nil-slice→null bluff as listT metrics. Fixed in `getWorkerMetrics` with `[]*worker.` expanded 49 → 56 evidence files: HCQA-044 (worker-m (inline worker for the probe — decouples HCQA-044 fr PROJ + HCQA-044-PRE-SESS (project+session for action HCQA-046 (action=complete → status:completed), HCQA error). helix-qa: 51/51 PASS (56 evidence files). run_all_ scripts/scan-secrets.sh: clean.
2026-05-13		

Date	Updater	What changed
	Production-bug sweep round 11	Round 11 uncovered 1 more real bug plus a related par `PUT /api/v1/workers/:id` returned HTTP 500 "null v constraint" — same TEXT[] NOT NULL pattern as BUG # defect: the UPDATE unconditionally replaced every colu (a partial update) would overwrite the existing hostnan `worker.DatabaseManager.UpdateWorker`: (a) default bare `SET col = $1` with `COALESCE(NULLIF($1, '')` `WHEN $4 > 0 THEN $4 ELSE max_concurrent_task` the existing column when the input is zero-value. helix- CREATE (create worker), HCQA-042 (PUT renames displ catches both bugs), HCQA-043-DEL (DELETE 200), HCQA evidence files). run_all_challenges: 12/12 PASS. go test -s clean.
2026-05-13	Production-bug sweep round 10	Round 10 uncovered 1 more real bug while probing pro returned HTTP 500 "ERROR: column path does not exist update attempt. Root cause: `internal/project/manag` `RETURNING` clause referenced `path` and `type` columns t (`internal/database/database.go:333`). Real schem `Project.Path` in Go) and `config` JSONB (which holds extracted in GetProject at line 114). The canonical "rena unreachable for the entire deployment. Fixed by mirror RETURNING `workspace_path` + `config`, then extractin from `config["metadata"]` in Go. helix-qa expanded 4 lifecycle), HCQA-040 (PUT renames + asserts new name, (DELETE 200), HCQA-041 (GET-deleted 404). helix-qa: 43 12/12 PASS. go test -short server+project+task: ok. script
2026-05-13	Production-bug sweep round 9	Round 9 uncovered 1 deep bluff in the memory subsyst returned 6 systems all hardcoded as `status: "availa` reported `systems_connected: 0` — direct contradicti `listMemorySystems` is pure hardcoded English/metad Server struct has no memory manager bound for any o ChromaDB/Qdrant/Mem0/Zep). CONST-035 violation: ha same backend's stats endpoint correctly reports zero co helper that returns "not_configured" until a real manag state correctly returns "not_configured" for all 6 (match description/features) remains as the documented set of runtime claim. helix-qa expanded 40 → 41: new "HCQA `status="available"` until real backing exists (any fut manager-wiring will fail this check). helix-qa: 39/39 PAS PASS. go test -short server+task: ok. scripts/scan-secrets
2026-05-13		

Date	Updater	What changed
	Production-bug sweep round 8	Round 8 uncovered 2 more real bugs by extending helix `tasks/:id/checkpoints` returned `"checkpoints":` slice→null JSON contract bluff (tasks, projects, workers, `[]map[string]interface{}{}` in `getTaskCheckpoi` returned HTTP 500 “task not found, not in failed state, c 500 is a server-error code that lies about the nature of t is a client-side state mismatch, not an internal bug). Int `task.ErrTaskNotRetryable` in `internal/task/mana` returns 422 Unprocessable Entity for the state error, 50 → 40 evidence files (HCQA-036 task-create-for-lifecycle, retry-422, plus 2 internal-step probes HCQA-038-PRE-ST PASS (40 evidence files; the 2 internal-step probes write run_all_challenges: 12/12 PASS. go test -short server+tas
2026-05-13	Sessions-create probe addition (round 7)	Coverage expansion: probed `POST /api/v1/sessions` (from the round-4 HCQA-031 create-project probe) + val (no bug surfaced — Mode validation correctly rejects “i planning/building/testing/refactoring/debugging/deploy `session.go`). New HCQA-035 probe stitches the just-cr sessions POST, asserts 201 + `session.project_id` rou 35/35 PASS. run_all_challenges: 12/12 PASS.
2026-05-13	Production-bug sweep round 6	Round 6 uncovered 1 more real bug, same pattern as ro `refresh` returned HTTP 401 “User not authenticated” f token. Root cause: `refreshToken` handler called `c.Get` `authMiddleware` (must stay public for register/login). Authorization header via `VerifyJWTWithDB`, mirroring path returns a freshly-issued JWT (200) for valid Beare missing header, 401 with “Invalid or expired token” for `TestRefreshToken_WithUserContext` unit test updat c.Get(“user”) path no longer exists; mockDB-unverifiabl for the unit test). helix-qa expanded 32 → 34: HCQA-033 JWT” rather than “different JWT” because iat is per-sec identical tokens), HCQA-034 (garbage Bearer returns 40 PASS. go test -short server+auth: ok. scripts/scan-secrets
2026-05-13	Production-bug sweep round 5	Round 5 uncovered 1 more real bug, same pattern as ro returned HTTP 500 with `ssh_config` NOT NULL constr optional `ssh_config` map. Same fix as `task_data`: `wo` defaults nil `sshConfig` to `map[string]interface{}{` (capabilities is TEXT[] NOT NULL — pgx serializes nil sl applies because the INSERT explicitly provides the valu `POST /workers` with empty body and asserts 201 + UU run_all_challenges: 12/12 PASS. go test -short server+wo

Date	Updater	What changed
2026-05-13	Production-bug sweep round 4	Round 4 uncovered 2 more real bugs: (8) `POST /api/v` ID: invalid UUID length: 12" — `internal/project/mar` `m.CreateProjectWithUser(..., "default-user")` "default-user" is exactly 12 chars, matching the error. H `*auth.User` from context and calls `CreateProjectWi` same POST route was registered under a `publicProje` `testing` — any unauthenticated caller could create pro building/testing/refactoring) against any projectId. This the createProject handler unreachable (it now requires authMiddleware). Consolidated under the authenticate entirely. helix-qa expanded 29 → 31: HCQA-030 (unauth (authenticated POST /projects creates project with UUID `TestCreateProject_ValidRequest` updated to accep `TestListProjects` pattern). helix-qa: 31/31 PASS. run_ server/...: ok. scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 3	Continuing the anti-bluff push uncovered 2 more real b `api/v1/users/me` returned a User stub with `is_activ` `created_at:"0001-01-01T00:00:00Z"` — `authMidd` that returns ONLY {ID, Username, Email} from JWT clai `VerifyJWTWithDB` exists for exactly this case (fetches t switched to `VerifyJWTWithDB`; defense-in-depth bonus authenticating with pre-deactivation JWTs. (8) `GET /ap` third instance of the nil-slice→null JSON contract bluff (`[]*worker.Worker{}` when nil. New helix-qa probes H stats sub-objects). HCQA-017 tightened to assert `is_act` have caught BUG #6 if it had existed before the fix. heli go test -short server+auth: ok. scripts/scan-secrets.sh: cl
2026-05-13	Production-bug sweep round 2	Continuing anti-bluff push uncovered 3 more real prod checks with tasks-CRUD lifecycle and projects/sessions l HTTP 500 "null value in column task_data violates not-n `internal/task/manager_db.go:CreateTask` forwar NULL; fixed by defaulting `parameters` to empty map a `task_data JSONB NOT NULL` now upheld). (4) `GET /a` `[]` for empty list — JSON contract bluff (Go's nil-slice→r `api/v1/projects` returned 401 "user_id not found in c called `c.GetString("user_id")` while `authMiddlewa` "user_id" key); the entire projects-list endpoint was unr switching to the established `c.Get("user")` pattern. (6 New helix-qa probes HCQA-019..027: tasks-CRUD full lif by-id → delete → get-deleted-404), plus list-projects-as-a

Date	Updater	What changed
		`c332f98`. helix-qa: 27/27 PASS. run_all_challenges: 12 p scripts/scan-secrets.sh: clean.
2026-05-13	Mistborn distribution + anti-bluff QA harness	Stood up the mistborn.local distribution stack and the h operator mandate "all tests and Challenges do work in a tested codebase really works as expected". Delivered: (1 (gitignored, key-based auth only — password never con `compose.mistborn.yml` 7-service podman stack (postg grafana); (3) `scripts/mistborn-{up,down}.sh` boots (4) `scripts/bridge/main.go` (helix-bridge) multi-pro `api_keys.sh` / `.env` loader contract — 3/5 providers re `qa/main.go` (helix-qa) anti-bluff QA harness with captu + body_head + duration_ms in per-check JSON); (6) `tes` `helix_qa_live_anti_bluff.sh` wired into `run_all_` `internal/server/handlers.go:getLLMProvider` re ANY id including `does-not-exist-xyz` — now correct from verifier + constitutional fallback set. New `TestGe` reproduces and guards. (8) helix-qa expanded 6 → 19 c database.connected==true, goroutines>0) + sequenced a garbage-bearer-401 → logout) with strict invariants (se `registration`, `user.username === registered us` existing self-bluffs in HCQA-004/005/006 (assertions of e corrected. Also-discovered + fixed: helix_qa orchestrato `orchestrator.go:186`; tests at `integration_test.g` `orchestrator_stress_test.go` migrated from `asse` `assert.False`. Browser test context-deadline bumped serves `database.connected:true` through mistborn- FAIL and no empty bodies; `run_all_challenges.sh`: packages 0 failures 0 races; HelixQA: 135 packages 0 fai benign matches in comments/template-substitution). M `5c44ba3`, all 4 remotes parity (github + gitlab + origin f
2026-05-15	Cascade Challenge runtime-validation widened + Task #274 isolation infeasibility documented (round 41 close-out[44])	(1) **Cascade Challenge runtime evidence widened fro** stress, chaos) × 5 submodules (HelixQA, Containers, Sec stub on port 8081 with captured wire evidence per Cha EventBus/HelixLLM DDoS). Total cascade Challenges wi only concern is now substantially mitigated — same 5-6 with consistent latency profiles. (2) **Task #274 build-tag** importing `digital.vasic.debate/*`; `internal/han` debate constants (`ModelHelixDebate`), structs (`debate` fields deeply interwoven with non-debate handlers. Bu into debate vs non-debate halves — multi-day refactor t Documented as truly infeasible in single-session scope. upstream repo OR refactor HelixAgent to remove debat

Date	Updater	What changed
2026-05-15	CONST-051(B) ↔ (C) tension documented + go.work attempt + stability re-evidence (round 41 close-out[43])	Operator-directed continuation. (1) **CONST-051(B) ↔ (C)** `architecture/CONST-051-tension.md`: documented HelixLLM post-Tasks #254/#273 (relative-path `replace` `build ./...`). Attempted `go.work` workspace fix at He `digital.vasic.docprocessor` module identifier colli `submodules/doc_processor` (same-named-different-c LLMProvider, VisionEngine. `go.work` removed to resto options for operator: (A) module-path renaming, (B) pro accept current state, (D) bootstrap script. Resolution req design. (2) **Stability re-evidence**: full Challenge runner 19/1 caught early — likely Groq rate-limit on conversati same transient pattern as close-out[24]). 6/6 verifier gates `make verify-compile` . (3) Task #274 forensics fron name URLs probed, none exist; operator action genuine
2026-05-15	Address remaining unfinished items — cascade runtime-validated + Task #274 forensics + Models pulled (round 41 close-out[42])	Operator-directed deep close. (1) **CONST-060 protocol v** constitution + 64/68 owned at parity; `submodules/mode` pulled via FF; 3 third-party drifts (LLama_CPP 188, Olla informational. (2) **Cascade Challenges runtime-valida** — proves structural cascade is NOT bluff-only: helix_qa containers DDoS **50/50 100% p50=2.2ms**, submodules/ev submodules/helix_llm DDoS **50/50 100% p50=1.6ms**. Ca "structural-only on 66 of 72" honest concern (4 of 66 cas runtime exercise; the remaining 62 share the same tem alternate org-name URLs (`HelixDevelopment/Debate` `HelixDevelopment/debate`, `vasic-digital/Debate` `vasic-digital/debate-orchestrator`, `vasic-digi` 12 sub-packages required (`agents`, `audit`, `comprehens` `reflexion`, `testing`, `tools`, `topology`, `validation`, r ("compiles but feature fake"). **Operator action genuine** remove the debate-dependent code (~6 files in `interna` the missing module source. Task #274 stays pending.
2026-05-15	Complete cascade: 54 submodules + regression gate + verify-compile + ledger refresh (round 41 close-out[41])	Operator-directed deep close of all actionable unfinishe **submodule capture**: ran `git submodule foreach` fet party (LLama_CPP 187 ahead, Ollama 33, HuggingFace_ `models` 1 ahead — pulled via FF, was the CONST-060 cas (2) **CONST-050(B) cascade gap audit** + **fix** via new `scr` gate — exposed that 44 owned submodules from Tasks NOT the 6-Challenge matrix. **54 submodules total now** from close-outs [30]/[33]/[36]/[40]), each commit-pushed to its or `make verify-compile` → `All packages compil` #273 refactor did not break HelixCode's compile). (4) **Co** features × **68 owned submodules audited** (was 12 pre-

Date	Updater	What changed
		verifier PASS. (5) **New regression gate**: `scripts/veri` submodule has 6 required Challenge files with bluff-sig `SKIP-OK` patterns); 2 exemptions documented (HelixAg **captured evidence**: verify-all-constitution-rules.sh **6/6** **covered** + run_all_challenges.sh **20/20 PASSED** + Helix(Task #252 (cross-org rename, operator), Task #274 (Deb Challenges-against-real-infra (needs services running), CONST-051(B)↔(C) tension (architectural design).
2026-05-15	**Task #273 COMPLETE** — HelixLLM 34 nested submodules migrated; full cascade closed (round 41 close-out[40])	**All 6 verifier gates GREEN against the full 68-strong** identical pattern from completed Task #254. HelixLLM submodules + updated go.mod replace directives in a si `../../vasic-digital/<Name>` for vasic-digital ones, `{Challenges,HelixQA,Security,Containers}` for p submodules added to HelixCode root (Config, Lazy, Wat Document, Filesystem, TOON), each cascaded with full missing, each pushed to its origin (Config `772f332`, Laz `e206a1b`, RateLimiter `5e3a0ea`, I18n `2bb4c74`, Recover `d13a5f2`, TOON `64176e3`). `submodule_owned.txt` fina `constitution` removed per CONST-059 since it's the ca **bluff captured evidence**: `verify-all-constitution` `implementable gate green — anti-bluff covena` owned submodules), G2 anti-bluff smoke, G3 mock-fron G5 .gitignore audit, G6 case-conformance all GREEN. Tw (operator-coordinated cross-org rename), Task #274 (De gone from upstream).
2026-05-15	Address unfinished features + issues — uncover real cascade gaps, fix 45+1 ungoverned submodules (round 41 close-out[39])	Operator-directed comprehensive audit of unfinished w `submodule_owned.txt` had only 12 entries while `.git` 5/6+0/6+20/20 sweeps in close-outs[35]-[38] were green ONL **CONST-035 false-success** in the verifier's input data. (2 submodules (3 HelixDevelopment + 42 vasic-digital) all missing same anchors. (3) 4 submodules (BackgroundTa missing `.gitignore` per CONST-053. (4) HelixLLM has style problem (tracked as Task #273, multi-session). (5) on `digital.vasic.debate → ./DebateOrchestrato` ran; the repo doesn't exist on GitHub; production code + 3 services actively imports it). **Fixes shipped:** 45+1 su block (~720 insertions per file × 3 files × 46 submodules 45 commits pushed to their origin remotes (panoptic `f5` `3f943c6`, HelixSpecifier `830aecc`, and 41 vasic-digital s HelixCode pointer-bumps for all 45 included in this con remaining FAIL is HelixLLM(34) tracked as Task #273. **F**

Date	Updater	What changed
		DebateOrchestrator repo gone (Task #274 candidate — remove the debate-dependent code).
2026-05-15	Anti-bluff re-invocation — CONST-060 protocol self-exercised + 6/6+20/20 re-evidence (round 41 close-out[38])	Operator re-invocation of canonical anti-bluff covenant CONST-060 (Fetch-before-edit Mandate), the FIRST actio `all --prune` (all 4 HelixCode remotes fetched), `git l` `e071e80` == upstream — no drift since close-out[37]), `git` `--oneline HEAD..@{u}` → **empty** (constitution still at exercised cleanly — captured evidence per §11.4.2 of its MUST produce captured evidence — the actual `HEAD..` **per §11.4.2 (captured-evidence requirement)**: re-ran runtime evidence: (1) `verify-all-constitution-rul` implementable gate green — anti-bluff covenant honou **20/20 PASSED, 0 failed** — ALL CHALLENGES PASSE itself remains in place: 14 anchors (CONST-035 / §11.9 + governance files, 0 verifier failures. The covenant is hor with positive runtime evidence captured during execut round 41 began.
2026-05-15	CONST-060 cascade — Fetch-before-edit Mandate codified + anti-bluff re-evidence capture (round 41 close-out[37])	Operator re-invocation of canonical anti-bluff covenant **fetch**+**pull** per CONST-049/060 (which is the new rule's upstream added §11.4.37 (Fetch-before-edit mandate, o session redundant-work incident on Project-Toolkit). **Ca** CONSTITUTION.md + CLAUDE.md + AGENTS.md AND to governance files updated this round). Each owned subr Challenges (`af4fba1`, 4 remotes), containers (`752d685`) github_pages_website (`b0ccb17`), DocProcessor (`aceb6` (`5af6b6b`), VisionEngine (`f229b64`), LLMsVerifier (`1aa` `.gitmodules` constitution pointer bumped to `1b0b03a` **evidence per §11.4.2**: (1) `verify-all-constitution-` after Task #254 close); (2) `run_all_challenges.sh` en PASSED!, Anti-Bluff Verification: COMPLETE; (3) `verify` CONST-060 anchor present across every owned submod PASS in this codebase carries positive runtime evidence
2026-05-15	**Task #254 COMPLETE** — all 46 HelixAgent nested own-org submodules migrated to parent root (round 41 close-out[36])	**G4 verifier-gate FAIL eliminated; verifier sweep is n** batches (close-out[35] + this commit's 8 follow-on batches HelixAgent to canonical parent-root locations: **41 vasic-** `vasic-digital/<name>`, **3 HelixDevelopment** at depe **Challenges** resolves to `../Challenges` (already at Hel (`3b97d1b1` → `55c9a9d1`) on its `origin` (HelixDevelopm submodules + updated go.mod replace directives. Repla `dependencies/vasic-digital/<Name>` (or HelixDeve the "consuming submodule reaches it via documented i

Date	Updater	What changed
		CONST-051(C) explicitly names. **One repo naming surp** submodule resolves to upstream git@github.com:vas different name) — preserved upstream name at parent blocked task: #252 (CONST-052 phased rename across u
2026-05-15	Task #254 pilot: CONST-051(C) HelixAgent migration — 5 of 46 nested submodules (round 41 close-out[35])	First measurable progress on Task #254 (the known G4 submodules migrated from helix_agent/<name> (nes canonical parent root at dependencies/vasic-digit Observability, Auth, Storage. **Workflow validated**: (1) g digital/<name>.git dependencies/vasic-digita deinit -f <name> + git rm <name> in HelixAgent; (from ./<Name> → ../dependencies/vasic-digital resolver" pathway CONST-051(C) explicitly names. Heli (HelixDevelopment/HelixAgent). HelixCode pointer-bur committed here. **G4 verifier count**: 46 → 41 nested ow across future close-outs following this proven pattern.
2026-05-15	Round-41 cascade capstone — post-cascade verifier sweep + ledger refresh (round 41 close-out[34])	Final capture after the 12-submodule CONST-051(A) cas gates GREEN, 1 FAIL = G4 known Task #254 (HelixAgent governance cascade PASS (14 anchors × 36 owned-subm (zero production bluff markers), G3 mock-from-produc G6 case conformance PASS. **(2) run_all_challenges.sh** evidence still holds. **(3) regenerate-coverage-ledger.s** owned submodules; 46 nested chains flagged; 30 VERIF outs ([22] through [34]); **72 new Challenge files** delivered a **owned submodules now have full CONST-050(B) mat** DocProcessor, LLMOrchestrator, LLMProvider, VisionE github_pages_website); remaining 2 excluded for valid constitution governance-only). All 4 HelixCode meta-re upstream) at parity for each pointer-bump close-out. **Th** **Challenge PASS + 5/6 verifier-gates GREEN + 12 owne** **positive runtime evidence per CONST-035 / Article X**
2026-05-15	CONST-051(A) panoptic + github_pages_website cascade — 12 of 14 owned submodules now covered (round 41 close-out[33])	Final two non-blocked owned submodules cascaded. **pa** panoptic) adds the 6 CONST-050(B) Challenges to its exi **github_pages_website** (commit pushed to HelixDevelo challenges/scripts/ directory with the 6 Challenge gracefully — structural contract still captured). HelixCo bumped per CONST-049 step 7. **12 of 14 owned submod** (HelixQA + Challenges + containers + Security + DocPro VisionEngine + LLMsVerifier + Models + panoptic + gith session: **72** (6 HelixCode + 6×11 submodules = 6 + 66). R reasons: **HelixAgent** (multi-session per Task #254, 46 n CONST-051(C)), **constitution** (governance-only submod

Date	Updater	What changed
		Cumulative round-41 close-out count: **[22]-[29] HelixCode Pano+GhPw = 33-12=12 cascade close-outs delivered.
2026-05-15	CONST-051(A) dependencies/ HelixDevelopment cascade — 6 submodules × 6 test types (round 41 close-out[32])	Mass cascade continuing operator's "do not stop" mand submodules cascaded with the 6 missing CONST-050(B) `cascade_const050b.sh`) — same anatomy, per-submo (`DOCPROCESSOR_*`, commit pushed), **LLMOrchestrator** (`LLMPROVIDER_*`), **VisionEngine** (`VISIONENGINE_*`), **L** (`MODELS_*`, pushed to `master` not `main`). Total: **36 new** single batch. All 6 submodules pushed to their respectiv 6 bumped in this commit per CONST-049 step 7. Cumula **containers** + **Security** + **6 dependencies/HelixDevelo CONST-050(B) matrix** (60 new Challenge files this sessi github_pages_website, panoptic, HelixAgent (multi-sess test cascade needed).
2026-05-15	CONST-051(A) Challenges + containers + Security cascade — 6 new test types each (round 41 close-out[31])	Continuing the operator's "do not stop until ABSOLUTE Three more owned submodules cascaded with HelixCo (commit `873c0b1`, pushed to all 4 remotes: github vasic upstream); **Containers** (commit `7f3c58a`, pushed to or (commit `56fdc16`, pushed to origin = github vasic-digita Challenges (ddos / stress / chaos / scaling / ui / ux) with s (`CHALLENGES_*`, `CONTAINERS_*`, `SECURITY_*`). All 6 are — they SKIP-OK cleanly when their target env vars are evidence when wired. Challenge counts per submodule 5→11. HelixCode meta-repo `.gitmodules` pointers for Cumulative round-41 submodule cascade: **HelixQA** + **C owned submodules now have full CONST-050(B) mat** Remaining 8 owned submodules: Assets, Dependencies panoptic, etc. — to follow in subsequent close-outs.
2026-05-15	CONST-051(A) helix_qa submodule cascade — 6 new test types in helix_qa (round 41 close-out[30])	Per operator's "do not stop until ABSOLUTELY EVERYT submodule-side CONST-050(B) cascade. helix_qa submo `challenges/scripts/` matching the round-41 HelixCo + `stress_sustained_load_challenge.sh` + `chaos_f` `scaling_horizontal_challenge.sh` + `ui_terminal` `ux_end_to_end_flow_challenge.sh`. Each ports the configurable knobs / captured wire evidence or honest specific defaults: `HELIXQA_HEALTH_URL=http://loca` discovers `helix_qa/bin/helixqa`. DDoS validated end **rate, p50=1.2ms, p95=3.8ms, p99=5.8ms, wall=1.19s**. O running (the common dev-box case). helix_qa pushed t (github.com:HelixDevelopment/HelixQA, `951ed50..1f`

Date	Updater	What changed
		pointer bumped to `1f03882` in this commit. HelixQA's C submodules to cascade: Challenges, Containers, Securit
2026-05-15	CONST-050(B) matrix-completion sweep + ledger regeneration (round 41 close-out[29])	Post-#266-closure verification capture. **(1) `scripts/ve`** G1 governance cascade PASS (14 anchors × 36 files), G2 markers in helix_code/{internal,cmd}), G3 mock-from-p mocks), G4 nested own-org submodule chains FAIL (He G5 .gitignore + sensitive-file PASS, G6 case-conformance protected). **5/6 green; 1 FAIL** = **known Task #254. (2) ba** **`run_all_challenges.sh`** end-to-end: **20/20 PASSED, 0** Bluff Verification: COMPLETE. This is the released evide completion. **(3) `scripts/regenerate-coverage-led`** (9 platforms × 30 features × 12 owned submodules; 46 r VERIFIED cells preserved). The Anti-Bluff Verification m chaos ✓ scaling ✓ UI ✓ UX ✓ (all 6 of Task #266's missing runner).
2026-05-15	UX End-to-End Flow Challenge — FINAL 6th of CONST-050(B) missing test types — closes Tasks #266 + #256 (round 41 close-out[28])	Task #270 closed; **Tasks #266 + #256 (6 of 6 missing tes** `challenges/ux_end_to_end_flow.sh` — drives the c **coherence** across the chain: discover → health → list-m friendly-recover → post-liveness. **6-step structure**: (1) k of 5 recovery hints (wizard/provider/key/list-models/he models exposes ≥1 selectable model (pick first); (5) delik `does-not-exist-xyz -prompt ping` → assert error i actionable token — bogus name / "model" / "provider" / models") AND **NOT user-hostile** (panic / goroutine / run 5-pattern blacklist on EVERY step's output); (6) post-erro survived the bogus journey, no leaked state). **Anti-bluff** schema assertions from journey coherence — what UI p The 5-pattern user-hostile blacklist (`panic:`, `goroutine` `segmentation fault`, `fatal error:`) catches the "sta invisible to schema-only assertions. **Configurable knob** `UX_BOGUS_MODEL`. Validated 6/6 with full captured evid verdict, **7 models listed** (first picked: `llama-3.2-3b`), b post-error operational. **Full runner now 20/20 PASS** (w (grand-parent) both closed in this commit. Anti-bluff ma populated: DDoS ✓ stress ✓ chaos ✓ scaling ✓ UI ✓ UX ✓.
2026-05-15	UI Terminal-Interaction Challenge — 5th of CONST-050(B) missing test types (round 41 close-out[27])	Task #269 closed (Task #266 5-of-6 installment). New te `ui_terminal_interaction.sh` — drives the built hel asserts on actual stdout. **6-step structure**: (1) binary-pr missing); (2) **-help flag-label assertions** (all 8 documen `-continue -health -list-models -model -non-i` **assertions** (`=== System Health Check ===` banner +

Date	Updater	What changed
		(4) **-list-models structural assertions** (≥1 entry with m `Status:` lines — catches schema-bluff where one field **runnability** (second `-help` still works — catches "first **invalid-flag graceful-exit** (bogus flag → nonzero exit C **Configurable knobs**: `HELIX_CLI_BIN` (default `helix_` `UI_MIN_MODELS` (1). **Anti-bluff value**: catches "CLI exit return-code gating misses. Validated end-to-end: 8 flag **model entries** with all 3 labels matched, re-runnable, **19/19 PASS** (was 18/18 after scaling). Only **UX Challeng**
2026-05-15	Horizontal-Scaling Challenge — 4th of CONST-050(B) missing test types (round 41 close-out[26])	Task #268 closed (Task #266 4-of-6 installment). New `te` — operator-safe horizontal-scaling Challenge with **topo** `SCALING_REPLICA_URLS` env var resolving to ≥2 reacha dispatch (reachable-replica detection + SKIP-OK if <2); (test (50 reqs × N replicas at concurrency 10, ≥95% pass tolerance, informational since direct-replica calls bypas (modulo uptime/timestamp via sed-strip — catches the cache" misconfiguration class which is a top-3 producti **Configurable knobs**: `SCALING_REPLICA_URLS` (comma `SCALING_CONCURRENCY` (10), `SCALING_MIN_PASS_PCT` Validated SKIP-OK path on bare run (default empty REP `replica` marker). Validated happy path against 3-replic replicas reachable, schema valid each, **150/150 (100%)** **body sha256 matched across all 3 replicas** (ebc01776 **18/18 PASS** (was 17/17 after chaos). Remaining 2 test ty
2026-05-15	Chaos Failure-Injection Challenge — 3rd of CONST-050(B) missing test types (round 41 close-out[25])	Task #267 closed (Task #266 3-of-6 installment). New `te` `chaos_failure_injection.sh` — operator-safe client host mutation per CONST-033 host-power-management (sustained-throughput): **shape of input is the chaos**. **5** shapes via raw /dev/tcp (bad verb, bad HTTP version, d line); (2) oversized header (`X-Chaos-Huge: <8 KiB>`); GET line, 4s idle, late completion); (4) abrupt connectio **legit-traffic control group** (100 reqs at concurrency 20 **Configurable knobs**: `HELIX_CHAOS_HOST/PORT`, `CHAOS` (50), `CHAOS_LEGIT_MIN_PASS_PCT` (95). **6-step structu** post-malformed liveness → oversized-header + post-hu control + threshold gate → post-chaos liveness + schema malformed input (server crash vs clean 4xx reject); 000 chaos starving legit users (pass-rate < threshold); zombi degraded). **Honest SKIP-OK** when server unreachable. server: pre-chaos 200, 5 malformed shapes survived, ov ignores X-Chaos-Huge per design), 5 slow-loris spawne post-chaos 200+valid status. curl-code formatting bug c

Date	Updater	What changed
		"000000" when curl's `-w %{http_code}` already wrote **Full runner now 17/17 PASS** (was 16/16). Remaining 3
2026-05-15	CONST-052 rename-safety guardrails + G6 strengthening (round 41 close-out[24])	Operator follow-up safety mandate 2026-05-15: "double codebase which is by convention non-snake-case — dir System and working building process!" Strengthened th `constitution-rules.sh` with explicit `NEVER_RENAME` system/package-manager files (Makefile, Cargo.toml, go for Java/Kotlin/Android/Apple/C#/Swift; **Android AOSP** vendor/, kernel-5.10/, packages/, prebuilts/, build/, devic breaks the AOSP build); **AOSP-mandated files** (Androic bootstrap.bash, BUILD, OWNERS); build/cache artefacts (.git/, .github/); coordinated owned-project names (helix atomic rename across GitHub+GitLab+all consumers). N `rename_safety_guardrails.md` documents the hard (full rebuild + all tests + run_all_challenges.sh + verify-a cascade.sh + reference-resolution regression test). Task correctly reports **12 protected** + **6 unprotected** renam Challenge runner stability check: **3 sequential runs** × **1** Groq rate-limit on live-LLM Challenge). Anti-bluff sweep gates green, 1 FAIL = known Task #254 (HelixAgent 46 r packages all green (`internal/llm`, `cmd/cli`).
2026-05-15	Stress Sustained-Load Challenge — 2nd of CONST-050(B) missing test types (round 41 close-out[23])	Task #266 2-of-6 installment delivered. New `tests/e2e` exercises the server's `/api/v1/health` endpoint with s seconds at target rps, per-second pass-rate snapshot, lat **Configurable knobs**: `STRESS_DURATION_SEC` (default 3 `STRESS_CONCURRENCY` (40), `STRESS_TIMEOUT_SEC` (5), `STRESS_MAX_LATENCY_DEGRADATION_PCT` (300). **6-step** baseline median latency; (2) schema sanity (catches 200 per-second wall-clock + pass-rate captured; (4) aggregat degradation mid-stress); (5) **post-stress latency vs base** zombie-slow" class that pass-rate gate alone would miss **evidence**: per-second curve, overall + worst-second pas latency degradation %. Honest SKIP-OK pattern. Validat 250 reqs, 100% overall, 100% worst-second, p50=0.5ms, cache-warm-up). Wall-clock fractional-seconds bug cau Full runner now **16/16 PASS** (was 15). Remaining 4 test
2026-05-15	DDoS Health-Flood Challenge — first installment of CONST-050(B)	Task #256 first installment delivered (1 of 6 absent test t `tests/e2e/challenges/ddos_health_flood.sh` — a with concurrent requests + asserts pass-rate threshold + vars: `HELIX_HEALTH_URL`, `DDOS_REQUESTS` (default 500 (5), `DDOS_MIN_PASS_PCT` (95). **Captured wire evidence**

Date	Updater	What changed
	missing test types (round 41 close-out[22])	clock seconds, latency p50/p95/p99. **5-step structure** (a schema sanity (real JSON with status field, not empty 20 flood liveness (catches "fell-over-during-flood" class). **H** §11.4.3 topology-dispatch. Validated end-to-end against concurrency → 100% pass rate, p50=1.0ms p95=3.9ms p **runner now 15/15 PASS** (was 14/15). Remaining 5 test t #266 — each follows the same skeletal pattern.
2026-05-15	scripts/verify-all-constitution-rules.sh — CONST-055 enforcement engine (round 41 close-out[21])	Task #265 closed. Implemented the canonical post-pull- as the **enforcement engine for every other rule** (with `scripts/verify-all-constitution-rules.sh` runs tree: G1 governance-cascade (§11.9 + CONST-047..059, 1 smoke (production code grep for "simulated/TODO impl G3 CONST-050(A) mock-from-production (forbids intern CONST-051(C) nested-own-org submodule chains; G5 CO tracked-sensitive-file scan; G6 CONST-052 case-conform without failing since rename is phased per Task #252). on violation. Paired meta-test `scripts/test-verify-` bluff requirement, §1.1 mutation pair): plants known vi reverts cleanly. **9 sub-tests PASS** (G2/G3/G5 baseline + r fixed two real findings: (a) `submodules/models` was m via Models commit `9ef33e4` pushed to its origin; (b) he a tracked sensitive file but is a legitimate test fixture wi `test` / `.env.test` / `.env.ci` patterns now allowlisted e GREEN, 1 FAIL = G4 HelixAgent's 46 nested own-org sub Meta-test caught a real bluff in my initial G5 plant-patte `.gitignore` correctly rejected — defense in depth; fixe past .gitignore, which is the actual failure mode G5 nee
2026-05-15	Anti-bluff verification sweep — CONST-035 captured runtime evidence (round 41 close-out[20])	Operator re-invoked the canonical CONST-035 / §11.4 / A already in place (14 anchors × 36 owned-submodule go answer per §11.4.2 (captured-evidence requirement): a captured runtime evidence. **Constitution submodule f** (upstream merge "drop obsolete §11.4.25-27 numbering **CONST-035 anti-bluff smoke**: `grep -rn "simulated\` `production this would" helix_code/internal h` **Conversational REPL Challenge** (6/6 PASS): live LLM r clear context-reset probe + token stats `tokens: in` `finish=stop`. **@-mention Challenge** (5/5 PASS): live se time-stamped string planted in a temp file — proves the **Full Challenge runner** (`tests/e2e/challenges/run_` ALL CHALLENGES PASSED! Anti-Bluff Verification: C packages (`internal/llm/...` + `cmd/cli/...`): 7 pack `7.081s`, `cmd/cli 0.055s`, etc.). Constitution pointer bu

Date	Updater	What changed
		step 7. Governance cascade verifier: 0 failures. The anti executed test + Challenge produced positive captured e
2026-05-15	containers/ apply_caps.py CONST-051(B) decoupling refactor (round 41 close-out[19])	Task #263 closed — last genuine-violation in the Task #2 `scripts/resource-policy/apply_caps.py` no longe `cli_agents/helix_code/helix_code/`, `/helix_cod` `helix_qa/tools/opensource/`, plus 12 others). Refac universal-only fragments (.git/, node_modules/, vendor/ research/, docs/specs/, docs/examples/). New `--skip-pa` `RESOURCE_POLICY_SKIP_PATHS_FILE` env var) loads p file at runtime via new `load_skip_paths_file()` hel exclude. Example file `containers/scripts/resource` `paths.example.list` shipped alongside the script. He `containers/resource-policy-skip-paths.list` (1 `python3 -m py_compile` clean; `--help` shows the ne cli_agents in apply_caps.py source returns 0 matches. c origin.
2026-05-15	scripts/regenerate-coverage-ledger.sh + first mechanical regeneration (round 41 close-out[18])	Task #261 closed. New generator script `scripts/reger` mechanises the coverage ledger per CONST-048 + §11.4. inventories supported platforms from `helix_code/Ma` android/aurora-os/harmony-os/containers/headless); (2 manual (detected 30); (3) inventories test-type presence walks each owned submodule's `.gitmodules` for neste runs `scripts/verify-governance-cascade.sh` and ledger preserving every prior `VERIFIED` cell (anti-bluff UNCONFIRMED:→VERIFIED; promotions remain opera maintains audit-trail table. First-run results: 30 feature preserved, **HelixAgent flagged with 46 nested own-or** Task #254). Generator caught two subtle bash bugs in ea `echo 0` doubled the count, and `||echo` semantics on m `'%s' | tr -d ' \n\r'` sanitisation. Smoke-tested wit script is idempotent; subsequent runs preserve VERIFIE Composes with §11.4.18 (doc-block at top of script) + §1 this in future).
2026-05-15	load_api_keys.sh genericisation + github_pages_website sub-audit (round 41 close-out[17])	Two closures bundled. **Task #262 (cosmetic genericisa** `load_api_keys.sh` header + function-docstring with g (project-agnostic per CONST-051(B))" + "downstream LI 4 submodule copies (HelixQA + Security + containers + to its origin). Function names `helixcode_*` kept as na unchanged; smoke-tested. **Task #264 (github_pages_we** was the cosmetic-only load_api_keys.sh (now fixed), the tooling — they describe HelixCode because the site IS H

Date	Updater	What changed
		`decoupling_review.md`. **No genuine-violation refact load_api_keys.sh fix.** HelixCode meta-repo commits a2 and the follow-up bumping github_pages_website's c54 fully closed across all 5 copies of the script.
2026-05-15	CONST-051(B) decoupling review first pass (round 41 close-out[16])	Task #255 first pass delivered. Published `docs/coverag` remediation decision for every owned-submodule file f audit. Decision taxonomy (3 classes): legitimate-cross-pr Challenge-bank entries / website content) → no action; Task #262 to genericise; genuine-violation (1 × containe paths) → Task #263 medium-scope refactor to config-inj Task #254 remediation. github_pages_website's build sc files legitimate-cross-project (no action), 4 files cosmetic follow-up task), 12 files pending sub-audit (1 follow-up honest decision document is the deliverable; inline edit operator's "per-file review" mandate.
2026-05-15	@-file mentions in REPL (round 41 close-out[15])	Task #244 closed. Modern-CLI-agent parity feature (Clau in the REPL and the CLI auto-attaches that file's content `helix_code/cmd/cli/main.go`: new helpers `atMenti` at start-of-text or after non-identifier char so `user@hos` `path.to.symbol` don't match) and `expandAtMentions` `path="..."> ... </file></attached_files>` bloc file cap (oversized files listed but not embedded — anti- Deduplicates repeated paths. Directories silently skippe sees the `@token` as-is and no scary error. CLI surfaces a unit-test functions in `cmd/cli/at_mentions_test.go` `challenges/at_mention.sh` (PASS) with 5 paired stru round-trip that plants a unique sentinel string in a temp @-mention content actually reaches the model (not a w (`llama-3.3-70b-versatile`): `attached: /tmp/.` in 1 sentence with token stats `in=93 out=33 time=35` `run_all_challenges.sh` next to `conversational_re`
2026-05-15	CONST-048 coverage ledger first publication (round 41 close-out[14])	Task #257 surface delivered. Published `docs/coverage` ledger for HelixCode. Lists every F01-F30 feature × 6 inv documented promise, no open bugs, docs in sync, four- `UNCONFIRMED:` markers per §11.4.6 no-guessing rule (n Includes test-type matrix (CONST-050(B)) showing 8 pre challenges/helixqa + partial full-automation/performan partial ui/performance). Includes CONST-051 audit sum cascade snapshot (all 14 anchors VERIFIED across Helix Generator script `scripts/regenerate-coverage-led` authored manually) — tracked as follow-up. The ledger

Date	Updater	What changed
		UNCONFIRMED: anti-bluff posture; 6 test types entirely review; HelixAgent's 46 nested own-org submodule cha trail row added inside the ledger itself per §11.4.12 doc-
2026-05-15	First CONST-051(C) remediation: panoptic moved out of Challenges to HelixCode root (round 41 close-out[13])	First concrete remediation of a CONST-051(C) (nested o Task #250's audit. `challenges/.gitmodules` was decl own-org submodule — forbidden by CONST-051(C). Ren `panoptic` + `git rm panoptic` inside Challenges (comm Challenges remotes — github + gitlab + origin (fans to g add in HelixCode meta-repo to add panoptic at `<root>` pinned panoptic at `73b13b0` (the same SHA Challenges source-code refactor needed in Challenges: its `pkg/pan` takes the panoptic binary path as a constructor parame CONST-051(B) decoupling-via-config-injection was alrea `vet ./...` in Challenges PASS post-removal. Governan across 36 governance files. Task #253 closes the first CO nested own-org submodules) is the much larger remain
2026-05-15	CONST-057/058/059 HelixCode-side cascade (round 41 close-out[12])	Closes the cascade-side gap from close-out[11] for the thre (§11.4.33/34/35 added by another agent during the sessi aware Closure-Status Vocabulary (Bug→Fixed / Feature `Fixed.md`) suffix; same migration-discipline tooling), C (every Reopened item carries `**Reopened-Details:*` vocabulary Reason values; reopens without evidence a Canonical-Root Inheritance Clarity (constitution-submo ARE extensions opening with inheritance pointer; no si Cascaded through HelixCode root governance (CONSTIT QWEN.md), inner helix_code/ module (helix_code/CONS 12 owned submodules per CONST-047 (Challenges 5a01 DocProcessor 364fc9c0; LLMOrchestrator c2f46a88; LL Models d8793574; VisionEngine e1268ea4; github_page aca15473; helix_qa 2a850ec0; Security 72f1c3bb). Verifi CONST-047..059) — 0 failures across 36 governance file
2026-05-15	CONST-054/055/056 codified + cascaded (round 41 close-out[11])	Three new universal anchors codified per operator ma Constitution submodule HEAD advanced: `6f30ce7` → 5 (upstream-added §11.4.33 type-aware closure status + § canonical-root inheritance clarity by another agent) → **Submodule-Dependency-Manifest**: every owned-by-u declaring its own-org dependencies; tooling `incorpora` CONST-051(C) path, reads manifest, recurses for each d `<root>/.helix-manifest.yaml` audit. **§11.4.32 / CON** whenever constitution submodule is fetched + pulled, t `constitution-rules.sh` BEFORE the new HEAD is ca

Date	Updater	What changed
		rule. **§11.4.36 / CONST-056 Mandatory install_upstrea** MUST be followed by `install_upstreams` invocation f `upstreams/`) is present with `*.sh` recipes; skipping sile three cascaded into HelixCode root governance (CONST QWEN.md), inner helix_code/ module (helix_code/CONS 12 owned submodules per CONST-047 in two passes: CO CONST-056 (Challenges dd12c5db — pushed to 4 remote LLMOrchestrator 200d1d73; LLMProvider 5b6cfb5c; LL VisionEngine e1bfb696; github_pages_website 71d3dab helix_qa da388ba8; Security fbda6e2c). Verifier extende failures across 36 governance files. Constitution submo **follow-up tasks** (multi-session): #252 phased renames p Challenges; #254 HelixAgent's 46 nested own-org submo #256 add 6 missing test types; #257 coverage ledger; CO + incorporate-submodule tool; CONST-055 verify-all-con (§11.4.33/34/35) still need HelixCode-side cascade as CO
2026-05-15	CONST-053 codified + cascaded into 12 owned submodules + CONST-051 audit complete (round 41 close-out[10])	**§11.4.30 / CONST-053 .gitignore + No-Versioned-Build** submodule (commit `6f30ce7` pushed to origin) per ope session). Six forbidden-from-version-control classes: bu generator products), cache files (`__pycache__`, `node_m` `*.pem`, `*.key`, `id_rsa*`, `secrets/`, `api_keys.sh`), gen bluff invariant: an ignore-line alone is not sufficient — Pre-commit attention required: inspect `git diff --st` violations abort the commit. CONST-053 cascaded throu owned submodules (Challenges 24e69a15 to 4 remotes; LLMOrchestrator 5531dc8a; LLMProvider 82bf94f4; LL VisionEngine 11ff7071; github_pages_website 546d44ac 53a534f3; Security a5125e2f). Verifier extended to 8-and CONST-047/048/049/050/051/052/053) — 0 failures acros **audit (Task #250) completed**, gap findings: (1) CONST- Challenges has 1 (panoptic, Task #253); HelixAgent has HelixCode refs — 7 submodules with refs, per-file revie test types — DDoS, scaling, chaos, stress, UI, UX absent (ABSENT (Task #257). All remediation work tracked as fi directive.
2026-05-15	CONST-052 codified + cascaded + install_upstreams BOTH-dir support (round 41 close-out[9])	Per operator mandate 2026-05-15 (5th major mandate t **Snake_Case-Naming Mandate** codified in constitution 45d3678 for `install_upstreams.sh` BOTH-dir suppor directory/submodule/file MUST use lowercase snake_ca (helix_code/, challenges/, containers/, helix_agent/, helix dependencies/) MUST be renamed in phased migration sense exceptions: language-mandated case (Java/Kotlin/

Date	Updater	What changed
		vendor third-party submodules, build artefacts. Test cov test + full test-type matrix + anti-bluff wire-evidence). u via `install_upstreams.sh` that resolves to `<dir>/up` (smoke test passed: syntax OK; resolution against existi with WARN). Cascaded CONST-052 anchor through Heli CLAUDE.md/AGENTS.md/CRUSH.md/QWEN.md), inner CONSTITUTION.md/CLAUDE.md/AGENTS.md), and all 1 5eab467d (pushed to 4 remotes); containers d35ef96d; I baa35eb7; LLMProvider 3525d142; LLMsVerifier 909a9a github_pages_website 293aece6 (pushed to 3 remotes); 32368c80 — all pushed to respective origins. Verifier ex CONST-047/048/049/050/051/052) — 0 failures across 36 **deferred** to a separate phased program (multi-session) brainstorming → phase-divided plan → fine-grained tas Spurious additions inside the constitution submodule it reverted before push — only HelixCode-side files receiv constitution submodule retains only §11.4.29 (the autho
2026-05-15	CONST-051 codified + cascaded + 2 production bluffs fixed (round 41 close-out[8])	**Three workstreams.** (A) **Two CONST-050(A) producti** `internal/memory/providers/chromadb_provider.` collection="default" (was a bluff "for now … we'd need now walks every existing collection via new `listColle` `fetchFromCollectionLocked` helpers and unions hits only accepted flat-string arrays, but real ChromaDB v1 helper is tolerant of both shapes and gracefully returns `helix_code/internal/memory/providers/anima_p` "Simple equality check for now / In a real implementati now supports documented operator vocabulary `$eq/$n` via filter-value-as-operator-map syntax; backwards-con fail closed. Both unit tests PASS. Committed as `a3fb18b` `manager.go` removed (commit `8df3376`). (B) **§11.4.28** / 2026-05-15: every owned-by-us submodule (orgs: vasic- ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix- part of the consuming project's codebase. Three invaria full decoupling/reusability — NEVER inject project-spec injection; (C) dependency-layout — all submodule deps `<root>/submodules/<name>/`; nested own-org submo exempt. Codified in constitution submodule commit 77 **cascade** into 12 owned submodules per CONST-047 + bu submodules at new heads (Challenges dded4f80 pushed pushed to 3 remotes; others to origin). Verifier `scripts` again to walk CONST-051 — now requires all 6 anchors CONST-050 + CONST-051) per file in every owned subm

Date	Updater	What changed
2026-05-15	CONST-048/049/050 cascaded into 12 owned submodules + verifier extended (round 41 close-out[7])	Per CONST-047 (Recursive Submodule Application Man universal anchors codified in round 41 close-out[6]. Wrot `cascade_const_048_049_050.sh` that for each of the (self-applies CONST-049 step 1); (2) appends CONST-048, CONSTITUTION.md, CLAUDE.md, AGENTS.md if missing (3) validates no merge-conflict markers introduced; (4) that submodule. **Result: 12 submodule commits lande** DocProcessor 3b46146; LLMOrchestrator 531d5924; LL Models fdca27e; VisionEngine 3e2ebaa; github_pages_w e56ea92; Security 951180e), all pushed to their respectiv github+gitlab+origin+upstream per its multi-remote co github+origin+upstream; others to origin). Surfaced on committed merge-conflict markers (lines 829, 1158, 121 `92dcec1ef9fa07529e955946c60c205efb2b90f2` patt challenges/CLAUDE.md). Resolved by removing the 3 m content (§11.4.26 step 5). Also extended `scripts/veri` CONST-048/049/050 in addition to existing §11.9 + CONS every owned submodule. Verifier result: 0 failures acro Meta-repo: 12 submodule pointers bumped + verifier ex as one atomic close-out.
2026-05-15	Groq stream double-emit fix + CONST-048/049/050 governance cascade (round 41 close-out[6])	Two distinct workstreams in one round. **(A) Groq strea** the UX layer): `internal/llm/groq_provider.go::Ge` chunks (Content=delta) AND a final-completion chunk (accumulated content). The renderers (streamToRender final full-content chunk was rendered on top of the alre a “1 2 3” model response. Fix: at FinishReason time, sen (metadata fields Usage/FinishReason/ProviderMetadata chunk; rendered content not duplicated). Live verified: `prompt "Reply with only: 1 2 3" --stream` → cl **governance anchors** per operator mandates 2026-05-1 **Coverage** (every feature/functionality/flow/use-case/ed platform MUST ship with full automation tests proving matching documented promise + no open bugs + docs i **CONST-049 Constitution-Submodule Update Workflo** §11.4.17 classification + verbatim mandate → validate resolution preserving union → cascade verification CO commit); **§11.4.27 / CONST-050 No-Fakes-Beyond-Unit-** placeholders/TODOs/FIXMEs permitted ONLY in unit te system; production code MUST NOT import mock paths full-automation + security + DDoS + scaling + chaos + st Challenges + helix_qa test types; Challenges + helix_qa s three codified in constitution submodule (Constitution.

Date	Updater	What changed
		for §11.4.25+§11.4.26 and 6fa762e for §11.4.27, both pu HelixCode root CONSTITUTION.md, CLAUDE.md, AGEN helix_code/CONSTITUTION.md, CLAUDE.md, AGENTS.m pointer bumped from 5cca25c → 6fa762e in this same compliant). The §11.4.26 pipeline was self-applied durir commits before the first edit landed). Governance casca submodules per CONST-047 will land in a follow-up rou
2026-05-15	OpenRouter live / models + Mistral/ DeepSeek F12 fast-path (round 41 close-out[5])	Continued the F12 readiness sweep from round-41 clos **default-model bluff closed (CONST-035 + CONST-036)** openrouter_provider.go led with deepseek-r1-fre valid model ID"). End-to-end probe confirmed only 1 of returned 404/429). Replaced with a runtime fetch of Op preferredOpenRouterDefaults list (round-41-probec models is unreachable, falls back to the verified seed lis HELIX_LLM_PROVIDER=openrouter ./bin/cli → 364 gpt-oss-20b:free auto-selected, REPL returned "four tps=11.9). (B) **F12 fast-path extended 13 → 15 provider** DeepSeekProvider (OpenAI-compatible thin wrappers existed in missing_types.go, added ProviderTypeDeepS parseCloudProviderType + wizard_cmd.go::build Path-B disallow list (Path B now narrowed to 3: vllm, lo HELIX_LLM_PROVIDER=mistral → mistral-small-la HELIX_LLM_PROVIDER=deepseek → deepseek-chat – TestSelector_RejectsNonCloudType pivoted from " conversational_repl Challenge: 6/6 PASS unchanged. Us providers explicitly. Gemini probe re-confirmed key inv Ollama/Llama.cpp not running locally so unprobed but
2026-05-14	Modern-CLI-agent UX complete: conversational REPL + canonical-loader normalisation (round 41 close-out[4])	Operator's "is HelixCode ready for practical use like mo to-end probe that surfaced and fixed FOUR more readin **env-var fallback** for OpenAI/OpenRouter/XAI (Anthrop all do); **(2) default-model auto-select** from provider ca exist on any cloud provider); **(3) conversational REPL** commands — does NOT accept LLM prompts; now bufi commands /help//exit//clear//reset + token-timin **providers** (was claiming only 4); **(5) canonical loader** <PROVIDER>_API_KEY for 28 names + survives caller's silently). New Challenge tests/e2e/challenges/conv and-runtime gates (catches both source-regression AND HELIX_LLM_PROVIDER=groq ./bin/cli → real conve response via Groq + token stats in=40 out=2 tota **HelixCode CLI now has full modern-CLI-agent UX pa**

Date	Updater	What changed
		superset on multi-provider + distributed-execution remote parity.
2026-05-14	F12 fast-path 4→13 providers + readiness UX cleanup (round 41 close-out[3] — modern CLI agent parity)	Per operator's "is HelixCode ready like modern CLI age recommendations" directive: extended the F12 direct-c Cloud (11): anthropic, bedrock, vertexai, azure, **groq, o** Local (2): **ollama, llamacpp** (via newOllamaFromEntry generic ProviderConfigEntry into typed OllamaConfig/L CLI: HELIX_LLM_PROVIDER=→clean F12 construction narrowed to 5 (deepseek, mistral, vllm, localai, lmstudi server). Pre-existing tests pivoted to match new contra TestNewCloudProvider_RejectsLocalProvider→TestNew User manual §2.4 updated. **TWO of my own bluffs sur** nonexistent docs/COMPLETE_HOWTO.md — corrected ZERO_BLUFF_USER_MANUAL.md §2.4; (2) pre-existing model llama3.2" — that UX now works after this extens **just-plug-in-API-key-and-go parity with modern sing for the 11 most popular cloud providers + the 2 mos any single competitor's provider support.** Meta-repo
2026-05-14	Constitution-inheritance cascade across 12 owned submodules (round 41 close-out[2])	Operator standing mandate (8th re-assertion this sessi recursive cascade. Surveyed all 12 owned-submodule g AGENTS.md = 36 files) and prepended the Helix-Consti 23 files received the prepend this round; 13 already ha Challenges, Containers, DocProcessor, LLMOrchestrato commit pushed to its own remote(s); meta-repo `8d0bd` pushed to all 4 remotes. **Pointer text is consumer-agn** HelixAgent, ATMOSphere)") so the same anchor works projects. **Project-specific tightenings SURVIVE** — e.g. preserved despite the universal constitution's broader inheritance precedence. CONST-047 recursive-cascade constitution-inheritance layer atop the existing §11.9 +
2026-05-14	HelixLLM/ submodules/HelixQA wire-up — round-39 deferred close-out (round 41)	Closed the last deferred item from the round-39 3-deep `submodules/HelixQA` was declared in `.gitmodules` b `pathspec 'submodules/HelixQA' did not match` `submodule add git@github.com:HelixDevelopme` helix_qa tip pinned at upstream HEAD. Cascade close-o meta-repo `9074a59` pushed to all 4 remotes. All gates G `no-silent-skips`: OK).
2026-05-14	challenges/ CLAUDE.md merge-conflict resolution +	While auditing for further bluffs, surfaced a **CRITICAL** merge-conflict markers (`<<<<<<< HEAD` at line 651, `===` `92dcec1ef9fa07529e955946c60c205efb2b90f2` at 1 governance manual. Resolved by removing only the ma

Date	Updater	What changed
	panoptic screenshot bluff (round 41)	side: §11.4.6–§11.4.15 forensic anchors; other side: §6.O preserved. Also tightened Testpanoptic_WebAppIntegra `hasScreenshots { t.Logf } else { t.Logf }` (pa `*.png` exist when panoptic exits cleanly. panoptic 73b1 all converged across 4 remotes.
2026-05-14	HelixConstitution submodule incorporated at constitution/ (round 41 continued)	Operator direct in-conversation mandate (2026-05-14): *fully the HelixConstitution (git@github.com:HelixDevelop for root definitions of the Constitution, CLAUDE.MD and* distinct from the earlier suspected prompt-injection in GitVerse upstreams contradicting §6.W. Added `HelixDe` Git submodule at `./constitution/`, pinned at HEAD 5 root governance trio (`CONSTITUTION.md`, `CLAUDE.md`, A apply unconditionally, (b) project rules EXTEND never wins for universal scope, project tightenings (e.g. §6.W only narrow. **Deliberately NOT done** for safety: did NO violate §6.W), did NOT add GitFlic/GitVerse remotes, did submodules, did NOT modify the constitution submodu repo remotes.
2026-05-14	Round-41 begin: 5 production-code ctx-honour fixes (CONST-035)	Continued anti-bluff sweep into the notification packag PagerDuty) all had `Send(ctx context.Context, ...` support but the implementations called `http.Post(..` `context.WithTimeout` or `context.WithCancel` had CONST-035 bluff at the API-contract level. Fixed all 5 wi `http.MethodPost, ...) + http.DefaultClient.D` engine.go reverted to http.Post, the tightened `TestDisc` (which now asserts `errors.Is(err, context.Cance` Commits `b741026` (Discord) + `e38278a` (Slack+Teams+T repo remotes. Cumulative round-40+41 ledger: **68+ anti fixes**; no-silent-skips gate GREEN; verify-governance-ca `test ./internal/notification/...` PASSes with al
2026-05-14	Anti-bluff round-40 final ledger: 23 fixes + 1 production-code defect surfaced & fixed	Final round-40 totals after the operator re-asserted CON **improvements** + **1 real production-code defect fix**. **P** `providers/character_ai_provider.go:702+733` — `config.Description` unconditionally and panicked o unknown duration by `TestCharacterAIProvider_Cr` the call in `defer func() { recover() }() + t.Lo` `error + comment "Either succeeds or fails g` (removed recover, removed err discard, asserted non-e stack trace, confirming the real defect. Fixed in product convention. Mutation-verified: test fails without prod fi TestMultiProvider_CloudConstructors (multi_provider

Date	Updater	What changed
		production_deployer (fail-loud on WriteFile error rathe **Cumulative round-40 ledger** (all commits pushed to al Step-3b SKIP-OK + 11× unit-test tightenings + 4× integra replacements + 1× TestMain hardening + 1× production challenges still PASS; helix-qa SKIP-OK'd on `#env-db-u` across 9 affected packages. Prompt-injection flag from e submodule added; §6.W respected).
2026-05-14	Anti-bluff phase-challenge gates: grep-based → exit-code-based (round 40 close)	Final round-40 batch surfacing the biggest bluff yet: **6 p** `go test ./... | grep -q "FAIL" && (echo "FA` 1) which silently PASS on "no test files" output, compile that doesn't print the expected grep token. Replaced wi `exit 1; }` which uses Go's actual exit code. **Files tight** (workflow+session), phase5 (mcp+memory+notificatio **TRIPLE BLUFF surfaced and fixed in phase6**: original `"FAIL" && exit 1` — script already cd'd to HelixCode FAILED ("No such file or directory"), the && shortcut sk downstream grep saw empty input and returned non-z phase6 reported PASS for years without actually runni replacement caused it to honestly FAIL once (proving th cd and the test actually runs. Commit `1aaa09d`. Cumula = 1 Step-3b SKIP-OK (`ee4ba5a`) + 9 unit test tightenings 2 integration test tightenings (`5c771fe`) + 6 phase challe PASS; mutation-verified on representative test (sed-repl
2026-05-14	Anti-bluff zero-assertion test sweep (round 40 continued)	Operator re-asserted CONST-035 verbatim 6th time and on our plate". Audited helix_code/internal/ for `func Te` asserted NOTHING (zero `assert.*` / `require.*` / `t.Fa` value via `_ =` …). Surfaced 9 zero-assertion-with-discar `08746b8` earlier in this round): TestAIIntegration_Get{C _PreInitReturnsNil + assert.Nil pinning the documented verified); TestNewClient_WithDatabase (replaced `_, _` failure-path asserts nil-client + "Redis" in error, success TestHealthMonitor_PerformHealthChecks (snapshot pr Metrics are NOT mutated — pins the `if provider.Sta` **more in this commit**: TestCharacterAIProvider_Timeou cancelled ctx — now pins in-memory Health no-error c TestChromaDBProvider_Store_DefaultCollection (mock- `usedDefaultCollection` — now require.NoError on S TestDiscordChannel_Send_ContextCancellation (was `_` http.Post ignores ctx; a future promotion to http.NewRe explicit contract update); TestAIIntegration_ListProvide asserts the comment's claim: empty without Initialize). remotes at `08746b8`. Prompt-injection flag: a `<system-`

Date	Updater	What changed
		`constitution/` submodule with 4 upstreams **includin** (“GitFlic, GitVerse, and all other providers are forbidde **added, no installer was executed**.
2026-05-14	Anti-bluff Step-3b: SKIP-OK when DB unreachable (round 40)	Operator re-asserted CONST-035 verbatim 5th time this anti-bluff manner. Verified the anchor cascade is alrea governance files PASS via verify-governance-cascade.s live-server Challenge to distinguish “feature broken” fr `e2e/challenges/helix_qa_live_anti_bluff.sh` q responds, inspects `database.connected`. If false (or th connection-failed signatures), emit `SKIP-OK: #env-db` `credentials/availability drift` and exit 0, mirro SKIP-OK pattern. This replaces the previous FAIL cascad this round when mistborn-postgres credential drift wa detection is deliberately narrow: only `database.conn` body signatures trigger the skip; validation errors, real FAIL. Verified against a degraded HC: challenge correct Commit `ee4ba5a`.
2026-05-14	Known-issue: mistborn-postgres SASL credential drift (round 39 follow-up)	After completing the round-39 CONST-047 3-deep casca expand `helix_qa_live_anti_bluff` evidence beyon **Surfaced a deployment-configuration drift, not a co** restarted via `scripts/mistborn-up.sh`, accepts user rejects password `helixpass` with `SASL auth FATAL:` `(SQLSTATE 28P01)`. The compose default `HELIX_POST` running HelixCode binary (PID 29282, `/tmp/helixcod` data volume — that binary was killed during this trou tracked file. `HELIX_DATABASE_*` env vars correctly ove attempt logs match the env vars I supplied), so the bug password. **Net effect on round 39**: the CONST-047 casc submodules + 25 idempotent + 1 deferred across 3 recu **meta-repo remotes** at `d8c1814`. The live-server probe `offline` next time mistborn-up isn’t running; helix-qa currently fails 3 (HCQA-015 / HCQA-016 / HCQA-017) wh environment-only and will resolve when the correct mi HC boot (no code change needed).
2026-05-14	CONST-047 recursive cascade — 3-deep close-out (round 39 continued[2])	Extended the recursive cascade one more layer: helix_a grandchildren) + helix_agent/challenges/panoptic (1 ow **cascaded, 23 idempotent skip (shared submodules a** **(HelixLLM/submodules/HelixQA not initialized on d** grandchild pointers + pushed to origin. helix_agent/Cha pushed. HelixAgent commit `c5ecc279` bumps both Hel commit `64d2632` bumps HelixAgent pointer + at parity

Date	Updater	What changed
		upstream). **Round 39 cumulative cascade reach: 12/12 newly-cascaded submodules + 25 idempotent + 1 def layers of recursion under vasic-digital and HelixDev** verbatim mandate "fully recursively everywhere" is no item being the uninitialized HelixLLM/submodules/Hel initialized in a future session.
2026-05-14	CONST-047 recursive cascade — HelixAgent layer (round 39 continued)	Continued the recursive cascade per CONST-047 into He digital + HelixDevelopment orgs). Per submodule: fetch anchor (if missing) + CONST-047 anchor (if missing), co **cascaded this round, 2 already had both anchors fro other), 0 failures**. HelixAgent commit `f5294b80` bump Meta-repo commit `4971fe7` bumps HelixAgent's pointe gitlab + origin + upstream). Cascade-verification script §11.9 + CONST-047 in every owned-submodule governa layer not yet covered by the script — it walks only the t **recursion)**: HelixAgent → HelixLLM (34 grandchildren — 36 additional submodules at the 3rd layer. Cascade s `helixagent-nested-cascade.sh`) reusable for the ne regression (post-cascade `run_all_challenges`: 12/12 build clean).
2026-05-14	CONST-047 recursive submodule cascade (round 39)	Operator re-asserted verbatim (2026-05-14): *"Make sure Submodules we control under our organizations (vasic-d everywhere with full bluff-proofing and comprehensive d tests and Challenges coverage! If it is not already this ma AGENTS.MD and CLAUDE.MD of the main project and al this rule (mandatory constraint) ALWAYS!"* — codified a **Mandate**. Authored full anchor in root `CONSTITUTION.` + owned-submodule baseline list + cascade requiremen in root `CLAUDE.md` and `AGENTS.md`; mirrored in `helix_` Then cascaded the condensed §CONST-047 anchor verb CONSTITUTION.md / CLAUDE.md / AGENTS.md (12 owr additions + 6 at root/inner-app = 42 governance-file add commits pushed to every configured remote of each su github+gitlab+origin+upstream; the others to origin wh Conflict resolution: 9 submodules had non-FF rejection prior session — reset to upstream tip, re-applied cascad bumps 12 submodule pointers + 6 root/inner-app gover and is now at parity across all 4 meta-repo remotes (git code touched this round — pure governance cascade. h unaffected (no probe changes; the rule is enforced by th `governance-cascade.sh` augmentation as a deferred

close-out[131] — rounds 191-305 catch-up (constitution cascade + helix_code/internal i18n + per-repo deep-doc + PAN-001 fix)

Date: 2026-05-19 **Last commit at close-out time:** 2d95892 (gov: round-304 — bump challenges pointer) **Theme:** Single batched CONST-044 catch-up narrative for ~115 rounds executed since close-out[130] (round 192). Per CONST-044 (Continuation Document Maintenance Mandate) the documentation drift across rounds 191-305 is itself a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result; this close-out remediates that drift in a single condensed entry. Per-round narration cadence resumes next round.

> Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completition and full usability by end users of the product!"*
>
> Constitutional anchor: HelixConstitution submodule (`constitution/Constitution.md` + `constitution/CLAUDE.md` + `constitution/AGENTS.md`) is the **canonical root** per CONST-059. This catch-up narrative extends the consumer-side `docs/CONTINUATION.md` and does NOT modify any canonical-root file.

What happened (thematic, not per-round)

1. **Constitution cascade rounds** — rounds 191 / 207 / 227 landed constitution-submodule §11.4.41-66 anchors as CONST-054 through CONST-067 (per CONST-049 fetch-before-edit pipeline + CONST-047 recursive submodule application). All anchors cascaded verbatim or by ID-reference into every owned-submodule CONSTITUTION.md / CLAUDE.md / AGENTS.md across vasic-digital + HelixDevelopment orgs. Verifier `scripts/verify-governance-cascade.sh` GREEN at each round's close.

2. **helix_code/internal i18n migration completed** — rounds 220-241 covered all 60+ packages of `helix_code/internal/` per CONST-046 (no hardcoded user-facing content). Each migrated package shipped: (a) `i18n/` resource bundles in 5 locales (en, sr, ja, es, de — fixture set), (b) `i18n.Get(locale, key)` call-sites replacing static literals, (c) audit-gate `scripts/lint-hardcoded-strings.sh` exit 0 across whole tree at round-241 close. Packages with no user-facing strings (pure internal infra) received NO-OP

marker comments. helix_code/cmd subtree + LLMsVerifier deferred to HXC-003 backlog.

3. **Per-repo deep-doc enrichment** — rounds 242-305 applied the enrichment template across 52 vasic-digital + 9 HelixDevelopment + 5 top-level submodules (~65 enrichment rounds). Per-repo template (Template C): README ~150-210 LOC (purpose + architecture + quickstart + capability matrix + Challenges section + governance pointer), `docs/test-coverage.md` (per-package coverage matrix + Challenge wire-evidence pointers), `challenges/runner/main.go` (binary that executes the bank), `challenges/*_describe_challenge.sh` paired-mutation scripts (assertion + intentional-mutation revert proving the assertion would fail without the fix), 5-locale fixture set under `challenges/fixtures/i18n/`. Pushed per-submodule across all configured upstream remotes per CONST-056.

4. **Real production bugs discovered + fixed during enrichment**:

    - **PAN-001** (round 298 discovery, round 302 fix): vasic-digital/Panoptic UTF-8 truncation bug — `truncateAt(s, n)` sliced at byte offset n splitting multi-byte runes mid-codepoint (Cyrillic/CJK fixtures rendered as mojibake). Fix: rune-aware truncation via `[]rune(s)[:n]` + size budget. Paired-mutation Challenge `truncation_describe_challenge.sh` proves it.
    - **Veritas stale CLAUDE.md** (round 288 discovery): documents removed `RemoveSource` API. Deferred — not blocking.
    - **LLMOps structural-bluff Challenge** (round 290): existing Challenge asserted on file-presence (grep), replaced with exit-code-driven `go test` invocation per the round-40 phase-challenge tightening pattern.

5. **Submodule deduplication finding** (round 277): `HelixDevelopment/Models` and `vasic-digital/Models` point at the same upstream repo. Documented; no consolidation work this round.

6. **Sibling-session commit-message race pattern** (rounds 228-285): repeated non-FF rejections on submodule pushes because sibling subagent sessions had landed concurrent commits. Mitigation per CONST-043 (no force-push, no `--force-with-lease` without explicit per-operation approval): fetch + rebase or re-apply onto upstream tip + re-push. Pattern documented in close-out narrative.

What's still open (rolled forward into the live backlog)

- **VEN-002**, **HXA-002**, **HXQ-001** — still BLOCKED on operator decisions
- **HXC-001** — In progress (round 343). Operator approved "agent defaults" for all 12 decisions; round 343 executed 3 build-verified batches renaming 13

owned-org leaf submodule dirs to snake_case: `HelixDevelopment/{Models→models, DebateOrchestrator→debate_orchestrator}` + 11 `vasic-digital/*` zero-go.mod-consumer leaves (`auto_temp, claritas, doc_processor, gandalf_solutions, hyper_tune, i_llm, leak_hub, ouroborous, plinius_common, veritas, vision_engine`). Deferred (submodule-go.mod-entanglement / parent-dir / cluster-C): ~37 remaining owned-org leaves consumed by helix_agent/helix_qa/HelixLLM go.mod, parent dirs `HelixDevelopment/→helix_development/` + `vasic-digital/` kept (GitHub-org handle), 59 `Upstreams/→upstreams/` dirs inside submodule trees. Each deferred batch needs dedicated per-submodule rounds to avoid entangling pre-existing uncommitted work.
- **HXC-003** CONST-046 migration backlog — helix_code/internal COMPLETE; remaining pending: LLMsVerifier ~1804 strings + cmd/helix_config ~241 strings + cmd/cli ~7 strings
- **Veritas CLAUDE.md** stale `RemoveSource` reference — deferred from round 288
- **Nested third-party submodule pointer drift** in `helix_qa/tools/opensource/*` — deferred per CONST-051(C) (third-party submodules exempt from own-org dependency-layout rules)

Next

- Resume per-round narration cadence (CONST-044 compliance)
- HXC-003 remaining CONST-046 migrations (LLMsVerifier is the biggest single chunk)
- Cascade detection follow-up for any newly-pulled constitution anchors (CONST-055 sweep on next constitution-submodule pull)
- Operator-blocked item triage (VEN-002 / HXA-002 / HXQ-001 / HXC-001)

Posture

- **Constitutional posture**: CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-047 recursive-submodule-application + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B/C) decoupling + dependency-layout + CONST-052 snake_case + CONST-053 .gitignore-hygiene + CONST-055 post-pull-validation + CONST-056 install_upstreams-on-clone + CONST-057 Type-aware closure vocab + CONST-058 reopened-source-attribution + CONST-059 canonical-root inheritance + CONST-060 fetch-before-edit all honoured per-round. This close-out[131] remediates the CONST-044 drift across rounds 191-305 in a single batched narrative; per-round cadence resumes next round.

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close-out[132] — round 398: Speed Programme launch + constitution §11.4.68/70-74 cascade + mirror reconciliation

Date: 2026-05-20 **Theme:** Three concurrent working points active this session, recorded per CONST-044 in the same commit that advances state (operator docs-sync mandate: *"Keep Issues and relevant documentation up to date and in sync … All working points we are doing MUST BE there as well!"*).

Working points (filed into the live backlog as Issues)

1. **HXC-006 — HelixCode Speed Programme** (`Feature`, `In progress`). Operator mandate (2026-05-20) to make all HelixCode + owned-submodule code 3-5x faster than competitor AI CLI agents without breaking features or weakening anti-bluff posture. Research corpus complete under `docs/research/speed/` (5 docs — bottleneck audit, competitive analysis, optimization techniques, overview, 6-phase / 31-task phased plan). **Phase 0 (measurement baseline) execution started** — establishing reproducible per-operation latency/throughput baselines so every later speed-up carries before/after captured evidence per CONST-035.

2. **HXC-007 — Constitution §11.4.68/70-74 cascade** (`Task`, `In progress`). Constitution submodule fetched + pulled first per CONST-049 (`584b3ee` → `34a82b3`); 6 new rules cascaded verbatim/by-ID into meta-repo governance files + 67 owned submodules per CONST-047; meta `.gitmodules` constitution pointer bumped. Stays open until the CONST-049 step-4 multi-upstream reconciliation converges all mirrors.

3. **HXC-008 — CONST-055 G1 governance gaps** (`Bug`, `In progress`). Post-constitution-pull validation sweep surfaced two **pre-existing** (not regression) gaps: (a) `scripts/verify-all-constitution-rules.sh` references a non-existent `submodules/models` path — actual dirs are lowercase `models` + `vasic-digital/Models`; (b) `helix_qa/CONSTITUTION.md` missing CONST-047..057, `VisionEngine/CONSTITUTION.md` missing §11.4.69.

4. **HXC-009 — Owned-submodule mirror-divergence reconciliation** (`Task`, `In progress`). Systemic divergence between vasic-digital ↔ HelixDevelopment GitHub/GitLab mirrors of several owned submodules. `helix_qa` reconciled via merge-first commit `a0397a4`; `VisionEngine`, `LLMProvider`, others under reconciliation per CONST-061 / §11.4.71 merge-first — NO force-push, NO history rewrite.

State

- 4 new Issues filed (HXC-006/007/008/009); none closed this session; `Issues_Summary.md` regenerated — 6 open (1 Bug, 3 Task, 2 Feature, 0 BLOCKED).
- Tracker HTML+PDF exports regenerated for all touched docs per §11.4.12/53/60/65.
- No production code touched in this docs-sync commit; submodule pointer changes from parallel sessions deliberately NOT staged (parallel sessions share the meta worktree).

Next

- HXC-006 Phase 0 baseline capture → Phase 1 of the speed phased plan.
- HXC-009 reconcile remaining divergent owned-submodule mirrors (VisionEngine, LLMProvider).
- HXC-008 fix the verifier path reference + cascade the missing CONST-047..057 / §11.4.69 anchors; re-run `verify-all-constitution-rules.sh` G1.

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close-out[133] — CodeGraph incorporation Phase D (CG14-CG16)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase D — governance covenant sync + documentation + CONST-047 recursive sweep. Phases A+B (install, scan, five-agent registration) already landed; this close-out completes the governance/docs phase per `docs/research/codegraph/incorporation-plan.md` §6.

Work performed

1. **CG14 — anti-bluff covenant gap fix.** Diffed root `QWEN.md` and `CRUSH.md` against the current `CLAUDE.md`/`AGENTS.md` CONST-0xx anchor set. Both were stale (last modified 2026-05-16, before the 2026-05-20 CLAUDE.md/AGENTS.md revisions): each carried CONST-035..059 but was **missing CONST-060 (Fetch-before-edit, §11.4.37) and the §11.4.68/70-74 anchor block** (Positive Sink-Side Evidence, Subagent-Driven-Default, Pre-Push Fetch+Investigate, Audio Top-Priority, Main-Spec Versioning, Submodule-Catalogue-First). Appended all six anchors verbatim from the canonical CLAUDE.md text to both files. Classification: universal (per §11.4.17) — consumer-side extension files mirroring the project-level anchors.

2. **CG15 — documentation.** Updated `tools/codegraph/README.md` (Revision 2) — added per-agent registration section with the cross-agent config table and the `Catalogue-Check: no-match 2026-05-20` metadata line (§11.4.74 — CodeGraph is third-party, not in the vasic-digital/HelixDevelopment catalogue). Added a CodeGraph integration section to `docs/COMPLETE_CLI_REFERENCE.md` (install/init/scan/per-agent registration) and removed a stray closing-tag artefact at the end of that file. This close-out note added per CONST-044.

3. **CG16 — CONST-047 recursive sweep.** Ran `scripts/verify-governance-cascade.sh` — **2 failures, both pre-existing and already tracked as HXC-008**: (a) `submodules/models` verifier path-list entry out of sync with on-disk lowercase `models` dir; (b) `helix_qa/CONSTITUTION.md` missing CONST-047..057. Neither failure is caused by the QWEN.md/CRUSH.md/codegraph Phase-D changes — the covenant-sync edits introduce no new cascade gap. Owned-submodule CodeGraph-config assessment: no owned submodule (`helix_qa`, `challenges`, `containers`, `security`) currently warrants its own CodeGraph MCP wiring — the meta-repo + inner `helix_code/` graphs already cover the first-party Go code the agents query; a per-submodule graph would add maintenance cost for marginal benefit. Recorded as assessed-no-follow-up; revisit if a submodule grows a substantial standalone codebase queried independently.

State

- `QWEN.md` + `CRUSH.md` now carry the complete CONST-035..060 + §11.4.68/70-74 anti-bluff covenant — fully in sync with `CLAUDE.md`/`AGENTS.md`.
- `tools/codegraph/README.md` Revision 2; `docs/COMPLETE_CLI_REFERENCE.md` gains a CodeGraph section.
- Cascade verifier: 2 pre-existing failures (HXC-008) unchanged — Phase D introduced no regression.
- HTML+PDF exports regenerated for touched docs per §11.4.65.

Next

- HXC-008 remains the path to clearing the 2 cascade-verifier failures.
- CodeGraph Phase C anti-bluff Challenges (CG10-CG13) remain the outstanding incorporation work if not yet landed by a sibling session.

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close-out[134] — CodeGraph incorporation Phase C (CG10-CG13)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase C — anti-bluff verification per `docs/research/codegraph/incorporation-plan.md` §5 + §6. Proves CodeGraph genuinely works end-to-end on the real HelixCode repo and reaches all five CLI agents (CONST-035 / Article XI §11.9).

Work performed

1. **CG10 — Layer A.** Replaced the `tools/codegraph/verify.sh` stub with the real end-to-end proof: `codegraph status` (asserts non-zero files/nodes/edges — 39 022 / 624 092 / 1 644 454 captured), `codegraph query Provider` (asserts real Go symbols), `codegraph context` (asserts real repo files), plus an anti-bluff token guard. Wrapped as Challenge `CG-CHALLENGE-01`. **PASS** — evidence `docs/research/codegraph/evidence/phase-c/cg-challenge-01-*`.

2. **CG11 — Layer B.** Challenge `CG-CHALLENGE-02` drives `codegraph serve --mcp` over stdio with real JSON-RPC: `initialize` → serverInfo, `tools/list` → 9 `codegraph_*` tools, `tools/call codegraph_search` → real graph data. JSON-RPC wire transcript captured (`cg-challenge-02-jsonrpc-wire.jsonl`). **PASS**.

3. **CG12 — Layer C.** Challenges `CG-CHALLENGE-03..07`, one per agent, with a shared `lib-agent-challenge.sh` driver: tier-1 drives the agent CLI non-interactively and asserts a real `.go` symbol path; tier-2 (§11.4.3 topology dispatch) falls back HONESTLY to connect-only + tool-discovery proof when an agent cannot be driven to a real answer. **Results:** Claude Code (primary) — **true end-to-end PASS**; OpenCode — true end-to-end PASS; Crush — true end-to-end PASS; Kimi CLI — connect-only PASS (LLM monthly-quota 429 blocked the answer; its MCP loader did enumerate all 9 codegraph tools); Qwen Code — connect-only PASS (bundled OAuth free tier discontinued upstream 2026-04-15). 3/5 agents true end-to-end, 2/5 connect-only — reported honestly, no false end-to-end claims.

4. **CG13 — helix_qa bank.** Registered the 7 Challenges as a helix_qa test bank `tools/codegraph/challenges/codegraph-integration.bank.yaml`. Per CONST-051(B) the bank lives in the HelixCode tree — NOT inside the project-not-aware helix_qa submodule (the bank hardcodes HelixCode Challenge paths, which would couple the reusable submodule to HelixCode). The helix_qa runner accepts arbitrary bank paths via `-banks`, so it is fully consumable: `helixqa run -banks tools/codegraph/challenges/`

codegraph-integration.bank.yaml. Runner tools/codegraph/challenges/run-all.sh executes all 7 with per-Challenge captured evidence.

State

- 7 Challenges under `tools/codegraph/challenges/` — all `bash -n`-clean with §11.4.18 doc blocks; bank run 7/7 **PASS** after the CG-CHALLENGE-03 arg-shift fix.
- `tools/codegraph/verify.sh` is now a real anti-bluff proof (no longer a stub).
- All runtime evidence under `docs/research/codegraph/evidence/phase-c/` — status JSON, query JSON, JSON-RPC wire transcript, 5 agent transcripts, per-Challenge run logs.
- helix_qa autonomous-session integration: bank registered + discoverable; direct execution via `run-all.sh` captures per-check evidence. Full autonomous-session orchestration left to a helix_qa QA run.

Next

- CodeGraph Phase D (CG14-CG16) already landed (close-out[133]).
- A helix_qa autonomous QA session may execute the `codegraph-integration` bank for orchestrated per-check evidence capture.

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close-out[135] — HelixCode Speed Programme COMPLETE (P5-T04 — all 6 phases / 31 tasks landed)

Date: 2026-05-20 **Theme:** Speed-programme final task P5-T04 — CONST-048 coverage ledger, release-gate sweep, and programme close-out. Closes **HXC-006** (HelixCode Speed Programme, Feature). Operator mandate 2026-05-20: make HelixCode + owned-submodule code 3-5× faster than competitor AI CLI agents without breaking any feature or weakening anti-bluff posture.

> Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completition and full usability by end users of the product!"*

Work performed (P5-T04)

1. **CONST-048 coverage ledger.** Created `docs/research/speed/05-coverage-ledger.md` (constitution §11.4.44 metadata header) — one row per task P0-T01..P5-T04 (mapped to opportunities O1–O21), each carrying the commit SHA, the captured before→after measurement, and the six CONST-048 invariants (`[AB][WC][MD][NB][DS][TF]`). Roll-up: **29 PASS** + **2 PARTIAL** + **0 DEFERRED** across all 31 tasks.

2. **Release-gate sweep.** `scripts/audit-const046-hardcoded-content.sh` ran exit 0 (speed work added no user-facing strings — no new hardcoded content). `scripts/verify-governance-cascade.sh` reported 2 failures = the already-tracked **HXC-008** pre-existing drift (stale `Models` path + `helix_qa/CONSTITUTION.md` missing CONST-047..057), NOT speed-programme regressions. Anti-bluff smoke `grep -rn "simulated\|for now\|TODO implement\|placeholder" helix_code/internal helix_code/cmd` (prod, non-`_test.go`) = clean. `make verify-compile` not re-run (P5-T04 is docs-only; compile state unchanged from `98315a14`).

3. **Two deferred findings filed** per §11.4.15/16: **HXC-011** (Bug — helix_qa runner emits hollow sub-µs `PASSED` metadata rows for desktop-platform bank cases without executing them — a §11.4 PASS-bluff in the QA runner) and **HXC-012** (Bug — data race in `helix_code/internal/llm/load_balancer.go` stat-collector goroutine under full-package `-race`). Both pre-existing; surfaced during the speed programme; reported honestly.

4. **HXC-006 closed.** All 31 tasks genuinely landed → HXC-006 updated to `Implemented (→ Fixed.md)` per CONST-057/§11.4.33 and migrated to `docs/Fixed.md` per §11.4.19. Issues_Summary.md + Fixed_Summary.md regenerated; HTML/PDF exports refreshed.

Speed-programme outcome

All 6 phases / 31 tasks landed across rounds — P0 (baseline harness), P1 (LLM & startup wins), P2 (context-build speed), P3 (interactive & agent-loop levers), P4 (profile-gated tuning), P5 (dev-experience + submodule cascade + close-out). Headline measured wins (each with pasted in-session evidence per CONST-035): lazy Ollama discovery 67µs→2.7ns, lazy CLI startup ~8.85×, cache pre-warm ~7.6×, regexp hoist ~7.4×, repo-map cache ~10.6× warm, parallel search ~4.39×, incremental tree-sitter ~21×, small-model routing ~5.87×, diff edits 94-99% token cut, fast-apply ~516×, tool parallelism ~5.99×, PGO −46% CPU-bound.

Honest PARTIALs (no green PASS claimed): P5-T01 build/test parallelism tuning is landed and correct but the suite-wall-time before/after delta was not captured as a pasted benchmark; P5-T02 is a partial single-provider `internal/llm` split

(Cerebras extracted to `internal/llm/providers/cerebras/`) — a full 18-provider extraction is infeasible without an import cycle.

State

- `docs/research/speed/05-coverage-ledger.md` committed (+ HTML/PDF exports).
- HXC-006 closed in `docs/Issues.md` (migration tombstone) → full record in `docs/Fixed.md`.
- HXC-011 + HXC-012 filed `Queued` in `docs/Issues.md`.
- Issues_Summary.md / Fixed_Summary.md regenerated; tracker exports refreshed.

Next

- HXC-011 (helix_qa desktop-platform hollow-PASS) closed round 402 — `Fixed (→ Fixed.md)`: the runner's `run` path on the `desktop` platform now genuinely executes a bank case's `shell:` action via `os/exec` and scores PASS/FAIL on the real exit code (reproduce-before-fix RED→GREEN; helix_qa commit `6b46df0`, meta pointer bumped). HXC-012 (load_balancer.go race) closed round 401. Both pre-existing speed-programme findings now resolved.
- The speed programme is complete; no further speed tasks queued.

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close-out[136] — round 403: HXC-008 governance-gap fix + HXC-007/HXC-009 verified-complete closure

What landed

1. **HXC-008 closed `Fixed (→ Fixed.md)`** (Bug). The two pre-existing CONST-055 G1 cascade-drift gaps surfaced in close-out[132] / CG16 are fixed:
    - 1. `docs/improvements/submodule_owned.txt` line 10 referenced `submodules/models` — a path that does not exist on disk; the CONST-052 batch-1 rename (`a1ea3c8`) had lowercased it to `models` (`.gitmodules` registers it lowercase). Corrected to `submodules/models`. The cascade verifier (`scripts/verify-governance-cascade.sh`) reads this file, so the false `FAIL: submodules/models — path does not exist on disk` line is now gone.
    - 1. `helix_qa/CONSTITUTION.md` was missing anchors CONST-047..057 (CLAUDE.md/AGENTS.md already carried them); `VisionEngine/CONSTITUTION.md` was missing §11.4.69 (likewise). The contiguous CONST-047..057 block (217 lines) was

cascaded into `helix_qa/CONSTITUTION.md` from the meta CONSTITUTION.md; the §11.4.69 anchor (80 lines) into `VisionEngine/CONSTITUTION.md`. Both submodules committed + pushed to all upstreams (helix_qa `1364d23`; VisionEngine `b3a13d8`, integrating upstream docs commit `bfe3b06` first per §11.4.71).
   - Verification: `scripts/verify-governance-cascade.sh` → `=== Result: 0 failures === PASS`; `scripts/verify-all-constitution-rules.sh` → `Gates run: 6 / Failures: 0` (G1-G6 all PASS).
2. **HXC-007 closed `Completed (→ Fixed.md)`** (`Task`). Constitution §11.4.68/70-74 cascade verified genuinely complete: meta `.gitmodules` constitution pointer confirmed at `34a82b3`; spot-check `grep -c "11.4.70\|11.4.74"`=6 across `helix_qa`, `submodules/llm_provider`, `challenges` CLAUDE.md.
3. **HXC-009 closed `Completed (→ Fixed.md)`** (`Task`). Owned-submodule mirror-divergence reconciliation verified complete: VisionEngine carries convergence merge `3485f5f` and pushes FF-clean to all 4 upstreams; helix_qa, LLMProvider, challenges, containers, DocProcessor all converged per CONST-061/§11.4.71 merge-first.

State

- All three items migrated to `docs/Fixed.md` per §11.4.19; `docs/Issues.md` sections retained as migration tombstones.
- `docs/Issues_Summary.md` + `docs/Fixed_Summary.md` updated (open count 6→3); all 8 tracker exports (4 HTML + 4 PDF) regenerated via `scripts/regenerate-tracker-exports.sh`.
- Meta-repo: verifier-input fix + tracker migrations + exports committed; helix_qa + VisionEngine meta pointers bumped.

Next

- No open Bug items remain. Open: HXC-001 (Task), HXC-003 (Feature), HXC-010 (Operator-blocked Task — CodeGraph end-to-end verification awaiting LLM-backend credentials).

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close-out[137] — round 463: HXC-003 closed `Implemented (→ Fixed.md)` — CONST-046 i18n migration campaign concluded

What landed

1. **HXC-003 closed `Implemented (→ Fixed.md)`** (`Type: Feature`, per CONST-057/§11.4.33). The CONST-046 i18n migration campaign — the longest-running Feature in `docs/Issues.md` — is **concluded**. The genuine user-facing (C) string-literal surface (UI text, prompts shown to the operator, error messages surfaced to end users, labels, helper text) is **exhausted across all 7 CONST-046 scope areas**:
   - 1. helix_code `internal/`, (2) `cmd/`, (3) `applications/` — confirmed exhausted rounds 461/462 (`applications` final-sweep × 110 literals at meta HEAD `72389451`).
   - 1. `LLMsVerifier` — round 452.
   - 1. `helix_qa`.
   - 1. all owned `vasic-digital/*` submodules + (7) all owned `HelixDevelopment/*` submodules — rounds 413/441.
2. **Scale + anti-bluff posture.** Across ~91-462 rounds, **tens of thousands of literals** were migrated through i18n seams — `nicksnyder/go-i18n/v2` Bundle/Localizer (Option D, design `f9dc102`, `pkg/i18n` foundation `e29b075`) plus locale-aware `.toml`/`.yaml` resource files. **Every migration round shipped paired-mutation anti-bluff tests** (per §1.1): a known un-migrated literal is planted and the test asserts the audit-gate reports FAIL — so a PASS certifies real i18n coverage rather than absence-of-error. The audit-gate `scripts/audit-const046-hardcoded-content.sh --fail-on-new` is enforced; each round re-ran `--update-baseline` so the snapshot shrank monotonically.
3. **Remaining audit hits are out of scope.** The ~55k audit-baseline hits that remain are **all OUT of CONST-046 scope** — (A) LLM prompt templates (strings addressed to a model, not a human), (B) wrapped-error developer-facing tech strings, identifier tokens, struct-tag keys, format-spec tokens, and test fixtures — classified file-by-file in `docs/audits/2026-05-20-internal-const046-classification.md` (Revision 2). CONST-046's invariant is satisfied: no user-facing text is hardcoded as a static literal; every such string is LLM-generated, i18n-loaded, or metadata-composed.

State

- HXC-003 migrated to `docs/Fixed.md` per §11.4.19; `docs/Issues.md` section retained as a migration tombstone.

- `docs/Issues_Summary.md` updated (open count 3→2; Feature open 1→0); `docs/Fixed_Summary.md` updated (Feature count 67→68; total closed 95→96; open snapshot 3→2).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via `scripts/regenerate-tracker-exports.sh`.

Next

- Open set is now exactly 2 items: **HXC-001** (CONST-052 rename programme — Task, In progress; ~37 submodule-entangled leaf renames + parent dirs + 59 Upstreams dirs deferred) and **HXC-010** (Kimi CLI + Qwen Code CodeGraph end-to-end verification — Operator-blocked Task; awaiting LLM-backend credentials). No open Bug or Feature items remain.

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close-out[138] — round 464: HXC-010 closed `Completed (→ Fixed.md)` — Kimi/Qwen CodeGraph end-to-end verified with operator-provided credentials

> Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completition and full usability by end users of the product!”*

What landed

1. **HXC-010 closed `Completed (→ Fixed.md)`** (`Type: Task`, per CONST-057/§11.4.33). The operator supplied OpenAI-compatible router credentials (`/home/milosvasic/api_keys.sh`), unblocking the two CodeGraph per-agent Challenges that had been `Operator-blocked` since round 463.
2. **§11.4.10.A pre-use leak-audit: CLEAN.** `git grep` + `git log -S` of the three relevant key values (`KIMI_API_KEY`, `OPENROUTER_API_KEY`, `SILICONFLOW_API_KEY`) confirmed none has ever been committed to a tracked file or git history.
3. **The originally-blocking backends remain blocked.** `KIMI_API_KEY` shares the **same exhausted account-level monthly billing-cycle quota** as Kimi’s bundled OAuth (`exceeded_current_quota_error` on `api.kimi.com/coding/v1`); Qwen’s bundled OAuth free tier is still discontinued;

OPENROUTER_API_KEY had insufficient credit (~$0.0007). Resolution: both agents driven against the **SiliconFlow** OpenAI-compatible router, which has credit and serves both target models with working tool-calling.

4. **Kimi CLI** — driven via an openai_legacy-type provider (config-file ~/.kimi/config-codegraph-or.toml carrying only a placeholder api_key); the real key injected at runtime via the OPENAI_API_KEY env var (kimi_cli/llm.py augment_provider_with_env_vars), model moonshotai/Kimi-K2.6. **Qwen Code** — driven via --auth-type openai with OPENAI_API_KEY/OPENAI_BASE_URL/OPENAI_MODEL env vars (key NEVER written into the tracked .qwen/settings.json), model Qwen/Qwen3-Coder-30B-A3B-Instruct.
5. **Both Challenge scripts updated** (tools/codegraph/challenges/cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh) to honour HELIX_CG_OPENAI_API_KEY + HELIX_CG_OPENAI_BASE_URL (+ optional HELIX_CG_QWEN_MODEL) for the credentialed path, falling back to the bundled quota-gated provider when those env vars are absent — anti-bluff preserved (the connect-only fallback is still reported honestly when no creds are present).
6. **Result — both true tier-1 PASS.** cg-challenge-05-kimi.sh → CG-CHALLENGE-05: PASS (true end-to-end — agent invoked codegraph_* and returned real graph data); cg-challenge-07-qwen.sh → CG-CHALLENGE-07: PASS (true end-to-end ...). Each transcript shows the MCP loader connecting to the codegraph server (9 codegraph_* tools), the agent invoking codegraph_search for symbol Provider, the ToolResult/tool_result returning 10 real .go symbol paths from the scanned HelixCode graph (first: docs/helix_qa/HelixQA_Integration/research/testdata/raw/pkg_llm_provider.go:40), and the agent answering with a real file path.

State

- HXC-010 migrated to docs/Fixed.md per §11.4.19; docs/Issues.md section retained as a migration tombstone.
- Evidence captured under docs/research/codegraph/evidence/hxc010/ (both transcripts + README); secret-scan of every transcript confirms **no API-key value present**.
- docs/Issues_Summary.md updated (open count 2→1; Operator-blocked Task 1→0); docs/Fixed_Summary.md updated (Task count 7→8; total closed 96→97; open snapshot 2→1).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via scripts/regenerate-tracker-exports.sh.
- All 7 CodeGraph anti-bluff Challenges (CG-CHALLENGE-01..07) now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, and Qwen Code.

Next

- Open set is now exactly 1 item: **HXC-001** (CONST-052 rename programme — Task, In progress). No open Bug, Feature, or Operator-blocked items remain.

close-out[139] — round 465: Phase 0 of the own-org submodule sync → flatten → rebuild → test programme (governance baseline)

> Verbatim 2026-06-03 operator mandate (preserved per CONST-049 §11.4.17): *"first we go to the latest main (or master) commit … Then on top of this we merge carefully all changes … all existing Submodules that are coming from vasic-digital and HelixDevelopment organizations into the root of the project … Everything MUST BE flat … HelixCode/submodules/SUBMODULE_NAME … rebuild EVERYTHING … bootup all containers … execute all existing tests and Challenges … all tests MUST PRODUCE real evidence … No false results or bluff(s)."*

What landed

1. **Programme decomposed into 6 phases** (0 governance baseline → 1 sync-to-latest+merge → 2 flatten+rewrite-refs → 3 rebuild → 4 boot-infra+run-all-tests/Challenges → 5 fix-every-issue). Design spec written + committed: `docs/superpowers/specs/2026-06-03-phase0-governance-baseline-design.md`. Each phase: design → approval → subagent execution → evidence-backed verification. Operator approved Phase 0 + "one phase at a time" + autonomous subagent-driven loop.
2. **Governance verification (subagent, read-only): PASS.** `scripts/verify-governance-cascade.sh` → 0 failures (evidence: `docs/migration/phase0-cascade-verify-2026-06-03.txt`). Anti-bluff mandate present fleet-wide by anchor ID. Minor gaps logged: `github_pages_website` lacks the verbatim §11.9 sentence (covered by anchor; verbatim insertion deferred to Phase 1 when it is pushed); `debate_orchestrator` was absent from the owned roster → **fixed** (added to `docs/improvements/submodule_owned.txt`).
3. **CONST-038 / §6.W reconciled** (operator decision 2026-06-03): the GitFlic/GitVerse remote ban is **LIFTED** — pushes now target all four providers (GitHub + GitLab + GitFlic + GitVerse). Edited in root `CLAUDE.md`, `CONSTITUTION.md`, `AGENTS.md` (QWEN.md/CRUSH.md did not carry the ban). Resolves the doc-vs-practice contradiction created by the "push to all upstreams incl. GitFlic/GitVerse" decision.
4. **Safety backup (subagent, read-only):** reversibility restore manifest written + round-trip validated (218 recursive submodules; main HEAD `34bc865e`) →

`docs/migration/phase0-restore-manifest-2026-06-03.txt`. This is the rollback anchor for the higher-risk Phases 1–2.

State

- Phase 0 mutations limited to documentation + evidence/manifest files in the **main repo** (no submodule content touched — pre-existing dirty submodules `helix_agent/helix_qa/filesystem` are explicitly left for Phase 1's careful sync).
- **HXC-001** (CONST-052 rename programme) is subsumed by Phase 2 (flatten + snake_case + rewrite-refs).

Next

- Phase 1: per own-org submodule — `git fetch --all` → checkout main/master → pull latest → carefully merge local work (e.g. `filesystem` `docs/ARCHITECTURE.md`) → `install_upstreams` → commit + push to all upstreams. Subagent-parallel across the 70 own-org repos, with working-tree quiescence (§11.4.84) + per-repo fetch-investigate-integrate (§11.4.71).

close-out[140] — round 466: Phase 1 complete — own-org submodule sync-to-latest + install_upstreams + push-to-all-mirrors

What landed

1. **`submodules/filesystem`** — the only own-org repo with real local content (a 232/-138 `docs/ARCHITECTURE.md` rewrite). Staged via `git add --renormalize` (a `.gitattributes` text-normalization quirk made plain `git add` a silent no-op), committed `1d8f3f9`, pushed + verified on GitHub + GitLab.
2. **All 69 checked-out own-org submodules processed** (6 concurrent subagent batches; conductor sandbox shell was unstable for git loops, subagent shells stable). Each: safety-checked clean, `install_upstreams` run from its root, current branch pushed to every configured remote. **Result: 67 OK / 2 PARTIAL / 0 SKIP-dirty.** The 2 PARTIAL are both `models` repos — failures confined to unreachable gitflic.ru/gitverse.ru mirrors; their GitHub+GitLab are current. Several lagging GitLab mirrors were resynced (UPDATED).
3. **Known defects logged** (for Phase 2 config-cleanup): (a) `HelixDevelopment/doc_processor` + `HelixDevelopment/llm_provider` carry unresolved `<<<<<<< HEAD` conflict markers in `upstreams/GitHub.sh` (breaks their install_upstreams); (b) `github_pages_website` configured remote (`HelixDevelopment-s-Code/Website.git`) disagrees with the

owned roster (HelixDevelopment-Code/Welcome.git); (c) models gitflic/gitverse mirrors unreachable.

State

- Meta-repo submodules/filesystem gitlink bumped to 1d8f3f9 in this commit. helix_agent/helix_qa third-party nested drift deliberately left uncommitted (not our work).
- **Phase 2 collision blocker discovered:** 5 own-org submodules are mounted under BOTH dependencies/vasic-digital/ and dependencies/HelixDevelopment/ pointing at the SAME upstream (models, doc_processor, llm_orchestrator, llm_provider, vision_engine). Flattening to submodules/<leaf> collides 5× → requires a de-dup/disambiguation decision before the move.

Next

- Phase 2 (flatten + rewrite-refs): resolve the 5 collisions, handle constitution specially (huge @constitution/ reference surface + find_constitution.sh), then git mv own-org submodules to submodules/<snake_case>, rewrite all refs (go.mod replace, .gitmodules, scripts, docs, imports), clean git configs, reference-integrity tests. Discovery (read-only ref map) runs first.

close-out[141] — round 467: Phase 2 — flatten own-org submodules to submodules/<leaf> (layout + go.mod + refs; module-resolution GREEN)

Operator decisions 2026-06-03: drop the 4 unused vasic-digital forks + the duplicate HD models mount; KEEP constitution at ./constitution/ (exempt from flatten). Push to all upstreams incl. GitFlic/GitVerse.

What landed (subagent-driven, build-verify-before-commit)

1. **Relocated 63 own-org submodules** → submodules/<leaf> via git mv; **dropped 5 mounts** (HD/models dup + vd/{doc_processor,llm_orchestrator,llm_provider,vision_engine}). No collisions; constitution untouched; dependencies/ now holds only third-party (LLama_CPP/Ollama/HuggingFace_Hub). github_pages_website (org HelixDevelopment-Code) and mcp_servers NOT moved — NEEDS-REVIEW (org/remote mismatch; resolve before moving).

2. **go.mod rewritten** for flat layout: `helix_code/go.mod`, `submodules/helix_agent/go.mod` (37 → ../<leaf>), `submodules/helix_qa/go.mod` (consolidated duplicate capital+lowercase replace blocks → one lowercase per module; fixed `LLMsVerifier` casing). **`go list -m all` GREEN** for all consumers; remaining build failures PRE-EXISTING + layout-independent (macOS-only `syscall.CLONE_NEWNET` in `helix_code/internal/tools/shell/sandbox.go` + go.sum staleness).
3. **Reference rewrite (2b):** 48 files — dependencies/{org}/<leaf>→submodules/<leaf> (handled BOTH snake_case AND CamelCase leaf forms, e.g. `LLMsVerifier`→`llms_verifier`). `.codegraph/config.json` already §11.4.79-compliant. **Residual rewritable own-org `dependencies/{org}/` refs: 0.**
4. **NEEDS-REVIEW scripts fixed:** `scripts/audit-const046-hardcoded-content.sh` + `scripts/propagate-governance.sh` retargeted to flat `submodules/*`.

State

- Module resolution verified GREEN before commit (rollback gate). Restore anchor: `docs/migration/phase0-restore-manifest-2026-06-03.txt`.
- **Open NEEDS-REVIEW (deferred, non-blocking):** `.gitmodules` still uses [submodule "dependencies/<org>/<leaf>"] section-name KEYS though every `path =` is submodules/<leaf> (functional; cosmetic git-config cleanup pending); `github_pages_website/mcp_servers` move pending org/remote-mismatch resolution; `helix_agent/helix_qa` third-party nested drift still uncommitted (intentional).

Next (Phases 3–5 — best run in a fresh full-context session, subagent-parallel)

- Phase 3: rebuild everything (Linux/containers via the `containers` submodule — the macOS sandbox-syscall blocker means real builds run in-container). Phase 4: boot all infra + run all tests + Challenges with real evidence (anti-bluff). Phase 5: fix every issue found, validated. Periodic: fetch+pull `constitution` and re-process on any new rule (constitution currently up-to-date @ d90ab87).

close-out[142] — round 468: Comprehensive-completion programme kickoff — grounded discovery + Wave 1 fix-first (subagent-parallel, anti-bluff)

> Operator mandate 2026-06-04: complete everything (no module/test broken/disabled, 100% coverage, leak/deadlock/race fixes, Snyk+SonarQube, lazy-init/semaphores/non-blocking, full docs/courses/website), running 3–5 parallel subagent-driven streams, no interactive/sudo processes.

What landed (conductor-committed; streams implement+prove, conductor integrates per §11.4.84)

1. **Grounded discovery** (read-only, 16-agent fan-out → `docs/CONSOLIDATED_UNFINISHED_WORK_AND_PLAN.2026-06-04.md`). HONESTY: only **4/16 modules verified** (helix_code, root meta, helix_qa, containers); the other 12 were **rate-limited** (16 concurrent heavy agents trips the account token-rate limit → safe band ≈5–10). All 4 assessed Go modules **build+vet clean**. Supersedes the ~95 stale root planning `.md` (Phase 2 retires them). Wave 2 (remaining-12 discovery) launched concurrently.
2. **Wave 1 fix-first (4 streams, RED→GREEN + §1.1 mutation-proven, re-verified by conductor on integrated tree):**
   - **S1B helix_qa** — 7 `title:`→`name:` in `banks/atmosphere.yaml` (was hard-failing `helixqa list --banks banks` by aborting the whole dir) + new `pkg/testbank/loader_test.go` regression guard.
   - **S1C root** — 6 `$(MAKE) -C HelixCode`→`-C helix_code` (case-sensitive-FS breakage) + new real non-tautological `scripts/bluff-detector.sh` (Checks 4+5, exit 0 clean). Finding: the "known LLMsVerifier dual-pin divergence" is **already resolved** (verify-llmsverifier-pin-parity exits 0) — stale Makefile/report note corrected.
   - **S1D root** — removed 2 no-assertion PASS-bluff e2e stubs (`tests/e2e/core/{simple,auth_simple}_test.go`) + `doc.go` pointing to real challenge coverage.
   - **S2C helix_code** — 7 i18n silent skips → hard deterministic assertions (`cmd/helix_config/main_i18n_test.go`).

New defects surfaced (tracked for next waves — NOT yet fixed)

- **helix_qa**: duplicate test-case id `CLI-CODEX-001` across `banks/cli-agents-comprehensive.yaml` + `banks/cli-agents-test-helixagent.yaml` (was masked by the S1B abort; now the next `list --banks banks` blocker).

- **helix_code product bug**: `internal/config` `saveConfigLocked` lacks `os.MkdirAll(filepath.Dir(path))` → fresh-install `helix-config reset --force` errors "no such file or directory". Real end-user defect; needs a scoped fix + RED test.
- helix_qa loader resilience (collect-all-errors vs abort-on-first) — recommended follow-up.

State / Next

- Verified GREEN on integrated tree: i18n test ok; e2e core stubs gone; `make bluff-detector` PASS exit 0; helix_qa S1B 3 PASS.
- **Next waves:** Wave 2 (12-module discovery, running) → completes the report (§11.4.118). Then: helix_qa duplicate-id + saveConfigLocked MkdirAll fixes; root `internal/{security,fix,testing}` dead-code disposition + `cmd/security-test` re-enable; containerized Snyk/SonarQube/`govulncheck`/`-race`; FAISS/consensus/capture stub completion; coverage-to-max; doc/course/website reconciliation.

close-out[142b] — round 468b: Wave 2 complete + Wave 3 (5 broken-build fixes) + Wave 4 partial — committed + PUSHED

- **Wave 2 discovery complete** (12 modules) → `docs/CONSOLIDATED_REPORT_ADDENDUM.wave2.2026-06-04.md`. Confirmed-broken builds found: doc_processor (committed conflict markers), challenges (missing containers dep), llms_verifier outer (go.sum gitignored). Plus secret leak (postgres-init.sql), website soft-skip, helix_code FAISS-sim + consensus-stub.
- **Wave 3 — 5 fixes committed + PUSHED to all upstreams** (each RED→GREEN, conductor-re-verified, fast-forward):
    - doc_processor `c1b9b85` (i18n conflict-marker resolution + bundle.go embed fix) → github.
    - challenges `8b423e3` (wire `digital.vasic.containers` replace + helix-deps.yaml) → github+gitlab.
    - llms_verifier `9302b5c5` (un-ignore+track go.sum, dead replace + cruft removal; offline build proven) → github+gitlab.
    - github_pages_website `edbca6c` (HTML soft-skip→honest exit-3 SKIP, podman script reconcile, untrack nginx logs) → github.
    - root `25d41351` (postgres-init.sql CONST-042 secret removal + .env.example sanitize).
- **helix_qa re-merge resolved + PUSHED** `84dac47`: merged origin/main (conduit/4K/subtitles work, was 2 commits ahead) on top of the S1B fix, then corrected 8 pre-flatten CamelCase go.mod replace paths → flat-layout (CONST-052). `4aaacd4..84dac47` → all 3 URLs. Builds clean.

- **Wave 4 partial:** W4C (helix_code `internal/config` saveConfigLocked+ExportConfig `os.MkdirAll` — fresh-install fix, RED→GREEN+§1.1) committed into root; W4E (llm_orchestrator `80ead87` dead-replace + artifact removal) PUSHED → github. **W4A (FAISS), W4B (consensus), W4D (llm_provider fallback) were RATE-LIMITED mid-work → their unverified edits REVERTED** (no evidence report; re-run pending in Wave 4b).
- **New defects queued** (not yet fixed): helix_qa duplicate-id `CLI-CODEX-001`; doc_processor `cmd/docprocessor` `runCLI` undefined + CLAUDE.md conflict markers; docker-compose.test.yml hardcoded test creds (CONST-042); helix_code FAISS-sim relabel + consensus bounded-timeout (W4A/W4B re-run); helix_code `internal/config` BackupConfig same MkdirAll gap.
- **Lesson:** safe concurrent heavy-agent ceiling ≈5; 16 trips the account rate limit. Run mutating waves at ≤5, revert any rate-limited mid-flight edits (no evidence = no commit).

close-out[142c] — round 468c: Wave 4b (re-run rate-limited streams) — committed + PUSHED

- **W4A FAISS** (helix_code `internal/memory/providers`): root-caused as a *misnomer* — the provider is a real pure-Go brute-force vector store (real cosine/euclidean math + JSON persistence), NOT a simulation. Honest relabel of 17 "simulation" labels → "pure-go-brute-force"; index `flat_simulation`→`flat_brute_force`; metadata `backend` key. §1.1-proven. (in root commit)
- **W4B consensus** (helix_code `internal/worker`): implemented full multi-peer `VoteTransport` interface + `InProcessCluster` real transport + **bounded-safety step-down** (was livelock: ticker re-ran startElection every tick, never elected/stepped-down). New `consensus_election_test.go` (3-node real election + bounded-safety + vote-dedup); §11.4.120 reconciled 2 stale tests; §11.4.115 polarity + `-race -count=3` clean. (in root commit) FLAG: `ssh_pool.go` still single-node (no transport injected) — degrades cleanly, distributed election latent until SSH transport wired (follow-up).
- **W4D llm_provider** `74106f5`: removed CONST-036 hardcoded Tier-3 fallback (DiscoverModels/GetCachedModels return nil honestly); §1.1-proven → github.
- **W4F helix_qa** `845066c`: re-ID duplicate `CLI-CODEX-001`→`CLI-CODEX-HELIXAGENT-001` (+ json twin); §1.1-proven → gitlab+github.
- **Newly-surfaced helix_qa bank blockers (queued):** `banks/atmosphere_video_4k.yaml` lines 66-69 `!!str`→`testbank.DocRef` parse error; 4 more cross-bank dup ids (PERF-001/002 + SEC-001/002 between aichat-bash-tools-comprehensive.yaml and performance/security-validation.yaml). Full `helixqa list --banks banks` not green until these clear.

- **Still queued:** doc_processor `cmd/docprocessor` runCLI undefined + CLAUDE.md conflict markers; docker-compose.test.yml hardcoded creds; helix_code config BackupConfig MkdirAll gap; ssh_pool consensus transport wiring. Then larger phases: containerized Snyk/SonarQube/govulncheck/-race, coverage-to-max, dead-code disposition, docs/courses/website.

close-out[142d] — round 468d: Wave 5 (queued blockers) — committed + PUSHED

- **W5A helix_qa** `913ff9c`: atmosphere_video_4k.yaml DocRef YAML shape (4 strings→{type,section,path} mappings) + re-ID PERF/SEC cross-bank dups → AICHAT-PERF/SEC. §1.1-proven → gitlab+github. **REMAINING bank blocker:** `FQA-ATV-001` dup across full-qa-androidtv-challenges.json vs full-qa-androidtv.{json,yaml} still blocks full `list --banks banks` — queued (W6).
- **W5B doc_processor** `3821e8a`: restored i18n-wired `runCLI` (was dropped by an auto-commit → "undefined: runCLI" broke vet + cascaded root TestAutomation_GoVet) + resolved 14 CLAUDE.md conflict regions (§9 no-loss, CONST-051(B) decoupled phrasing kept). Whole module go `vet ./...` exit 0, 10 pkgs pass. §1.1-proven → github. FLAG: `newTranslator` honestly NoopTranslator (docprocessor_cli_* IDs only in active.en.yaml which BundleTranslator skips) — real flat-bundle localization is a pkg/i18n follow-up.
- **W5C docker-compose.test.yml** (root commit): externalized POSTGRES_PASSWORD (`:-helixtestpass` throwaway) + HELIX_AUTH_JWT_SECRET (`:?` required) + DB-URL passwords to env (CONST-042). leaked literals helixcode123 + test-jwt-secret removed; YAML valid; §1.1-proven.
- **W5D helix_code** `internal/config` BackupConfig (root commit): added os.MkdirAll matching W4C — completes the fresh-install config-write fix trio (saveConfigLocked + ExportConfig + BackupConfig). §1.1-proven.
- **Next (W6+):** helix_qa FQA-ATV-001 dup (→ fully green `list --banks banks`); ssh_pool consensus transport wiring; pkg/i18n flat-bundle reconciliation; root `internal/{security,fix,testing}` dead-code + `cmd/security-test` re-enable; then containerized Snyk/SonarQube/govulncheck/-race (podman, §11.4.76), coverage-to-max, docs/courses/website.

close-out[142e] — round 468e: Wave 6 (5-stream subagent-driven batch) — committed + PUSHED

Operator directive 2026-06-04: run 3–5 parallel subagent-driven streams continuously. 6 streams dispatched (4 background implementation + 1 background docs + conductor main-stream docs); conductor-commits model held cleanly (subagents prove RED→GREEN + §1.1 + captured evidence, never git;

conductor re-verifies on the integrated tree per §11.4.108 + commits/pushes, staging disjoint paths per §11.4.84). Concurrency steady at 5 with NO rate-limit this run.

- **W6A helix_qa** `6417753` (→ all 3 upstreams): cleared 3 mutually-masking bank defects — deleted stale clone `full-qa-androidtv-challenges.json` (260 dup FQA-ATV-* ids), fixed `helixagent-cli-agent-tests.yaml:33` YAML parse error, re-IDed 6 `validation-androidtv-focus.json` ids → FQA-ATV-FOCUS-*. `list --banks banks` now EXIT 0, 2601 cases. New `pkg/testbank/loader_test.go` guard. RED→GREEN + §1.1.
- **W6C doc_processor** `c81359e`: `NewBundleTranslator` now loads the flat `active.<locale>.yaml` surface (docprocessor_cli_*); conductor wired `newTranslator()` → real embed-backed `BundleTranslator` — closes the §11.4.108 user-visible layer (CLI renders real localized text, not id-echo). §11.4.120 reconciled the production-wiring test. RED→GREEN + §1.1.
- **W6B helix_code** `d985e3ae` (root): wired real in-process multi-node consensus election through the SSH pool (`consensus.go` +GetNodeID/Reconfigure; new `ssh_pool_consensus.go` WireConsensusCluster over a real InProcessCluster; new test w/ RED_MODE polarity). Cross-host `RemoteVoteTransport` is an HONEST §11.4.6 typed-error seam (NOT a fake stub). `-race -count=3` clean. RED→GREEN + §1.1.
- **W6D llm_provider** `2420e29`: fixed a REAL production data race + cache corruption in `DiscoverModels` (asymmetric defensive-copy contract); §11.4.120 reconciled 32 provider tests that were secretly hitting the live internet API (non-deterministic flaky-green = §11.4.50/§11.4.98 bluff) → assert honest offline contract + zai live-discovery via local httptest. `go test -race` full module EXIT 0, 52 ok, 0 races. RED→GREEN + §1.1.
- **W6E** + **W6F** `31d0d3be` (root): authored `docs/ARCHITECTURE.md` (was CLAUDE.md §11-cited but missing; flags §3.2 staleness — flat→`submodules/` relayout, stale `container-*` targets, snake_case `applications/`) + `docs/DOC_RECONCILIATION_PLAN.2026-06-04.md` (106 root `.md` inventory, Rule-9 over-claim artifacts, SQLite-SSOT-superseded trackers, retire/consolidate plan — execution operator-gated). §11.4.65 `.html`+`.pdf` siblings generated.

Root commits this round: `31d0d3be` (W6E/W6F docs + W6A/W6C pointer bumps), `d985e3ae` (W6B worker consensus), `<this>` (W6D llm_provider pointer bump + CONTINUATION 468e). All pushed github + gitlab.

New backlog surfaced by W6 reports: W6B `RemoteVoteTransport` cross-host impl + `AddWorker` consensus auto-wire; W6C real `sr` flat-bundle; W6D hardcoded `ModelsEndpoint` ignores baseURL in ~30 provider constructors (blocks hermetic live-discovery testing — only zai is baseURL-derivable) + pre-existing `pkg/providers/ollama/ollama_test.go:327` “always pass” comment.

- **Next (W7+):** operator-gated WIRE-or-DELETE of root `internal/{security,fix,testing}` (~2965 LOC) + disabled `cmd/security-test`; then containerized Snyk/SonarQube/govulncheck/`-race` (podman §11.4.76); stress+chaos suites (§11.4.85) for the W6 fixes; coverage-to-max; doc-archival execution per the W6E plan; submodule-internal doc sprawl (helix_code 84, llms_verifier 159).

close-out[142f] — round 468f: Wave 7 (4-stream batch) — committed + PUSHED; W7D deletion VETTED + deferred to next `continue`

Operator directive: continue parallel subagent-driven streams; then “wrap up at this W7A point so we can continue tomorrow by sending `continue`.”

- **W7C doc_processor** `d6eda34`: Serbian flat CLI bundle `active.sr.yaml` (18 `docprocessor_cli_*` keys, natural Serbian Latin, placeholders preserved, no UNCONFIRMED terms). RED→GREEN + §1.1; §11.4.120 reconciled one W6C assertion (sr→done English-fallback was valid only pre-Serbian-bundle).
- **W7B llm_provider** `143237f`: §11.4.85 stress+chaos suite for the W6D discovery cache (8 tests: sustained 400-sample, 24-goroutine concurrent, boundary, 4 chaos classes incl. the caller-mutation-does-not-corrupt regression-guard). `-race -count=3` clean, **no production defect** (W6D defensive-copy fix holds). §1.1 (revert copy → DATA RACE + 270 leaked sentinels). `qa-results/` gitignored (CONST-053).
- **W7A helix_code** `d7a72b96` (root): §11.4.85 stress+chaos suite for the W6B consensus election (120 sustained + 12 concurrent + 1/2/7-node boundary + 4 chaos classes via a test-only faultVoteTransport). `-race -count=3` EXIT 0, 0 races, NEVER-list asserted (no deadlock/leak/panic/livelock + ≤1 leader/term). **No production defect** (W6B bounded-safety holds). §1.1 (quorum→fake-win mutation FAILs hard-errors test).
- **W7D root dead-code** — INVESTIGATED (read-only) → **DELETE-approved by operator, execution DEFERRED to next `continue`.** Full forensic + deletion plan in `docs/W7D_DEAD_CODE_DISPOSITION.2026-06-04.md`. Verdict CONCLUSIVE + SAFE: the live entry points (`helix_code/cmd/{cli,security_test,security_fix}`) all import the INNER `helix_code/internal/{theme,security,fix}`; root `internal/{security,fix,testing,theme}` + `cmd/security-test` have ZERO live consumers, are older (2025-11/2025-12) + abandoned, and are superseded by the inner app + `submodules/security` (canonical own-org). NOT the reverse — we are NOT deleting the real implementation.

Root commits this round: `83f0d4d2` (W7B/W7C pointer bumps), `d7a72b96` (W7A worker stress+chaos), `<this>` (W7D disposition doc + CONTINUATION 468f). doc_processor `c81359e→d6eda34`, llm_provider `2420e29→143237f`. All pushed github + gitlab.

- **NEXT SESSION (continue) — execute in order:** (1) **W7D deletion** per `docs/W7D_DEAD_CODE_DISPOSITION.2026-06-04.md` §3 (final ref-sweep → `git rm` the 5 root targets → root `go build/vet` clean → commit+push). (2) Then resume the W7+ backlog: containerized Snyk/SonarQube/ govulncheck/`-race` (podman §11.4.76); W6D-surfaced `ModelsEndpoint`-baseURL sweep (~30 providers, unblocks hermetic live-discovery testing); W6B `RemoteVoteTransport` cross-host impl + `AddWorker` consensus auto-wire; coverage-to-max; operator-gated mass doc-archival per `docs/DOC_RECONCILIATION_PLAN.2026-06-04.md`; submodule-internal doc sprawl.

close-out[142g] — round 468g: W7D deletion + LIVE infra proof + §11.4.140/141 cascade COMPLETE + real bug fixes — committed + PUSHED

Operator directive: resume under constitution `60e2d66` per §11.4.140/141 + §11.4.126 default autonomous loop; multi-wave subagent-driven (§11.4.70/103). All evidence under `qa-results/*-evidence.txt` (gitignored, CONST-053). Conductor re-verified every subagent claim (§11.4.108) — caught one "3/2 vs 3/3" miscount + one branch-mismatch false-alarm + one incomplete (C1) + stray theme-regen artifacts.

- **W7D deletion** `437714ac` (→ github+gitlab): `git rm` of the entire superseded zero-consumer root Go cluster (~2978 LOC: `internal/{security,fix,testing,theme}` + `cmd/security-test`). Proven SAFE: baseline `go build+vet` GREEN → zero importers (`go list` reverse-dep) → post-deletion build names none of the deleted pkgs. (Root `./internal/...` + `./cmd/...` WAS exactly these 5 pkgs — root is a thin governance module; all live code is inner `helix_code/` + submodules.)
- **LIVE RUNTIME PROOF** (marquee, `qa-results/W2-INFRA-evidence.txt`): PG+Redis booted via podman (machine 5.8.2), `bin/helixcode` (make build, 43.6 MB arm64) connected to real PG+Redis, `GET /health`→**HTTP 200** `{"status":"healthy"}`, `/api/v1/health`→200, auth→401. Gotcha: host port 6379 held by qemu (Android emulator) → used 6380.
- **§11.4.140/141 cascade COMPLETE** — batches 1-6 across **65 owned governance targets** (52 `submodules/*` + 13 already in batch1-2 + `github_pages_website`), each 3/3 governance files, every commit ls-remote-confirmed on its own branch (master|main), root pointers bumped in `e710ddb1`(4)+`b18b549f`(8)+`b3475890`(53). Verifier (F1) confirms 140/141

ALL-PASS fleet-wide. Excluded: `constitution` (had them), `mcp_servers` (THIRD-PARTY modelcontextprotocol/servers — reset, §11.4.51 exempt), `cli_agents/bridle` (unrelated drift).

- **G14 docs_chain pre-build gate** `a9ccc1c6`: `verify-all-constitution-rules.sh` runs `docs_chain verify` (§11.4.106), exit0=PASS/non0=FAIL/ engine-absent=honest-SKIP; §1.1-proven (marker → STALE FAIL → restore → PASS). Same commit: `verify-governance-cascade.sh` COVENANT114_ANCHORS extended → §11.4.103-141; voice `errors.Is` SKIP fix (real bug — == bypassed the no-mic SKIP).
- **Real bug fixes** (all RED → GREEN + §1.1, conductor-verified ok): config `${VAR:default}` expansion (`81f3c482`/C1 — root-caused: Viper does no shell expansion → redis.host literal → “too many colons”); deployment `generateDeploymentID` nanosecond-collision → atomic counter; deployment perf-check `numCPU*10=110%` + throughput=0 → real micro-benchmark + timed CPU spin (`81f3c482`). helixqa banks: **2761 cases, `list-banks` exit 0**.

Root commits this round (all → github+gitlab): `437714ac` (W7D), `e710ddb1` (batch1 ptrs), `a9ccc1c6` (voice+G14+verifier-ext), `b18b549f` (batch2 ptrs), `b3475890` (batch3-6 ptrs / cascade COMPLETE), `81f3c482` (config+deployment fixes), `<this>` (CONTINUATION 468g).

- **READY-BUT-UNCOMMITTED (next `continue`):** `.docs_chain/contexts/{issues,fixed}.yaml` authored (doctor exit0; verify=STALE until `docs_chain sync` generates exports — activate by running sync then commit). `helix_code/internal/config/redis_host_expansion_test.go` RED guard is committed.
- **NEXT SESSION — execute in order:** (1) **§11.4.103–121 backfill** (19 H2 anchors) into all 66 targets ×3 governance files = 198 files (same mechanical cascade as 140/141; anchor text in `constitution/Constitution.md`) + root `CONSTITUTION.md` §11.4.103-141 (39 anchors). (2) **verifier defects** (F1): add `exit $FAILURES` to `verify-governance-cascade.sh` + rewrite stale `docs/improvements/submodule_owned.txt` paths → `submodules/<name>` (69 false-failures). (3) **docs_chain sync** → activate issues/fixed contexts → commit. (4) **§11.4.65 export-regen** of the cascaded submodule `.md` (the 6 cascade waves added §-anchors to CLAUDE/AGENTS/ CONSTITUTION but did NOT regen `.html`/`.pdf` siblings — CONST-066 follow-on). (5) **live e2e challenge runner** (`helix_code/tests/e2e/challenges`, owns PG/Redis). (6) Then W6D `ModelsEndpoint`-baseURL sweep; W6B `RemoteVoteTransport` cross-host.
- **OPERATOR DECISIONS NEEDED (§11.4.122 — did NOT auto-act):** pre-existing uncommitted **deletions** of XSS security content — `cli_agents/bridle` deleted xss-scanner README (425 lines); `submodules/helix_agent` deleted `xss_prevention_challenge.sh` (445 lines) + edited `cli_agents/continue`. Restore (`git checkout`) unless intentional. Also: non-

deterministic theme-color regen writes to the W7D-deleted root `internal/theme/` path (latent — a generator has a stale relative output path).

468g addendum — late findings (ACCURATE state; corrects an optimistic earlier note)

- **XSS content RESTORED** per operator decision: `cli_agents/bridle` README (8939 B) + `submodules/helix_agent/challenges/scripts/xss_prevention_challenge.sh` (18127 B) both `git checkout`-restored. (`submodules/helix_agent` nested `cli_agents/continue` pointer could NOT be restored — its recorded ref `3d264768` no longer exists upstream; pre-existing nested drift, unrelated.)
- **§11.4.103–121 backfill — 52 submodules GENUINELY GREEN** (verifier shows ZERO submodule anchor-gap FAILs; pointers bumped `8b94df0b`). Verifier `scripts/verify-governance-cascade.sh` reports **72 FAILs** = **68 stale-path false-positives** + **only 4 REAL anchor gaps**. The 4 real gaps (NOT yet fixed — 3 finisher agents truncated on the fiddly per-anchor marker-shape matching):
    1. `docs/improvements/submodule_owned.txt` lists 68 owned entries with WRONG paths (bare names + `dependencies/...`) → all 68 "path does not exist" false-FAILs. Fix: rewrite each to `submodules/<name>` (cross-check `.gitmodules`). PURE WIN (no anchor work).
    2. `github_pages_website` CLAUDE.md/AGENTS.md/CONSTITUTION.md missing ~12 anchors (103,105,107,108,110,112,113,114,115,117,118,119) — add in the verifier's exact marker shapes, commit+push+bump pointer.
    3. root `CONSTITUTION.md` missing §11.4.103–141 (39 anchors) — root CLAUDE.md/AGENTS.md HAVE them; copy equivalent blocks in the verifier's exact per-anchor marker shapes (`## §11.4.NNN –` 103-134; `### §11.4.NNN` 122-134 wait—read the verifier's expected-marker list, shapes vary 135-141).
    4. `scripts/verify-governance-cascade.sh` lacks `exit $FAILURES` (always exits 0 → not a real gate). Add it WITH items 1-3 so the gate flips straight to green (avoid leaving a known-red gate).
- **e2e challenge runner — BLOCKED on Ollama (environmental, NOT a code defect):** `helix_code/tests/e2e/challenges/cmd/runner` ran all 6 challenges live against real PG+Redis; all 6 FAILed with `Ollama API unavailable: localhost:11434 connection refused`. The runner correctly has NO simulation fallback (good anti-bluff design). To get real e2e proof: start `ollama serve` + `ollama pull <model>` OR configure a cloud provider key (`api-keys.yaml`). OPERATOR/ENV action needed. (Invocation: `cd helix_code/tests/e2e/challenges && go run ./cmd/runner -providers ollama -models <m> -verbose` — there is NO `-all` flag; omit `-challenge` to run all 6.)

- **docs_chain issues/fixed contexts** authored (`.docs_chain/contexts/{issues,fixed}.yaml`, UNCOMMITTED) — `doctor` exit0; `verify`=STALE until `docs_chain sync` generates exports. Activate: run `docs_chain sync` → commit contexts + generated exports.
- **NEXT SESSION precise order:** (1) verifier-green finish — the 4 items above (item 1 first = removes 68 false-FAILs; then 2+3 anchor backfill in exact marker shapes; then 4 exit-fix; re-run → 0 FAILs). (2) e2e: start Ollama/cloud key → re-run runner for real §11.4.5/69/107 proof. (3) docs_chain sync → activate+commit issues/fixed contexts. (4) §11.4.65 export-regen across the ~65 cascaded submodules (the cascade added `.md` anchors but did NOT regen `.html`/`.pdf` siblings — CONST-066 follow-on; per-submodule commit+push). (5) backlog: W6D `ModelsEndpoint`-baseURL sweep; W6B `RemoteVoteTransport` cross-host.

468g FINAL state (accurate — session ended at account 1M-context credit ceiling)

The §11.4.103-121 H2-anchor requirement: the verifier wants each anchor as a literal `## §11.4.NNN` H2 heading. The earlier B-batch agents added several anchors (103,105,107,108,110,112-119) only as INLINE refs, which the stale `submodule_owned.txt` paths masked (verifier checked wrong paths). With paths now FIXED (`bfcc0562`), the verifier honestly shows the real gap. Progress this round: - root `CONSTITUTION.md` §11.4.103-141 backfilled → PASS (`bfcc0562`). - `github_pages_website` → PASS (`65157cf`). - `submodule_owned.txt` 68 paths corrected → 0 false-positives (`bfcc0562`). - **16 submodules** H2-anchor-fixed + pushed + pointers bumped (`5c41eebd`): llms_verifier mcp_module memory messaging middleware models normalize observability optimization ouroborous panoptic planning plinius_common plugins security vision_engine. - ~27 submodules (B1-B4 set: agentic auth … document, filesystem … leak_hub, rag … watcher) already had correct H2 anchors — PASS. - **9 submodules STILL NEED the H2-anchor fix** (D1 batch hit the account credit ceiling, produced nothing): **challenges containers debate_orchestrator doc_processor event_bus helix_agent llm_ops llm_orchestrator llm_provider**. For each: read its 3 files' `## §11.4.NNN` missing lists from the verifier output, append those anchors as `## §11.4.NNN — <title>` H2 blocks (match the existing §11.4.122 H2 format; titles from root CLAUDE.md §11.4.103-121), commit governance-only, push on its branch, bump root pointer. - THEN add `exit $FAILURES` to `scripts/verify-governance-cascade.sh` (item 4) and re-run → must reach 0 failures → THEN the gate enforces. - **RESUMPTION (paste into fresh session):** "Read docs/CONTINUATION.md close-out 468g FINAL + qa-results/D2-FIX-evidence.txt for the proven H2-anchor pattern, then: (1) run `bash scripts/verify-governance-cascade.sh 2>&1 | tail` to get the exact per-file `## §11.4.NNN` missing lists for the 9 remaining submodules (challenges containers debate_orchestrator doc_processor event_bus helix_agent llm_ops llm_orchestrator llm_provider); (2)

add the missing anchors as `## §11.4.NNN —` H2 headings (match existing §11.4.122 format, titles from root CLAUDE.md), commit governance-only + push each on its branch (ff-only, §11.4.113) + bump root pointers; (3) add `exit $FAILURES` to the verifier; (4) re-run → 0 failures; (5) then e2e (start Ollama), docs_chain sync, §11.4.65 export-regen. Anti-bluff §11.4: conductor-verify every subagent claim via ls-remote before bumping pointers."

Active Work — Xiaomi MiMo Integration (2026-06-19)

Status: IMPLEMENTED

Completed: - Xiaomi MiMo provider implementation (`helix_code/internal/llm/xiaomi_provider.go`) - ASR + TTS endpoint support - Factory, key recognition, hosted catalogue, verifier registration - reasoning_content field fix - Integration tests (4/4 live API) - Stress + chaos tests (9/9) - Feature status documentation - HelixQA test bank (8 tests) - E2E challenge script - Dev config entry - 3 recordings in `/Volumes/T7/Downloads/Recordings/` - Pushed to GitHub + GitLab (20 commits)

In Progress: - LLMsVerifier submodule integration (agent running) - HelixAgent submodule integration (agent running)

Known Issues: - V2 models deprecated 2026-06-30 (auto-routes to V2.5) - Rate limits: 100 RPM on free tier

Next Steps: - Complete LLMsVerifier + HelixAgent submodule integration - Push submodule pointer updates - Doc exports (HTML/PDF/DOCX)

close-out — 2026-06-22: §11.4.166 semgrep-mandate REPEAL + meta-repo de-wiring + §11.4.55 sweep-debt batch

Operator decision (2026-06-22): semgrep is no longer a mandatory requirement. All work below is FACT — already committed + pushed before this CONTINUATION update; the conductor reviews + commits this doc edit.

What landed

- **§11.4.166 REPEAL (constitution `4ff4985` → `2953179`):** §11.4.166 replaced with a visible REPEALED tombstone across `Constitution.md` + `CLAUDE.md` + `AGENTS.md` + `QWEN.md` + `GEMINI.md` (metadata Revisions bumped, html/pdf regenerated). Removed `scripts/semgrep`, `scripts/hooks/`

`semgrep_precommit.sh`, the `submodules/semgrep` git submodule, `docs/semgrep`, `docs/.semgrep`, and `.docs_chain/contexts/semgrep_status.yaml`. Pushed to all 6 constitution upstreams. Independent §11.4.142 review GO.
- **Meta-repo de-wiring (`26dd71b0` → `8dd43e9d`):** removed the semgrep MCP server from `.mcp.json`; deleted `.docs_chain/contexts/semgrep_status.yaml` + `docs/semgrep/Status.md`; dropped the retired 11.4.166 from `scripts/verify-governance-cascade.sh` expected-anchor list (anchor ceiling now 165); bumped the constitution submodule pointer. Pushed github + gitlab.
- **KEPT (NOT a semgrep dependency, per §11.4.124):** `buildscript_provenance_guard_test.go` + the 4 `// nosemgrep` annotations in `helix_code/internal/llm` are a real command-injection security guard, independent of semgrep — retained.
- **§11.4.55 sweep-debt batch (`8dd43e9d` → `e3a71d97`, 14 files; independent §11.4.142 review GO; pushed github + gitlab):**
    - **G8** — §11.4.120 obsolete-details gate reconciliation: `REASON_VOCAB` now mirrors §11.4.90 (incl. `not-reproducible`); paired meta-test positive case added; `docs/Fixed.md` HXC-044 Triple-check token fixed; gate now PASSes.
    - **#22** — `CLAUDE.md §3.2`: `assets/` corrected to tracked-dir-not-submodule; root-level submodule lines corrected to the `submodules/` grouped layout.
    - **G12** — `Issues_Summary` + `Fixed_Summary` regenerated, both `--check` PASS.
    - **G14** — html/pdf siblings regenerated via docs_chain fixed + governance contexts + a direct Issues_Summary export.

State

> **SUPERSEDED — historical record of 2026-06-22 only. Do NOT act on this gate list.** Current gate state (verified 2026-07-28 against `qa-results/full_retest/verify_rules_20260727T184002Z.log`): **16 gates, 1 failure — G7 alone.** G1, G8, G11, G12 and G13 **all PASS**; G14 is a SKIP-OK (docs_chain engine absent). The G11 DB divergence recorded below was resolved (G11 now reports "committed DB validates + md⟷db byte-identical in sync"). See the Status summary at the head of this document and `RESUME.md` rev 5.

- **Sweep delta: 6 failing gates → 5** (G8 + G12 now PASS, no regression from the semgrep removal). *(figure as of 2026-06-22)*
- **Remaining failing gates** (all operator-gated / out of scope for this batch): G1 (fleet cascade, out of scope), G7 (docs/qa evidence backlog), G13 (Sources-verified footers), G11 (the newly-surfaced DB divergence below). *(as of 2026-06-22 — G1/G11/G13 have since PASSED; only G7 still fails)*

- **G11 DISCOVERED (operator-gated, NOT touched):** `docs/workable_items.db` has a large PRE-EXISTING bidirectional divergence from `Issues.md`/`Fixed.md` (over 40 items, incl. a type conflict HXC-036 md=Bug db=Task). docs_chain's issues-context correctly refused a silent merge (authority `items_db`). Resolution direction is an operator decision; the tracked SSoT DB was left pristine.

Next

> **SUPERSEDED — these "next" items are from 2026-06-22 and are mostly DONE.** G11 was reconciled (gate PASSes, DB ⟷ md byte-identical); G13 PASSes at 82/82; G1 PASSes. Only **G7** remains, and its disposition is now DECIDED (operator, 2026-07-28): clear it by real retrospective QA runs, the 3 security commits first — not a baseline bump.

- **Operator decision needed (G11):** choose the reconciliation direction for `docs/workable_items.db ↔ Issues.md/Fixed.md` (incl. the HXC-036 type conflict) before any sync runs. *(RESOLVED — G11 PASSes)*
- G7 / G13 / G1 remain operator-gated. *(G13 + G1 now PASS; G7 decided 2026-07-28)*
- **In flight (separate background stream):** container distribution config update per CONST-045 — `thinker.local` + `amber.local` enrolled, `nezha.local` distribution-disabled.

2026-07-27 — Test-suite remediation cycle (PAUSED mid-flight, operator session limit)

> **SUPERSEDED HEADER — the state line below describes 2026-07-27 mid-cycle, NOT today.** Verified 2026-07-28: meta HEAD is **`0a4eb8d0`** (not `66d6fb29`), `git log @{u}..HEAD` is **EMPTY** — the work described here **was committed and pushed**. helix_agent `0165ab1d`, helix_qa `88ef0579`, challenges `072724af`. Constitution checkout **`32d75788`**, pin **`731bf1d3`** — **6 commits behind, not** 79. The working tree holds **~26 porcelain entries** (volatile) of newer in-flight work. Read the Status summary at the head of this document and `RESUME.md` rev 5 for live state.

Base (as of 2026-07-27): meta `66d6fb29` · helix_agent `36597718` · helix_qa `a729b51` · constitution checked-out `b00ab1c` (pin `7295a189`, 79 behind). **Everything below was UNCOMMITTED at the time of writing (2026-07-27).** It has since been committed and pushed — see the banner above. **Canonical resumption entry point:** `RESUME.md` (**now rev 5**; was rev 2 when this section was written). **Captured evidence:** `docs/qa/test_remediation_20260727T124352Z/README.{md,html,pdf}`.

Done

- **18 packages closed with captured runtime evidence**, each root-caused before fix; paired §1.1 mutation wherever a gate was touched. No assertion weakened, no bare `t.Skip()` added.
- **3 PRODUCTION defects fixed** (highest review priority — see below):
  1. `internal/clis/agents/master/master.go` — `ListAvailable()` returned `nil` not `[]` → JSON `null` to clients.
  2. `docker/mcp/docker-compose.mcp-servers.yml` — build contexts off by one dir since the grouped-submodule move; all 7 core MCP services would fail `compose build`.
  3. `internal/services/provider_registry.go` — credential-free `zen` provider materialised inside `NewProviderRegistryWithoutAutoDiscovery` registries. Security-relevant: an approved-providers-only deployment would silently route prompts to an unapproved public endpoint.
- **Governance:** 42 anchors §11.4.193–234 cascaded into all 6 carriers — verifier ceiling raised FIRST so the fix could not recreate the blind spot; now 277 PASS/0 FAIL, mutation-proven stronger than the old `grep -F`. G12 reconciled per §11.4.120. 12 carrier HTML/PDF exports regenerated (CONST-066). `helix_code/.env` 644→600 (CONST-042).
- **Review remediation:** independent review of the main-stream changes returned GO; all 5 warnings + 3 nits closed with evidence (incl. W1, where the in-code rationale I had written was factually wrong in 5 files).

State

- **The inherited "24 failing packages" figure was stale in BOTH directions** — two entries were already green, `internal/verifier` had newly regressed, `internal/search` had recovered. Re-derive; never trust a prior run's log.
- **Dominant defect class: tests whose verdict depended on ambient host state** (§11.4.50) — SonarQube being down, `init.defaultBranch`, CLI binaries on PATH, Ollama models pulled, port 8100 ownership. They passed on the authoring machine and failed here; **bringing the platform live is what exposed them.**
- **Sweep is NOT yet green.** `tests/automation` fails one subtest: 507 of 2422 tracked Go files are not gofmt-clean (pre-existing, whitespace-only, previously hidden because the gate only scanned its own 4 files).
- **Pending §11.4.142 gap:** the subagent-authored changes (incl. all 3 production fixes) have NOT had independent review — author self-

verification does not satisfy it. A review agent was dispatched and stopped mid-run at pause.

- Still-open harness bluff: `internal/testutil/checkHTTP` treats any status < 500 as "available", green-lighting a 404-answering neighbour instead of skipping.

Next

1. Independent review of the subagent changes (§11.4.142) — three production changes are the priority.
2. `gofmt -w $(git ls-files '*.go')` in helix_agent — **separate commit** (507 files; folding it in makes the fix commit unreviewable).
3. Full sweep: `-count=1` throughout, `-tags=nogui` for the inner module (without it, `applications/*` produce FALSE `[build failed]` from missing X11 headers — the inherited log's mistake).
4. Only if green: §11.4.84 quiescence, commit, tag `helix-code-1.2.0-dev-0.0.1` on main repo + every owned submodule with the identical prefix (§11.4.151), ff-only push (§11.4.113 — force-push absolutely forbidden). Operator authorised the tag CONDITIONAL on green.

Operator decisions blocking closure

> **PARTIALLY SUPERSEDED.** Four decisions were taken on 2026-07-28 (see the Status summary): `helixcode-infra` becomes sole infra owner (`HELIX_AUTOBOOT_INFRA=false`); G7 cleared by real retrospective QA runs, security commits first; `helixcode-server` secret env names reconciled; tag only when genuinely green. The pin figure below is also stale — it is **6 behind, not 79**.

- **Port collision:** registry says HelixAgent owns 8100; `configs/development.yaml` says 7061; LLMsVerifier occupies 8100. Beyond tests — `helixagent --generate-agent-config=opencode` emits `baseURL: http://localhost:8100/v1`, so shipped CLI-agent configs point users at the wrong service. *(STILL OPEN — verified 2026-07-28:* `llm-verifier` *holds :8100 and* `helixagent` *holds :7061, exactly as described.)*
- **Constitution pin** 79 commits behind (clean FF) — advancing needs a commit. *(STALE FIGURE — now* `731bf1d3` *vs checkout* `32d75788` *=* ***6 behind****. Still open: the 2026-07-27 decision to advance it in this release is not yet executed. Note the pin predates the merge that brought §11.4.235, which is why the six root carriers still top out at §11.4.234 — an open §11.4.157 lockstep gap.)*
- **Canon defect:** §11.4.64 / §11.4.205 / §11.4.206 cited but never defined in `constitution/Constitution.md`.

- **CONST-053:** `reports/latency/p99-baseline-2026-03-16.txt` + `releases/.version-data/helixagent.last-hash` are tracked files rewritten by tests. Exclude from staging; never `git add -A`.
- **`.docs_chain/contexts/{issues,fixed}.yaml`** still point the legacy summary generators at Markdown rather than the DB — inert only because the engine is absent; would reproduce the corruption G12 now blocks.